IELTS ACADEMIC READING

Volume 7

10 complete Academic Reading tests — three full passages and 40 questions each — with a complete answer key and passage evidence for every answer

IB

10
FULL TESTS

400
QUESTIONS

Full
ANSWER KEY

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Introduction

Welcome to **IELTS Academic Reading — Volume 7**, a printable practice book of 10 complete Reading tests, compiled and prepared by Teacher Ibodullaev Bekhruz.

Every test follows the real exam format: three passages of increasing difficulty and 40 questions covering the full range of IELTS question types — True/False/Not Given, Matching Headings, Matching Information, Multiple Choice, Summary and Sentence Completion, and more. Answer circles and blank lines are left for you to write your answers, just like on test day.

Part One contains all 10 tests with questions only, so you can practise without seeing any answers. **Part Two** contains the complete Answer Key for each test, with the keywords from each question and the exact sentence in the passage that proves the correct answer.

How to use this book:

- Sit each test in one go, under timed conditions — 60 minutes for all three passages.
- Write your answers directly in the circles and blanks provided.
- Check your answers in Part Two and read the passage evidence for every question you got wrong.
- Use the band-score table on the answer key pages to estimate your result, then revisit weak question types.

Good luck with your preparation, and follow **@bekhruzposts** for more free IELTS materials.

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IELTS Academic Reading — Test 1

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Carnivorous Plant

A They attract insects and then eat their flesh. Is that any way for a plant to behave? The naturalist and author of Origin of Species, Charles Darwin, was fascinated by carnivorous plants. In 1860, soon after he came across his first carnivorous plant the sundew, Drosera he wrote, 'I care more about Drosera than the origin of all the species in the world.' He spent months running experiments on the plants. He dropped flies and bits of meat on their leaves and watched them slowly fold their sticky tentacles over their prey. He thought it incredible that brushing a leaf with a single strand of human hair was enough to bring about a response. Yet sundews, he observed, ignored raindrops. To react to such a false alarm, he reasoned, would obviously be a great evil to the plant. This was no accident. This was adaptation.

B Darwin expanded his studies from sundews to other species in his book Insectivorous Plants. He was amazed at the quickness and power of the Venus flytrap. He showed that when one of its leaves snapped shut, it formed itself into a temporary 'stomach', secreting enzymes that could dissolve the prey. He noted that a leaf took more than a week to reopen after closing, and reasoned that the interlocking spines along the margin of the leaf allowed tiny insects to escape, saving the plant the expense of digesting an insufficient meal.

C Today, biologists using 21st-century tools to study cells and DNA are beginning to understand how these plants hunt, eat, and digest and how these strange adaptations came about in the first place. Alexander Volkov, a plant physiologist at Oakwood University in Alabama, believes he has figured out the Venus flytrap's secret. 'This,' Volkov declares, 'is an electrical plant.'

D When an insect brushes against a hair on the leaf of a Venus flytrap, the movement sets off an electric charge. The charge builds up inside the tissue of the leaf but is not enough to stimulate the snap, which keeps the Venus flytrap from reacting to false alarms, such as raindrops. An insect, however, is likely to brush a second hair, adding enough electric charge for the leaf to close. Volkov's experiments reveal that the electric charge travels down fluid-filled tunnels in a leaf, which opens up pores in cell membranes. Water rushes

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from the cells on the inside of the leaf to those on the outside, causing the leaf to rapidly flip in shape from convex to concave, like a soft contact lens. As the leaves flip, they snap together, trapping an insect inside.

E The bladderwort plant has an equally sophisticated way of setting its underwater trap. It pumps water out of tiny air sacs or bladders, lowering the pressure inside. When a water flea or some other small creature swims past, it bends hairs on the bladder, causing a flap to spring apart. The low pressure sucks water in, carrying the creature along with it. In one five-hundredth of a second, the flap swings shut again. The cells in the bladder then begin to pump water out again, creating a new vacuum.

F Many other species of carnivorous plants act like living flypaper, catching animals on sticky tentacles. Pitcher plants use yet another strategy, growing long tube-shaped leaves into which insects fall. Some of the largest have pitchers up to 30cm deep and can consume whole frogs unlucky enough to fall into them. Sophisticated chemistry helps make the pitcher a death trap.

G Nicholas Gotelli, of the University of Vermont, is trying to figure out what evolutionary forces pushed these plants towards meat. Carnivorous plants clearly benefit from eating animals; when scientists feed pitcher plants extra bugs, the plants get bigger. But the benefits of eating flesh are not the ones you might expect. Carnivorous animals, like ourselves, use the carbon in protein and the fat in meat to build muscles and store energy. Carnivorous plants, however, take nitrogen and phosphorus from the flesh in order to build light-harvesting enzymes. Eating animals, in other words, lets carnivorous plants do what all plants do: grow by taking energy directly from the sun.

H Unfortunately, they do a really bad job of it. That's because they have to use a lot of energy to make the equipment they need to catch animals - the enzymes, the pumps, the sticky tentacles, and so on. A pitcher or a flytrap is not very good at photosynthesis because, unlike plants with ordinary leaves, they do not have flat solar panels that can absorb lots of sunlight. Gotelli suspects that only under special conditions are the benefits of being carnivorous greater than the costs. The poor soil of bogs and swamps, where many carnivorous plants grow, offers little nitrogen and phosphorus, so carnivorous plants enjoy an advantage there over 'conventional' plants. Also, bogs are often flooded with sunshine, so even an inefficient carnivorous plant can carry out enough photosynthesis to survive. 'They're stuck, and they're making the best of it,' says Aaron Ellison of Harvard University.

I Unfortunately, the adaptations that enable carnivorous plants to survive in harsh habitats also make them extremely sensitive to environmental changes. Chemical fertilizers used in agriculture and pollution from power plants are adding extra nitrogen to many bogs in North America. Carnivorous plants are so finely adapted to low levels of nitrogen that this extra fertilizer is overloading their systems. Humans also threaten

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carnivorous plants in other ways. The black market trade in exotic carnivorous plants is strong, but even if this can be prevented, carnivorous plants will continue to suffer from other dangers. Their habitat is disappearing, to be replaced by shopping centers and houses. The suppression of wildfires by government agencies allows other plants to grow quickly and outcompete the Venus flytraps. Good news, perhaps, for flies. But a loss for all who delight in the inventiveness of evolution.

Questions 1–5

Complete the notes below. Write **ONE WORD ONLY** *from the passage for each answer.*

Carnivorous Plants

DARWIN'S EXPERIMENTS

Drosera

- He put insects and small amounts of meat onto leaves and observed what happened.
- He was amazed that one human hair triggered a response.
- He understood why the plant did not respond to (1) _____.

Venus Flytrap

- He demonstrated how leaves close and then, for a short period, act like a (2) _____.
- He noted the structure of leaves ensures small insects are released.

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Venus Flytrap

- If insects come into contact with leaf hairs, their movement produces an electric charge.
- The charge enters fluid-filled tunnels, causing (3) _____ in the cell walls to open.
- Water moves between cells and then the leaf changes from convex to concave, trapping the insect inside.

Bladderwort

- Hairs bend when insects swim past, resulting in opening of a flap.
- Water is sucked in.
- When water is pumped out of bladder cells, a (4) _____ builds up inside.

Pitcher Plants

- The plants have tube-shaped leaves that insects fall into.
- Some plants are big enough to capture and eat (5) _____.

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Questions 6–13

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

6 Pitcher plants increase in size after they have digested a lot of insects.

7 Carnivorous plants produce light-harvesting enzymes with the nutrients they extract from animals.

8 Pitcher plants and Venus flytraps are more efficient at photosynthesis than plants with ordinary leaves.

9 Venus flytraps are better adapted to the soil of swamps and bogs than other carnivorous plants.

10 Carnivorous plants frequently find it difficult to photosynthesise in bogs due to a lack of sunlight.

11 Scientists have campaigned to reduce the amount of nitrogen that is released into the soil by agricultural practices.

12 A lot of exotic carnivorous plants are sold illegally.

13 Preventing wildfires is beneficial to the Venus flytrap.

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Reading Passage 2

Drawing Lessons from History

A Numerous ancient civilisations collapsed or disappeared, leaving behind grand ruins reminiscent of what poet Shelley described in his sonnet Ozymandias. The term 'collapse' refers to a significant reduction in population and/or political, economic, and social complexity of a society across a wide area for an extended period.

B The awe-inspiring ruins left behind by these past societies captivate us all. As children, we marvel at them when we first discover them through pictures-vast stone structures, intricate carvings, or towering temples that defy imagination. As we grow older, many of us plan vacations to witness these remarkable sites firsthand, standing in silent reverence before the remnants of a world that once thrived. We are drawn to their magnificent and haunting beauty, as well as the enigmas they present. The grandeur of these ruins attests to the previous wealth and power of their builders, yet their sudden disappearance challenges our understanding of human resilience. These builders vanished, abandoning the remarkable structures they had painstakingly constructed. How did a once-mighty society end up collapsing, leaving behind only echoes of its former glory?

C For a long time, it has been suspected that the mysterious disappearance of many such societies was, to some extent, triggered by ecological issues unintentional destruction of the environmental resources on which their societies relied. This suspicion of unintended ecological suicide, or ecocide, has been substantiated by discoveries made in recent decades by archaeologists, climatologists, historians, palaeontologists, and palynologists (pollen scientists). The processes through which past societies have undermined themselves by damaging their environments can be categorised into eight areas, which vary in significance from one case to another: deforestation and habitat loss, soil degradation, water management challenges, overhunting, overfishing, the impacts of invasive species on native ones, human population growth, and increased human activity.

D Historical collapses generally followed similar patterns resembling variations on a theme. These patterns often include resource depletion, environmental degradation, internal strife, or external pressures, yet the specific combination and intensity of these factors varied across civilisations. Writers are often tempted to draw analogies between the trajectory of human society and that of an individual's life, speaking of a society's birth, growth, peak, old age, and eventual demise. However, this metaphor proves inaccurate for many past societies: they experienced rapid declines after reaching their peak in terms of population and power, and these rapid declines must have come as a surprise and shock to their inhabitants. It is evident that not all past societies followed this trajectory uniformly to its conclusion: different societies collapsed to varying degrees and in somewhat distinct

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ways while many societies did not collapse at all. The diversity in collapse mechanisms suggests that while certain systemic vulnerabilities may be universal, their expression is deeply contingent on local conditions, cultural responses, and the interplay of multiple stressors over time.

E Currently, many individuals believe that environmental problems eclipse other global threats. These environmental problems encompass the same eight factors that contributed to past societies' downfalls, plus four new ones: human-induced climate change, the accumulation of toxic chemicals, energy shortages, and the complete utilisation of the Earth's photosynthetic capacity. However, the seriousness of these modern environmental dilemmas is vigorously debated among the scientific community. Are the perceived risks overstated or undervalued? Will contemporary technology resolve our issues, or is it producing new challenges faster than it can remedy old ones? When one resource (such as wood, oil, or ocean fish) is depleted, can we trust that new alternatives (like plastics, wind and solar energy, or farmed fish) will emerge? Isn't the rate of human population growth slowing down, suggesting we might soon reach a manageable global population?

F Questions like these highlight why the well-known collapses of past civilisations carry more significance than merely being romantic mysteries. There may be practical lessons to be learnt from these historical failures. However, there are also differences between the modern world and its problems, and those ancient societies and their problems. It would be naive to assume that studying the past will yield straightforward answers directly applicable to today's world. We have certain advantages over past societies that reduce our risks, including our powerful technology (i.e. its beneficial effects), globalisation, modern medicine, and enhanced awareness of both past and distant modern societies. At the same time, there are also factors that heighten our risks in comparison to earlier civilisations: again, our potent technology (i.e. its unintended destructive effects), globalisation (where a problem in one area can affect the entire world), the reliance of millions on modern health care for survival, and our vastly larger human population. Perhaps we can still learn from the past, but only if we carefully contemplate its lessons.

Questions 14–16

Choose the correct answer, **A**, **B**, **C** *or* **D**.

14 In describing the appeal of monumental ruins, the writer highlights

A the curiosity of historians about the original purpose of these ruins.

B the vast geographical areas covered by these structures.

C their profound impact on the field of archaeological research.

D the mixture of beauty and mystery these sites represent.

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15 Recent findings regarding lost civilisations have

A upended established theories.

B triggered a heated debate among scientists.

C emerged from a range of disciplines.

D confirmed a major cause of environmental destruction.

16 In discussing how previous societies fell, the writer mentions that

A the pace of collapse showed some resemblance across societies.

B the decline of societies was usually a slow and predictable process.

C all societies peaked in terms of population and power before collapsing.

D ordinary people could occasionally affect the course of events.

Questions 17–21

Choose **YES** *if the statement agrees with the claims of the writer,* **NO** *if the statement contradicts the claims of the writer, or* **NOT GIVEN** *if it is impossible to say what the writer thinks about this.*

17 The impressive relics of ancient societies were influenced by Shelley's sonnet Ozymandias.

18 There is a widespread belief that environmental issues pose the principal hazard to the contemporary world.

19 The presence of poisonous substances in high concentrations is a relatively modern issue.

20 Scientists have reached a consensus about how severe the present ecological concerns are.

21 We should reflect cautiously when comparing the past with the present.

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Questions 22–26

Complete each sentence with the correct ending, **A–F**, *below.*

LIST OF ENDINGS

A are not applicable to all societies.
B provides a hopeful perspective.
C is embodied in physical structures.
D is potentially both positive and negative.
E will not address current issues directly.
F have increased compared to the past.

Q	Sentence start	Answer
22	Evidence of remarkable achievements of certain past civilisations	
23	Analogies between the development of an individual and that of a society	
24	The environmental challenges that societies encounter	
25	In modern society, powerful technology	
26	Reflections on ancient events and tendencies	

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Reading Passage 3

Neanderthal Technology

A We think of our prehistoric ancestors as people of the ice and snow, living in caves, and for many of the west European Neanderthalers that is a just picture of their life. But where there were no caves, further to the east on the Russian steppe, for example, open-air sites with some sort of constructed shelter were the only option. We now know much more about the cave sites than the open-air ones because, historically, it was the cave sites of Western Europe that were first explored by archaeologists and also because open-air sites are harder to find - many of them have disappeared under deep mud deposits or under the rising postglacial seas. Caves, moreover, aid the survival of archaeological material and can preserve the records of remote millennia.

B In south-west France, the limestone caves of the Périgord region made ideal homes for the Neanderthal people. There were good supplies of flint to hand for axes and the like, and the caves were often sited in small river valleys that offered protection against the worst of the weather. The Neanderthalers liked south-facing caves, for obvious reasons of sunshine and wind avoidance, and caves at some height above the valley floor offered refuge from floods and good game-watching vantage points. The Périgord region during the last ice age was, in fact, an exceptionally benign habitat for humans. It enjoyed a rather maritime climate with cooler summers that permitted the extension of tundra and steppe over its higher plateaux, and its year-round high levels of sunshine favoured the growth of the ground plants needed by reindeer, bison and horse. Winters were mildish for the ice age, animals never needed to migrate far from summer to winter, and men never needed to travel far from home to find abundant supplies of meat.

C In Central and Eastern Europe, where caves were unavailable, such open-air sites as have been discovered were mostly located near water - both because this was a good area to be for people and animals, and also because the sedimentation potential of lakes and stream courses has aided archaeological preservation - whereas erosion has presumably blown away sites which were out in the open. Some of the open-air sites in Germany, Central Europe and Russia have provided valuable information about Neanderthal man and his way of life. From Moldova, for example, comes evidence that has been interpreted as the remains of wind-break structures, or even a large tent: a ring, up to about 8x5m in size, of mainly mammoth bones enclosing a dense concentration of stone tools, animal bones and ash.

D From the west European caves more evidence of built structures is available, and some of it goes back a long way in time. In the Grotte du Lazaret, near Nice, at a date during the last ice age but one, claims for some sort of skin tent within the cave have been advanced,

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on the basis of arrangements of large stones out from the cave wall that might have supported timber struts for a covering of skins up to the rock face above. At Lazaret, what might be openings in the hypothesised tents seem to point away from the cave mouth, and finds of wolf and fox foot bones, without the rest of the skeletons, inside these 'tents' have been thought to indicate the use of animal pelts as bed coverings. The two patches of ash at Lazaret that mark ancient fires, with stone tools around them evidently made and used on the spot, are edged with small marine mollusc shells, prompting the excavator to suggest that seaweed had been used as bedding around the fires. The cave of Baume-Bonne in the Basses-Alpes region of France, another early site, boasts ten square metres of cobbles brought up from the local river and laid down, as though to take care of a puddle area in the cave, with the smoothest and roundest surfaces of the stones uppermost, and there are other similar cases.

E The ash encountered in concentrations at some sites testifies to the Neanderthal people's use of fire: not surprising, since use of fire was, by Neanderthal times, an already ancient accomplishment of evolving humanity, and survival in the sub-arctic conditions faced by the Neanderthalers is inconceivable without control of fire. Fire gave warmth, light, heat for cooking and defence against predatory animals. A charred piece of birch from Krapina in Croatia, is thought to be the remains of a fire-making twirl stick. But Neanderthal hearths, in the sense of specially constructed places for fire, are fewer and harder to identify with certainty than the mere ash piles that are a regular feature of their sites. They seem often to have just lit a small fire (40-50cm across) on the existing ground surface of the cave, without preparation. Judging from the shallow penetration of heat effects under the ash, this fire was only of a short duration. Sometimes the fires were larger in size, up to one metre across, and quite irregular in shape. It is not always easy to decide how much additional structure some fires possessed: claims of stone circles to contain the fire run up against the fact that stones tend to litter the cave floors everywhere and those around a fire can quite accidentally look as though they were arranged in a circle.

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Questions 27–31

Reading Passage 3 has five sections. Choose the correct heading for each section from the list of headings below.

LIST OF HEADINGS

- **i** Evidence of outdoor dwellings
- **ii** Learning to make fire
- **iii** A perfect place to live
- **iv** Examining the cave contents
- **v** Contrasting two types of home
- **vi** A vital source of power

Q	Section	Answer
27	Section A	
28	Section B	
29	Section C	
30	Section D	
31	Section E	

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Questions 32–36

Look at the following findings and the list of places below. Match each finding with the correct place, **A–E**. *You may use any letter more than once.*

LIST OF PLACES

A The Périgord region
B Moldova
C The Grotte du Lazaret
D The cave of Baume-Bonne
E Krapina

Q	Finding	Answer
32	a burnt piece of wood	
33	evidence of efforts to prevent pools of water forming	
34	the remains of sea creatures	
35	a circular arrangement of animal bones	
36	evidence suggesting the use of animal fur for warmth	

Questions 37–39

Complete the summary below. Write **ONE WORD ONLY** *from the passage for each answer.*

The use of fire

Neanderthalers could not have survived without fire because the conditions they lived in were (37) ————————. Most evidence of purpose-built fires takes the form of ash piles, features of which suggest that the fires lasted a (38) ———————— time. It is hard to be certain about the size and structure of the fires, though they were certainly needed to protect the occupants from dangerous (39) ————————, among other things.

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Question 40

Choose the correct answer, **A**, **B**, **C** *or* **D**.

40 The purpose of the writer of this article is to

A argue that Neanderthal homes were bigger than originally thought.

B explain why Neanderthal people migrated to Western Europe.

C discuss what is known about Neanderthal settlements.

D track the progress of early Neanderthal development.

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IELTS Academic Reading — Test 2

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Reading Passage 1

Wood: a Valuable Resource in New Zealand's Economy

During the settlement of New Zealand by European immigrants, natural timbers played a major role. Wood was easily accessible and relatively cheap. A tradition of wooden houses arose, supported by the recognition that they were less likely to collapse suddenly during earthquakes, a not infrequent event in this part of the world. But in addition to demand from the domestic market, there was also a demand for forest products from overseas.

Early explorers recognised the suitability of the tall, straight trunks of the kauri for constructing sailing vessels. The kauri is a species of coniferous tree found only in small areas of the southern hemisphere. So from the early 1800s, huge amounts of this type of wood were sold to Australia and the UK for that purpose. For a period, the forestry industry was the country's major export earner, but the rate of harvest was unsustainable and, by the beginning of the 20th century, indigenous timber exports were rapidly declining.

From the 1940s, newly established plantations of an imported species of tree called radiata pine supplied timber and other wood products in increasing quantities. By the 1960s, plantation-grown timber was providing most of the country's sawn timber needs, especially for construction. Today, less than two percent of timber is cut from indigenous forests, and almost all of that is used for higher-value end uses, such as furniture and fittings. As the pine industry developed, it became apparent that this type of wood was also well suited for many uses. It makes excellent pulp, and is frequently used for posts, poles, furnishings and mouldings, particleboard, fibreboard, and for plywood and 'engineered' wood products. Pine by-products are used in the chemical and pharmaceutical industries and residues are consumed for fuel. This amazing versatility has encouraged the development of an integrated forest-products industry which is almost unique in the world.

Exporters of wood products have largely targeted the rapidly growing markets of South and East Asia and Australia — 80 percent of exports by value go to only five markets: Japan, Korea, China, the United States and Australia. The product mix remains heavily based on raw materials, with logs, sawn wood, pulp and paper comprising 75 percent of export value. However, finished wood products such as panels and furniture components are exported to more than 50 countries.

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In New Zealand itself, the construction industry is the principal user of solid wood products, servicing around 20,000 new house starts annually. However, the small size of New Zealand's population (just over four million), plus its small manufacturing and remanufacturing base, limit the forestry industry's domestic opportunities. For the last few years local wood consumption has been around only four million cubic metres. Accordingly, the development of the export market is the key to the industry's growth and contribution to the national economy in decades to come.

In 2004, forestry export receipts were about 11 percent of the country's total export income, their value having increased steadily for ten years, until affected by the exchange fluctuations and shipping costs of recent years. The forestry industry is New Zealand's third largest export sector, generating around $3.3 billion annually from logs and processed wood products. But it is generally agreed that it is operating well below its capacity and, with the domestic market already at its peak, almost all of the extra wood produced in future will have to be marketed overseas. That presents a major marketing challenge for the industry.

Although the export of logs will continue to provide valuable earnings for forest owners, there is broad acceptance that the industry must be based on value-added products in future. So the industry is investigating various processing, infrastructure and investment strategies with a view to increasing the level of local manufacturing before export. The keys to future success will depend on a variety of factors: better international marketing, product innovation, internationally competitive processing, better infrastructure and a suitable political, regulatory and investment environment. The industry claims that, given the right conditions, by 2025 the forestry sector could be the country's biggest export earner, generating $20 billion a year and employing 60,000 people.

One competitive advantage that New Zealand has is its ability to source large quantities of softwood from renewable forests. Consumers in several key wood markets are becoming more worried about sustainability, and the industry is supporting the development of national standards as well as the recognition of these internationally. However, New Zealand is not the only country with a plantation-style forestry industry. Chile, Brazil, Argentina, South Africa and Australia all have extensive plantings of fast-growing species (hardwood and softwood), and in the northern hemisphere, Scandinavian countries have all expanded their forests or controlled their use in the interests of future production.

Finally, in addition to competition from other wood producers, New Zealand faces competition from goods such as wood substitutes. These include steel framing for houses. This further underlines the necessity for globally competitive production and marketing strategies.

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Questions 1–6

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 Settlers realised that wooden houses were more dangerous than other types of structure.

2 During the 1800s, New Zealand exported wood for use in boat-building.

3 Plantation-grown wood is generally better for construction than native forest wood.

4 Compared to other types of wood, pine has a narrow range of uses.

5 Demand for housing in New Zealand is predicted to fall in the next few years.

6 In future, the expansion of New Zealand's wood industry will depend on its exports.

Questions 7–13

Answer the questions below. Write **NO MORE THAN TWO WORDS AND/OR A NUMBER** *from the passage for each answer.*

Apart from exchange rates, which factor has had a negative impact on New Zealand's forestry exports? 7 _____

Which part of New Zealand's economy does the forestry industry rank third in?
8 _____

According to the New Zealand forestry industry, what could be the size of its workforce by 2025? 9 _____

What kind of timber product is available in large amounts from renewable forests in New Zealand? 10 _____

Which aspect of timber production are New Zealand's main customers increasingly concerned about? 11 _____

Outside the southern hemisphere, who are New Zealand forestry's main competitors?
12 _____

Which group of products is New Zealand's forestry industry now having to compete with?
13 _____

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Reading Passage 2

Intelligent Behaviour in Birds

Many people are aware of the intelligence of chimpanzees and other mammals. However, birds also demonstrate intelligent behaviour.

A For centuries, many scholars maintained that humans were the only intelligent organism on Earth. Many traits were considered to be exclusively human examples of acumen – for example, language, tool use, deception, awareness of self and others. However, exciting new research on a number of animals, particularly birds, has called into question the uniqueness of these traits, forcing us to reconsider this opinion. In 1964, people were amazed when naturalist Jane Goodall first discovered chimpanzees making and using tools. But ornithologists, people who study birds, were not overly surprised. Almost 20 years earlier, a renowned ornithologist had shown that tool use was commonplace in populations of woodpecker finches residing on the Galápagos Islands. These tiny birds routinely used twigs to extract grubs from under bark.

B Since then, the catalogue of tool-using animals has grown. At least three Australian bird species make tools similar to those of the woodpecker finch, and when white-winged choughs come across shellfish they have been known to use rocks as hammers to crack open the recalcitrant shells. Other birds show a more sophisticated level of insight. For example, black kites have been reported dropping bait into lakes to bring fish to the surface of the water, thereby making them easier to catch. A kite may also pick up a smouldering stick from an area recently burned by a bushfire and drop the stick on a patch of unburned grass. The bird then feasts on the small animals that flee from the subsequent fire.

C Most tool-using behaviours are a means of extracting food, which may provide a clue as to how the mental abilities needed for tool use evolved. The predominant explanation is based on the proverb that 'necessity is the mother of invention'. Essentially, brain tissue is energetically expensive, so animals should have evolved only the necessary intellectual capabilities required to overcome the challenges they face in their environment. Consider a hypothetical duck grazing on a seemingly endless supply of grass. Being particularly intelligent will not help the duck eat more grass. In contrast, other species, such as birds of prey, live in a more challenging environment, where food may be distributed erratically, hidden from view or highly mobile. The food itself may be quite intelligent. So, if there are not enough resources to feed all individuals, then only the smartest in each generation will live and reproduce.

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D New Caledonian crows boast many different tools in their tool kit. They use a hooked tool made by removing all but one of the side branches from a twig. They fashion serrated rakes (using their beaks as scissors) from stiff, leathery pandanus leaves. They also make probes by modifying their own moulted feathers. Each tool is used in slightly different ways to pull grubs from deep within tree trunks. The crows carry their favourite tool from one foraging site to the next. They also store their tools for later re-use in a secure place on their perch. Problem-solving abilities have traditionally been thought to be beyond the reach of animals. Nevertheless, birds are coming up with innovative solutions all the time. Recently, New Caledonian crows were observed moulding a piece of wire, something they had never seen before, into a hook and then using it to retrieve food.

E Literally hundreds of such reports have accumulated in back copies of scientific journals. Recently, a team of biologists from McGill University in Canada collated them and compared the frequency and size of innovations with the size of the birds' forebrain (the brain-area responsible for higher-order information processing) relative to the hindbrain. The team uncovered a clear relationship: birds with relatively large forebrains are able to invent fresh solutions to ecological challenges, and to exploit the discoveries and inventions of others, more often than birds with relatively small forebrains.

F Intelligence in birds may also arise as a result of selection to overcome the dynamic challenges of communal living. Since this involves competition between group members, to be successful, a social animal may need to be able to reflect on its own intentions, as well as those of others. The consequence of being part of a community may be the evolution of a distinctly 'political' brain.

G What better way to exercise a political brain than to be deceitful! Perhaps the best example of deception among birds comes from the white-winged choughs. Choughs are cooperative breeders – that is, they form a communal group consisting of one breeding-pair and up to 15 non-breeding 'helpers'. However, because young choughs have so little enthusiasm for foraging, or gathering food, they are often too hungry to help. And because it is socially unacceptable to be part of a group and provide little help, young choughs often act deceptively. For example, when an adult is watching, a young chough will place some food in the mouth of a hungry chick – but it does not release the food. Instead, it waits until the adult departs and then eats it. A chough can also help the group by preening the chicks. Interestingly, it is more likely to preen the chicks if another bird can see it do so. A chough that has been sitting totally still on the nest while the rest of the group is foraging out of sight will comically spring up and frantically start to preen the chicks as soon as some of its group members come into view. It is likely that these young choughs are only motivated to help when others are watching because they are concerned about their social status. Choughs need other choughs to like them as they cannot breed without them.

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Questions 14–20

Reading Passage 2 has seven paragraphs. Choose the correct heading for each paragraph from the list of headings below.

LIST OF HEADINGS

i The theory linking capacity for tool use in birds and survival
ii The influence of humans on tool use
iii The theory linking cognitive ability and living in a society
iv Reviewing long-held beliefs
v Intelligence helps birds to remember
vi How some birds trick each other
vii Physiological evidence of birds' intelligence
viii Several examples of birds who use tools
ix One species' multiple tool-using techniques

Q	Paragraph	Answer
14	Paragraph A	
15	Paragraph B	
16	Paragraph C	
17	Paragraph D	
18	Paragraph E	
19	Paragraph F	
20	Paragraph G	

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Questions 21–26

Look at the following characteristics and the list of birds below. Match each characteristic with the correct bird, **A**, **B**, *or* **C**. *You may use any letter more than once.*

LIST OF BIRDS

A White-winged choughs

B Black kites

C New Caledonian crows

Q	Characteristic	Answer
21	keeping tools that they like to use	
22	drawing out their prey by frightening it	
23	the use of tools to remove the outer covering from food	
24	using food to attract their prey	
25	the use of unfamiliar materials to make tools	
26	engaging in certain activities for the benefit of observers	

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Reading Passage 3

Jean Piaget 1896–1980

Seymour Papert looks at the work of pioneering Swiss philosopher and psychologist

Jean Piaget spent much of his professional life listening to children, watching children and poring over reports of researchers around the world who were doing the same. He found, to put it most succinctly, that children don't think like grown-ups. After thousands of interactions with young people often barely old enough to talk, Piaget began to suspect that behind their cute and seemingly irrational utterances were thought processes that had their own kind of order and their own special logic. Einstein called it a discovery 'so simple that only a genius could have thought of it.'

Although not an educational reformer, Piaget championed a way of thinking about children that provided the foundation for today's education-reform movements. It was a shift comparable to the way modern anthropology displaced stories of primitive tribes being 'noble savages' and 'cannibals'. One might say that Piaget was the first to take children's thinking seriously.

He has been revered by generations of teachers inspired by the belief that children are not empty vessels to be filled with knowledge (as traditional pedagogical theory had it) but active builders of knowledge – little scientists who are constantly creating and testing their own hypotheses about the world. And though he may not be as famous as Sigmund Freud or even B F Skinner, his influence on psychology may be longer lasting. In 1920, while doing research in a child-psychology laboratory in Paris, Piaget noticed that children of the same age made similar errors on intelligence tests. Fascinated by their reasoning processes, he began to suspect that the key to human knowledge might be discovered by observing how the child's mind develops.

On his return to Switzerland he began watching children play, scrupulously recording their words and actions as their minds raced to find reasons for why things are the way they are. In one of his most famous experiments, Piaget asked children, 'What makes the wind?'. A typical dialogue would be:

Piaget: What makes the wind?
Julia: The trees.
Piaget: How do you know?
Julia: I saw them waving their arms.
Piaget: How does that make the wind?
Julia: (waving her hand in front of his face): Like this. Only they are bigger. And there are lots of trees.

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Piaget recognised that five-year-old Julia's beliefs, while not correct by any adult criterion, are not 'incorrect' either. They are entirely sensible and coherent within the framework of the child's way of knowing. Classifying them as 'true' or 'false' misses the point and shows a lack of respect for the child. What Piaget was after was a theory that the wind dialogue demonstrated coherence, ingenuity and the practice of a kind of explanatory principle (in this case by referring to body actions) that stands young children in very good stead when they don't know enough or don't have enough skill to handle the kind of explanation that grown-ups prefer.

Piaget was not an educator and never laid down rules about how to intervene in such situations. But his work strongly suggests that the automatic reaction of putting the child right may well be counter-productive. If their theories are always greeted by 'Nice try, but this is how it really is...' they might give up after a while on making theories. As Piaget put it, 'children have real understanding only of that which they invent themselves, and each time that we try to teach them something too quickly, we keep them from inventing it themselves.'

Disciples of Piaget have tolerance for – indeed a fascination with – children's primitive laws of physics: that things disappear when they are out of sight; that the moon and the sun follow you around; that big things float and small things sink. Einstein was intrigued by Piaget's finding, especially by the idea that seven-year-olds insist that going faster can take more time – perhaps because this, like Einstein's own theories of relativity, runs so contrary to common sense.

Although every teacher in training still memorises Piaget's successive stages of childhood development, the greater part of Piaget's work is less well known, perhaps because schools of education regard it as 'too deep' for teachers. Piaget never thought of himself as a child psychologist. His real interest was epistemology – the theory of knowledge – which, like physics, was considered a branch of philosophy until Piaget came along and made it a science.

Through epistemology, Piaget explored multiple ways of knowing. He acknowledged them and examined them non-judgementally, yet with a philosopher's analytic rigour. Since Piaget, the territory has been widely colonised by those who write about women's ways of knowing, Afrocentric ways of knowing, even the computer's ways of knowing. Indeed, artificial intelligence and the information-processing model of the mind owe more to Piaget than its proponents may realise.

The core of Piaget is his belief that looking carefully at how knowledge develops in children will elucidate the nature of knowledge in general. Whether this has in fact led to deeper understanding remains, like everything about Piaget, controversial. In the past decade, Piaget has been vigorously challenged by the current fashion of viewing knowledge as an

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intrinsic property of the brain. Ingenious experiments have demonstrated that newborn infants already have some of the knowledge that Piaget believed children constructed. But for those, like me, who still see Piaget as the giant in the field of cognitive theory, the difference between what the baby brings and what the adult has is so immense that the new discoveries do not significantly reduce the gap, but only increase the mystery.

Questions 27–31

Choose the correct answer.

27 In the second paragraph the writer mentions the example of modern anthropology to illustrate

A the universality of Piaget's insights into the workings of the mind.
B the similarity between children's thought processing in different cultures.
C how Piaget's work represents a crucial turning-point in our approach to education.
D how Piaget's work has aided our understanding of man's evolution from primitive origins.

28 According to the writer, what point is illustrated by the dialogue about the wind?

A The factual accuracy of what children say is of minor significance.
B Children want to learn about scientific principles.
C Children's reasoning processes can be amusing to adults.
D Children often pretend that they know the answers to questions.

29 Piaget believed in the importance of

A preventing children from making false assumptions.
B giving children honest feedback on their hypotheses.
C showing children how to formulate their own ideas about the world.
D maintaining children's confidence in their ability to interpret the world.

30 What does the writer suggest in the seventh paragraph?

A Children's sense of their surroundings changes as they get older.
B Children are able to grasp certain complex ideas as well as adults are.
C Even apparently irrational ideas can be worthy of interest.
D Sometimes the simplest explanations are the best.

31 The writer's main purpose is to

A outline Piaget's contribution to a range of scientific fields.
B summarise how education has benefited from Piaget's findings.
C discuss Piaget's role in the development of 20th-century psychology.
D express doubts about a number of Piaget's theories.

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Questions 32–36

Complete the summary using the list of words, **A–I**, *below.*

Piaget's Theories on Children's Cognitive Development

LIST OF WORDS

A correct
B theories
C brain
D simple
E teachers
F psychology
G logical
H thought
I philosophers

Piaget maintained that children's mental processes were far more (32) _____ than they might appear. He encouraged the view that a child was not a 'blank slate' waiting to be filled with information, but rather a systematic builder of knowledge who regularly tries out his or her own (33) _____ about the world. Piaget's impact on the area of (34) _____ could well outlast that of more celebrated pioneers of this discipline. Despite doubts cast over his ideas by the current view associating knowledge exclusively with the (35) _____, the effects of his work are still strong today. His principles are still widely used in the professional development of (36) _____.

Questions 37–40

Choose **YES** *if the statement agrees with the claims of the writer,* **NO** *if the statement contradicts the claims of the writer, or* **NOT GIVEN** *if it is impossible to say what the writer thinks about this.*

(37) Piaget's early work in Paris involved innovative research techniques.

(38) Piaget gave clear guidelines as to how adults should give information to children.

(39) Piaget made a significant contribution to the field of epistemology.

(40) We still have much to learn about the nature of knowledge.

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IELTS Academic Reading — Test 3

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

The Early History of Olive Oil

A Olive oil is produced from the fruit of the olive tree, which is a member of the Oleaceae plant family. The trees require some cold weather during the year, but also tolerate hot, dry conditions, and do not like moisture when they are flowering. They actually produce better when subjected to these stressful conditions, and as a result, olive trees have traditionally been grown on land where little else will survive.

B Archaeologists today are divided over exactly where the first domestication of the olive occurred: some say it was in the area which is now Iran, Syria, Jordan and Egypt, while others contend it was in mainland Greece or on the island of Crete. The one thing that can be said with certainty is that cultivation began at least 6,000 years ago and spread slowly westward across the lands bordering on the Mediterranean Sea. Olive oil was used for a variety of purposes during these early times, including as a pharmacological ointment and in rituals for anointing royalty.

C The ancient Greeks believed the olive tree was a priceless gift from the goddess Athena and used its oil in sacred religious rituals. In fact, the Greek poet Homer called olive oil 'liquid gold', and during the 6th and 7th centuries BC Greek law forbade the cutting down of olive trees and made it punishable by death. The ancient Middle Eastern ruler King David valued his groves of olive trees and his olive oil warehouses so much that he posted guards around the clock to protect them.

D Over the years, olive oil developed other uses. Its employment in cooking dates at least as far back as the 5th century BC, as described by the Greek philosopher Plato. Its use as an aid to beauty and health later became ingrained in many Mediterranean cultures. The Romans, for example, are said to have used generous amounts on their bodies to moisturise their skin after bathing. With the spread of the Roman Empire, olive oil became a major commodity and its trade promoted commerce throughout the ancient world. It is generally believed that in the 1st-2nd century BC, olive trees were taken to North Africa and then to Spain, which was later to become the world's largest producer of olive oil. Artefacts found at various Mediterranean archaeological sites include olive oil storage vessels with olive plant residue still in them. Historical evidence still in existence in the

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form of wall paintings and ancient manuscripts (including the works of the Roman naturalist and philosopher, Pliny the Elder) all record production techniques and the various uses of olive oil.

E Making olive oil in those early days was a laborious process accomplished without mechanisation. Processing or milling the fruit involved several distinct stages to extract the liquid. The olives were harvested from the trees by hand or by beating the fruit from the trees with long sticks. The olives were then rinsed and crushed to separate out the large seed found in the centre of each. The remaining seedless flesh was put in woven bags and pressed. Hot water was then poured over the bags to separate the oil from the solid bits of olive. The liquid produced in this process, consisting of oil and water, was drained into stone basins or tanks, where it was allowed to settle and separate. In cold weather a bit of salt was added to speed up the process. As much oil as possible was drawn off the water, but the result was still not pure oil. Therefore, this impure mixture was allowed once more to settle in vats and then separated in order to refine the product.

F The waste water from the milling process, which is called amurca, is a bitter-tasting and foul-smelling liquid. In many ancient civilisations it was often simply discarded, causing serious pollution because of its acidity and high salt content. However, in the Roman period it was regarded as a very useful substance. When spread on surfaces, amurca forms a hard finish and therefore it was often applied to the floors of grain storage buildings where it hardened, keeping out water, mud and pests. When boiled down amurca was applied to leather to soften it so that it was easier to shape into articles of clothing and shoes. It could also be eaten by farm animals and was, in fact, fed to livestock suffering from malnutrition. According to ancient texts, amurca was also utilised in moderate amounts by farmers as a fertiliser or as a pesticide helping them to protect their crops from insects and even small rodents.

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Questions 1–6

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1. In the cultivation of olives, a period without rain is advantageous.
2. The most fertile fields are usually chosen for growing olives.
3. In ancient Greece, the olive tree was said to have divine origins.
4. Olive oil was more costly to buy in Greece than gold.
5. Plato mentions the use of olive oil in the preparation of food.
6. North African farmers initially resisted the introduction of olive trees.

Questions 7–9

Complete the flow-chart below. Write **ONE WORD ONLY** *from the passage for each answer.*

Ancient olive oil processing

olives are harvested by picking them or (7) _____ the trees

↓

milling stage

↓

olives are washed and crushed and seeds removed

↓

olive flesh is placed in (8) _____ and pressed

↓

water is poured over the mixture

↓

resulting liquid is given time to settle and separate, and (9) _____ is used to aid the process

↓

oil is drawn off and separation repeated

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Questions 10–13

Complete the notes below. Write **ONE WORD ONLY** *from the passage for each answer.*

Amurca

In ancient times, this waste liquid was usually thrown away, which led to (10) _____

However, Romans had practical applications for Amurca:

- when dried, created hard surface, so used on (11) _____ of certain buildings
- used when making (12) _____ into goods to wear
- fed to livestock
- used on farms as a (13) _____ to stop insects or animals damaging crops

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Reading Passage 2

Playing with Science

A The connection between science and play is not often discussed, perhaps because scientists take it for granted, or because they are a little self-conscious about it and therefore try to hide it. In this context, play might describe a number of different activities. Play might involve the exploration of new uses for everyday objects just for curiosity; it could also include word jokes, or the playing of jokes or tricks upon other scientists. The element of play in science is thus an elusive and difficult topic, but it is fundamental to the experience of scientists.

B One example of this connection is the jigsaw puzzle. Such puzzles present their players with two-dimensional fragments, each with a characteristic shape, from which to reconstruct an overall picture. Guessing the solution of a scientific problem has many similarities to completing a jigsaw puzzle. The scientists inspect each piece of data for a possible fit with its neighbours and, bit by bit, construct a whole argument. People who are good at jigsaw puzzles are able to guess which piece will fit even before trying it. In the same way, the best scientists are those who make the best guesses.

C But scientists do not play only at out-guessing each other, or at putting together pieces to complete a puzzle. They also regard their work as 'playing' with 'toys'. Joseph Lambert, a professor of chemistry at Northwestern University in the US, shed some light on this tendency. In a letter he wrote the following: When I grew up, every kid put in some serious sandbox time, and it often involved building... complex sand structures around which fantasies were composed... The organic chemistry labs at Yale... and chemical transformations took place... The odours were pleasant, and the physical process of working with our hands, as with sand, was satisfying... by the end of the year, I knew that I wanted to be an organic chemist, as I realised one could play in the sandbox for a living. Indeed, many scientists amuse themselves by 'playing' with various 'toys' of their trade perhaps coming up with ingenious devices to get a particular job done, or diverting a piece of commercial equipment for novel scientific purposes. The apparatus put together by Robert Millikan and Harvey Fletcher to measure the charge of the electron, which involved a perfume atomiser bought at a local pharmacy, is a classic example of such inspired tinkering.

D Whereas society often keeps a lid on playfulness, science encourages and nurtures it. Take, for example, the funny names that scientists have given to various chemical substances: tristane, windowpane, penguinone or megaphone. Each class of organic molecules includes a few such humorous names. Similar fun is had in other fields of science.

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E Hoaxes are a further aspect of the playfulness of scientists. A relatively recent example concerns the 'Plate of Brasse', which stated England's claim to California and was supposedly left by the English navigator Sir Francis Drake during his visit in 1579. A brass plate thought to be Drake's was discovered in the 1930s. But in 1977, Helen Michel and Frank Asaro, of the Lawrence Berkeley National Laboratory, showed that the copper and zinc used in the plate were of a higher quality than would have been available in the 1500s. They concluded that Drake's plate was most likely crafted much closer to the time when it was first brought to light.

F Shortly afterwards, staff at the University of California announced that the artefact was devised as a practical joke by a group of friends of Herbert E Bolton, who directed the Bancroft Library at the University's Berkeley campus. Bolton, it seems, was intrigued by tales of Drake having installed a plate to record his visit to California, and often urged his students to look for it. Some of Bolton's friends decided to play a joke on him, but things misfired after Bolton went public and announced that the relic was authentic before he could be told about the joke.

G Sometimes hoaxes serve a useful purpose. Some years ago, Nathan Lewis, a professor of chemistry at California Institute of Technology, and a graduate student were doing experiments in the laboratory of a senior professor, Harry Gray. Another co-worker had the habit of going through their data and rushing to Gray with his interpretation. Lewis decided to set a trap for the co-worker. He recalls: I manufactured an NMR spectrum. We left it out as bait. [Our colleague] took it and wrote up a paper on how important this result was. He was ready to go right to [the Journal of the American Chemical Society]... We didn't let him mail it [but] this stopped him temporarily from taking our data and interpreting it before making sure that it was right.

H But is the playfulness of science usually so helpful? One might argue that to play a practical joke is a waste of time. So why do it? Perhaps play is an inherent part of the human condition. The psychologist's answer might be that scientists tend to play because science presents them with too much seriousness. Another possibility might be that scientists like to play because they tend to be very young. Some disciplines, mathematics especially, have a reputation for the narrow window of creativity in youth. But it may also be that there is some cognitive value to the playful element in science. Playing with ideas, after all, is what science is all about. A playful, childlike attitude may be extremely fruitful, and scientists should not be too embarrassed to acknowledge that play is often what motivates them.

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Questions 14–20

Reading Passage 2 has eight paragraphs, **A–H**. *Which paragraph contains the following information?*

Q	Information	Answer
14	a description of how a test showed some evidence to be fake	
15	reasons why scientists may be unwilling to admit that they play	
16	the result of a trick going wrong	
17	the similarity between a skill used in play and one used in science	
18	how a group of scientists stopped a colleague misusing their results	
19	some examples of the playful nature of scientific language	
20	an example of an everyday object put to a scientific use	

Questions 21 and 22

Which TWO of the following reasons are given as possible explanations of scientists' play behaviour?

A Play may provide relief from their work.
B Scientists sometimes spend time as teachers.
C The age of scientists predisposes them to play.
D Scientists get ideas from children's questions.
E Scientists are secretive about their work.

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Questions 23–26

Complete the summary below. Write **NO MORE THAN THREE WORDS** *from the passage for each answer.*

Scientists play tricks

The passage gives two examples of tricks arising from the playfulness of scientists. Researchers at Caltech found that a colleague had been stealing their (23) _____ and used a hoax involving a fake NMR spectrum to trap the offender.

Another story concerns a plate which a famous (24) _____ is said to have left behind on a visit to California. The (25) _____ that the plate was made of later revealed that it was a fake, a hoax carried out by friends of the director of the (26) _____.

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Reading Passage 3

Does Class Size Matter?

A Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. Unlike competing proposals for reform, class-size reductions rarely elicit huge outcries or involve structural change. The testing of educators, by contrast, generally arouses the anger of unions. Similarly, establishing special 'charter' schools involves privileging some schools over others, with the credits provided usually coming out of the budgets of struggling local schools. With its uncomplicated appeal, class-size reduction in the U.S. has lately gone from being a subject of primary academic interest to a policy juggernaut with over twenty states aiming at decreasing class sizes.

B Do small classes improve school achievement? To answer this, investigators have attempted to analyse existing data, such as records at the U.S. Department of Education. These reveal that there were steep drops in pupil-teacher ratios between 1969 and 1997, but no significant gains in academic performance. But do these findings mean that class size makes no difference? Not necessarily. For instance, schools strive for more than just high test scores; they also usually try to keep their drop-out rates low. And, indeed, the drop-out rate for older students fell considerably over that period. Because drop-outs generally come from the low end of the achievement distribution, a reduction in the drop-out rate could be expected to pull down average test scores. Another reason for discounting those data is the difficulty of ensuring a level playing field. In a perfect world, U.S. students would all come from well-off families, with two highly educated English-speaking parents who are involved in their children's schooling. Teachers would all be creative and have complete mastery of the subject matter. The reality is very different.

C Over the past 35 years, some studies of existing data have produced evidence that smaller classes benefit students, but most of these studies were poorly designed. The exception was the Tennessee study called Project STAR (Student Teacher Achievement Ratio). Frederick Mosteller of Harvard University has called it 'one of the greatest experiments in education in United States history'. Students entering kindergarten were randomly assigned to one of three kinds of classes: a small class of 13 to 17 students, a regular-size class of 22 to 26 or a regular-size class with both a teacher and a full-time teacher's aide. The students remained in whatever category they had been assigned to throughout the third grade, after which they joined a regular classroom in the fourth. To ensure that teaching quality did not differ, teachers were randomly assigned to small and regular-size classrooms. Few teachers received any special training for working with small classes, and there were no new curricular materials.

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D At the end of STAR, researchers analysed the data. Jeremy Finn of New York University and Charles Achilles of Eastern Michigan University found evidence for 'an array of benefits of small classes'. They calculated that students in smaller classes were outperforming their counterparts in regular-sized classes by the first grade and that this advantage persisted even after students returned to larger classes. They also found that the effect was stronger for black and Hispanic minority groups - a significant finding for policy-makers. Eric Hanushek of Stanford, however, criticises some of STAR's key conclusions. He argues that STAR does not prove that gains persist long after students return to regular classes. It was debatable how much later improvement stemmed from other factors, such as a supportive home. Nor does he accept that the benefits accumulate, with participants widening the gap with their peers in larger classes year by year. Hanushek and others have also shown that during the study too many children moved from regular to small classes, probably because school personnel caved in to parent demands. And Hanushek also asserts that STAR failed to ensure good randomisation of teacher and student assignments. However, these points do not undermine STAR's basic findings.

E The largest public class size reduction programme so far, California's, stands more as a warning than as worthy of emulation. That state is trying to reduce classes in kindergarten through grade three despite a shortage of teachers that is most acute in low-income areas. This is exacerbating the disparity in resources available to rich and poor schools in California, because more affluent areas can attract the best teachers. Indeed, some of the extra teachers needed are being recruited from the poorer schools. Researchers found a statistically significant achievement advantage in reading, writing and mathematics for students in classes that had been reduced to 20. What is more, the effect did not vary for students of different backgrounds.

F Wisconsin's Student Achievement Guarantee in Education (SAGE) was a five-year pilot study to do some of the groundwork for a major project. Class sizes were reduced in only 14 schools, but it was noteworthy for targeting schools at which 30% of the students were below poverty level, compared with California's across-the-board approach. SAGE lowered the average pupil-teacher ratio in kindergarten through third grade to 12-15:1 from 21-25:1. Analysts have studied the results of first-grade students in these schools and similar first-grade students elsewhere and found the results accord with those from STAR. STAR and SAGE have made it hard to argue against reducing class sizes. But the California initiative shows that reductions made with too little forethought can yield minuscule gains. Administrators need solid information before they can make sensible policy decisions.

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Questions 27–31

Reading Passage 3 has six sections, **A–F**. *Which section contains the following information?*

Q	Information	Answer
27	detailed criticism of the methodology of a project	
28	a comparison of the data from class-reduction projects	
29	the level of public interest in the issues of class-size reduction	
30	details of action taken to protect the validity of a project	
31	reasons why class composition changed during a project	

Questions 32–40

Classify the following statements as referring to **A**, **B** *or* **C**.

OPTIONS

A Project STAR

B The California Project

C SAGE

Q	Statement	Answer
32	The student composition of each class was left to chance.	
33	A long-term improvement in performance was claimed.	
34	Similar results were obtained for all social groups.	
35	The project was a preliminary to a more comprehensive study.	
36	Several different class types were involved in the project.	
37	A special group of schools was selected to take part.	
38	Classroom assistants were used as part of the project.	
39	The project was responsible for aggravating existing problems.	
40	Certain groups of pupils within the sample were identified as having benefited.	

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IELTS Academic Reading — Test 4

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

The Development of Plastics

The first plastics were developed as a substitute for natural rubber. Chemically, rubber is a polymer-a compound containing large molecules that are formed by the bonding of many smaller, simpler units, repeated over and over again. The same bonding principle-polymerization-is the basis of the creation of a huge range of plastics by the chemical industry.

The first plastic was developed as a result of a competition in the USA. In the 1860s, $10,000 was offered to anybody who could replace ivory-supplies of which were declining-with something equally good as a material for making billiard balls. The prize was won by John Wesley Hyatt, with a material called celluloid. Celluloid was made by dissolving cellulose, a carbohydrate obtained from plants, in a solution of camphor dissolved in ethanol. This new material rapidly found other applications in the manufacture of everyday products such as knife handles and detachable collars and cuffs. But perhaps the best-known celluloid product was photographic film, without which the film industry could never have taken off at the end of the 19th century.

Celluloid can be repeatedly softened and reshaped by heat, and is known as a thermoplastic. In 1907, Leo Baekeland (1863-1944), a Belgian chemist working in the USA, invented a different kind of plastic, by causing phenol and formaldehyde to react together. Baekeland called it Bakelite, and it was the first of the thermosets-plastic that can be cast and moulded while hot, but cannot be softened by heat and reshaped once they have set. Bakelite was a good insulator, and was resistant to water and acid. With these properties it was soon being used in the manufacture of electrical switches as well as a variety of domestic items.

As the century went on, the range of newly developed plastic increased. Chemists began looking for other small molecules that could be strung together to make polymers. In the 1930s, chemists in Britain discovered that the gas ethylene would polymerize under heat and pressure to form a thermoplastic they called polythene. Polypropylene followed in the 1950s. Both are used to make bottles, pipes and plastic bags. A small change in the starting material-replacing a hydrogen atom in ethylene with a chlorine atom-produced rigid PVC

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(polyvinyl chloride), a fireproof plastic suitable for drains and gutters. By adding certain chemicals, a soft form of PVC can be produced, suitable as a substitute for rubber in items such as waterproof clothing.

A closely related plastic is Teflon or PTFE (polytetrafluoroethylene). It produces very little friction, making it ideal for products such as non-stick frying pans. Polystyrene, a hard, clear material like glass, was developed during the 1930s in Germany, and its applications included food containers and toys. Expanded polystyrene is rigid and is widely used in packaging and insulation. Polyurethane, developed in the same country, was commonly produced as a foam which was very useful in the production of insulating materials.

In the 1930s, the first of the man-made fibres was created-nylon. Its inventor was a chemist called Wallace Carothers (1896-1937), who worked for the Du Pont company in the USA. He found that under the right conditions two particular chemicals would form a polymer that could be pumped out through holes and then stretched to form long glossy threads that could be woven like silk. Its first use was to make parachutes for the US armed forces in World War II. In the postwar years, it completely replaced silk in the manufacture of stockings.

Many other synthetic fibres joined nylon, including Orion, Acrilan, and Terylene. Today most garments are made of a blend of natural fibres, such as cotton and wool, and man-made fibres that make fabrics easier to look after.

Despite its enormous usefulness, plastic has its drawbacks. In fact one its great strengths-its indestructibility-is its greatest disadvantage. Beaches all over the world, even on the remotest island, are littered with plastic bottles that nothing can destroy. Nor is it very easy to recycle plastics, as different types of plastic are often found in the same items and call for different treatments.

Plastics can be made biodegradable by incorporating into their structure a material such as starch, which is attacked by bacteria and causes the plastic to fall apart. Other materials can be incorporated that gradually decay in sunlight-although bottles made of such materials have to be stored in the dark, to ensure they do not disintegrate before they have been used.

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Questions 1–7

Complete the table below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

Name	Date	Country of origin	Properties	Common uses
Celluloid	1860s	USA	Can be softened and reshaped by heat	Billiard balls (original use); cutlery, clothing, (1) _____
(2) _____	1907	USA	Can't be softened after setting, good insulator, resistant to water and acid	Household objects, (3) _____
Polythene	1930s	(4) _____	Can be softened and reshaped by heat	Bottles, pipes, plastic bags
Polypropylene	1950s			bottles; pipes; plastic bags
Rigid PVC			Is (5) _____	external piping
Soft PVC				outdoor clothing
Polystyrene	1930s	Germany	Resembles (6) _____	food containers; toys
Polyurethane		Germany	Usually manufactured as a (7) _____	insulation

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Questions 8–13

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

8 The chemical structure of rubber is very different to that of plastics.

9 John Wesley Hyatt was an industrial chemist.

10 Celluloid and Bakelite react in the same way to heat.

11 If an object is made of several plastics, these prove hard to break down and reuse.

12 Adding starch to plastic makes it more durable.

13 Containers which are designed to decompose need particular storage conditions.

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Reading Passage 2

Lean Production Innovation - in Manufacturing Systems

A After the First World War, car makers Henry Ford and Arthur Sloan of General Motors moved world manufacturing from centuries of craft production into the age of mass production. Largely as a result of this, the United States soon dominated the world economy. After the Second World War, and approximately a hundred years after Japan opened up to the modern world, Eiji Toyoda and Taiichi Ohno pioneered the concept of lean production at the Toyota car company. And now, although superimposing the method on existing mass production systems causes pain and upheaval, manufacturers around the world are trying to embrace this innovative system.

B Perhaps the best way to describe lean production is to compare it with the two other major manufacturing systems: craft production and mass production. The craft producer uses highly skilled workers and simple but flexible tools to make exactly what the customer asks for one item at a time. A present day example of this method is the customised production of a few exotic sports cars. The concept of craft production remains very popular, but the problem with it is obvious. Goods produced by the craft method - as cars once exclusively were - cost too much for most of us to afford. So at the beginning of the twentieth century, mass production was developed as an alternative method. The mass producer uses narrowly skilled professionals to design products which are then made by unskilled or semi-skilled workers, using expensive, single-purpose machines. These churn out standardised products in very high volumes. Because the machinery costs so much, and is so intolerant of disruption, the mass producer keeps standard designs in production for as long as possible. The result is that the customer gets lower costs, but at the expense of variety, and by means of work methods which most employees find boring and dispiriting. By contrast, the lean production system combines the advantages of craft and mass production, while avoiding the high cost of the former and the rigidity of the latter. Towards this end, companies appoint teams of multi-skilled workers to all levels of the organisation, and use highly flexible and increasingly automated machines to produce goods in enormous volume and variety.

C Lean production is so called because, compared with mass production, it uses less of everything - half the human effort in the factory, half the manufacturing space, half the investment in tools, and half the engineering hours to develop the new product. It also results in far fewer defects.

D Perhaps the most striking contrast between mass and lean production systems lies in their production standards. Mass producers set a limited goal for themselves: 'good enough', which translates into an acceptable number of defects, a maximum acceptable

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number of inventories, and a narrow range of standardised products. Lean producers, on the other hand, are unwilling to compromise standards in any of these areas.

E Although cost reduction is the primary objective of the lean production system, it must meet three other intermediate objectives in order to achieve this: quantity control, quality assurance and respect for humanity. Firstly, the system must be able to adapt to daily and monthly fluctuations in demand. Secondly, each separate process must supply only good units to the subsequent process. Thirdly, in as far as the system uses human resources to attain its cost objectives, respect for human needs must be cultivated. It should be emphasised that none of these three objectives can be achieved separately.

F The continuous flow of lean manufacturing production relies on two practical mechanisms: just-in-time and autonamation. Just-in-time means, for example, that in the process of assembling the parts to build a car, components from the preceding process should arrive at the next part of the line at exactly the right time and in the correct quantities. If just-in-time is fully realised throughout the company, then superfluous inventories are completely eliminated from the factory, making stores or warehouses unnecessary. However, relying solely on a central planning approach to control schedules for all stages of the production process simultaneously is very difficult in the case of cars, which consist of thousands of parts. So the lean system looks at the production flow in reverse; in other words, employees go to the preceding process to withdraw the necessary quantity of units at the appropriate time. The preceding process must produce only sufficient quantities of units to replace those that have been withdrawn, and in turn withdraws the requisite number of components from the process that precedes it.

G Autonamation is the automatic checking for abnormalities in the production process. In order to realise just-in-time perfectly, only units which are in perfect condition must be allowed to flow to the next process, and this flow must be regular and uninterrupted. In other words, quality control must coexist with just-in-time procedures throughout the system. Autonamation involves building in a mechanism to prevent the multiplication of defects in machines or product lines. For example, in Toyota factories almost all the machines have been fitted with stopping devices, and the concept of autonamation has been extended to manual production lines. If something abnormal happens there, the worker pushes a button to stop the whole line, and lights, which are hung so high in the factory that they are visible to everyone, indicate the position of the problem.

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Questions 14–20

Reading Passage 2 has seven paragraphs, **A–G**. *Choose the correct heading for each paragraph from the list of headings below.*

LIST OF HEADINGS

i Global resistance to lean manufacturing
ii The historical context
iii Procedures for controlling quality
iv The pros and cons of different production systems
v The impact on profits
vi Procedures for controlling supply
vii The origin of the term
viii A crucial difference in levels of quality
ix Working conditions
x Interdependent strategies for controlling expenditure

Q	Paragraph	Answer
14	Paragraph A	
15	Paragraph B	
16	Paragraph C	
17	Paragraph D	
18	Paragraph E	
19	Paragraph F	
20	Paragraph G	

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Questions 21–26

Complete the sentences below. Write **NO MORE THAN THREE WORDS** *from the passage for each answer.*

A small number of unusual cars are still produced by the (21) _____ method.

Lean production requires staff who are (22) _____.

Lean production employs fewer people, and uses less (23) _____, equipment and time.

Storage facilities are not needed if a procedure known as (24) _____ is implemented in the lean production method.

Autonamation is a procedure for spotting any (25) _____ in the products on a production line.

At Toyota factories, (26) _____ are suspended above manual production lines in order to show where production has to be halted.

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Reading Passage 3

How Does Watching Sport Influence the Brain?

A During the epoch when the poet Homer unveiled his grand epic, the ancient Greeks initiated a festival where men contended in a race of roughly 200 metres. The victor would be rewarded with a symbolic olive branch. This event, known as the Olympics, has transcended its humble origins to embody the human quest for progress. The captivating nature of these games raises an intriguing question: what exactly draws us in as spectators? Is it the thrill of competition, the beauty of human movement, or something happening within our brains?

B In 1996, three Italian neuroscientists-Giacomo Rizzolatti, Leonardo Fogassi, and Vittorio Gallese conducted research on the premotor cortex of monkeys and made a groundbreaking discovery. They revealed a cluster of cells that act as a repository for motor actions, similar to 'grammar of movement'. These cellular networks represent the 'sentences' of full-body motions that our brains meticulously retain and refine. Take the art of swinging a golf club as an example. For those who only watch the Masters Tournament on television, playing golf may appear deceptively effortless. However, for novices, smoothly manoeuvring the asymmetrical metal club presents a nearly impossible challenge. This is because novices rely on conscious effort, utilising brain regions adjacent to the premotor cortex. Experts, by contrast, instinctively execute perfectly balanced movements, as their actions are deeply ingrained and intertwined with neurons in the premotor cortex. Their swing occurs with the serenity of a refined autopilot, requiring no conscious control. This neural distinction between novices and experts provides insight into how long-term training can physically reshape the brain.

C These neurons in the premotor cortex not only help explain why some athletes seem to have exceptional skills, but they also exhibit a wondrous characteristic that prompted Rizzolatti, Fogassi, and Gallese to bestow upon them the exalted label of 'mirror neuron'. They point out that mirror neurons are activated when a primate performs a particular action, such as grasping or holding an object, or when it witnesses another individual performing a comparable action. Humans possess an even more intricate mirror neuron system that reflects the external world in our brains, enabling us to internalise the behaviours of others. Nonetheless, these cells necessitate what scientists refer to as 'goal-orientated movements' to activate. If we stare at a photograph, a static image of a runner striding ahead, our mirror neurons remain utterly silent. It is solely when the runner is in action, be it in running, moving, or sprinting, that they are ignited. Movement, not stillness, is what speaks to our brains.

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D Electrophysiological studies have shown that when we observe a golfer or a runner in action, our own premotor cortex mirror neurons respond as if we were participating in the activity. This neural mirroring was initially identified in 1954 by French physiologists Gastaut and Bert, who noticed specific alpha and mu brain wave patterns in humans. The mu signal plays a role in neural mirroring, remaining active during periods of bodily stillness but dissipating when we undertake active pursuits like playing a sport or changing TV channels. Notably, the mu signal also remains subdued while we observe someone else in action, such as on television, which alludes to the influence of mirror neurons. This suggests that our brains do not simply passively register what we see, but actively simulate it, engaging systems normally reserved for action.

E Rizzolatti, Fogassi, and Gallese have coined the term 'direct matching hypothesis' to delineate the concept of mirror neurons. They asserted that to comprehend the movements of sports stars, we need to align the visual representation of the observed action with our own motor representation of the same action. According to this theory, watching an Olympic athlete can evoke a resonance in the observer's motor system. The observer's 'motor knowledge' is harnessed to decipher the observed action. However, mirror neurons extend beyond being merely neural foundations for our sporting inclinations. They also enable observers to enhance their athletic abilities through the process of observation. It turns out that watching a remarkable golfer improves our own golfing skills, and witnessing a skilled sprinter actually bolsters our running speed. This ability to learn through observation is a vital skill. From language acquisition in infancy to understanding facial expressions, mimicking plays a fundamental role in our consciousness. The most accomplished athletes possess a premotor cortex capable of envisioning victorious movements, supplemented by the physical attributes required to translate those movements into reality. In this way, action and imagination coalesce to drive performance.

F However, how many among us watch sport in order to enhance our athletic abilities? Most spectators tune in for the emotional roller coaster, the human drama it encompasses. This emotional connection finds its roots in mirror neurons, which allow spectators to perceive athletes' victorious movements and to share in the emotion of their triumphs. This is because they are directly associated with the amygdala, a key area of the brain responsible for emotional processing. During the Olympics, mirror neurons in individuals around the world resonate in unison, and watching sport unifies people. The majority of us may never run a mile in under four minutes or hit a home run. Yet, as we gather around our televisions, we all briefly get a taste of the feeling of executing something flawlessly. In those fleeting moments, sport becomes more than entertainment-it becomes a shared human experience.

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Questions 27–32

Reading Passage 3 has six paragraphs, **A–F**. *Which paragraph contains the following information?*

Q	Information	Answer
27	reasons why watching sport can bring emotional satisfaction	
28	an explanation of why beginners struggle with sports tasks	
29	a factor that can work alongside mirroring in sports competitions to maintain peak performance	
30	a comparison between the mirror neurons of human beings and primates	
31	mention of integrating visual and motor processes to understand athletic movements	
32	reference to the first discovery of brain activity associated with mirror neurons	

Questions 33–35

Choose the correct answer, **A**, **B**, **C** *or* **D**.

33 What point does the writer make about professional athletes in Paragraph B?

A They meticulously plan each phase of their movements.
B They engage in regular practice to retain and refine their skills.
C They actively consider and analyse the actions of their peers.
D They perform their actions without conscious contemplation.

34 In this passage, the writer mentions the example of a remarkable golfer to illustrate

A the critical role of imitation in the formation of consciousness.
B the improvement of sporting ability through watching sport.
C the stimulation of a resonance in the observer's motor system.
D the motivation of the observer's interest and love for golf.

35 The principal motivation for watching sport mentioned by the writer is

A to improve our physical ability by analyzing the athletes' technical strategies.
B to unite viewers from different nations.
C to experience intense positive emotions by immersing oneself in sport.
D to become a sports professional.

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Questions 36–40

Choose **YES** *if the statement agrees with the claims of the writer,* **NO** *if the statement contradicts the claims of the writer, or* **NOT GIVEN** *if it is impossible to say what the writer thinks about this.*

36 The emergence of Homer's epic poetry occurred in the same historical period as the beginning of the Olympic Games.

37 The Italian scientists concluded that the motor actions encoded within the premotor cortex have no discernible link to unrelated collections of neurons.

38 Spectators can achieve flawless golf swings merely by watching the Masters Tournament without physical practice.

39 Some neuroscientists believe that certain types of still images might activate mirror neurons under specific experimental conditions.

40 The mu system is suppressed while we are dynamically involved in a particular activity.

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IELTS Academic Reading — Test 5

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Rubber

T and W Musgrove discuss the origins and early uses of rubber

The plants that produce rubber are spread right across the globe, and grow in many different habitats. One might think it likely, therefore, that humankind has known about rubber for thousands of years. Yet, unlike other crops of economic importance, rubber led a relatively anonymous life until the last 150 years or so. The Indians of South America appear to be the first people to have understood the properties of rubber, and the Aztecs of what is now Mexico were the first to be recorded using the substance; a wall painting dating back to the sixth century depicts a scene of a tributary offering of crude rubber. With the arrival of Columbus in the Americas and the resulting Spanish influx, further evidence starts to appear concerning the Native American use of rubber. Antonio de Herrera y Tordesillas describes a ritual game played with a rubber ball at the court of the Aztec Emperor Montezuma II, and the Mayan and Toltec people are known to have taken part in similar activities. Rubber was also used to make raincoats, shoes, jars, torches and musical instruments, all of which must have been made from the indigenous Castilla elastica, as the Para plant now favoured for rubber cultivation does not grow in the Mexican region.

The first description of latex (liquid rubber) extraction is made by Juan de Torquemada, who noted that if a receptacle was not at hand the Native Americans would place the latex on their bodies to allow it to solidify. However, no real interest in rubber was shown by any European until Charles de la Condamine, a French mathematician, published an account of his journey to South America in 1735. The journey was undertaken on behalf of the Paris Academy of Sciences to measure an arc of the meridian line on the equator, but the journey home was to turn out to be more significant than the true purpose of the trip. Condamine explored Brazil and Peru and discovered how the local people used one single piece of coagulated latex to make boots. The boots were impervious to water and, when smoked, looked like real leather. In 1747 the first description of the rubber tree and latex tapping was made by a military engineer and amateur botanist, François Fresneau, who was posted to French Guiana. The publications of Condamine and Fresneau created considerable excitement among French scientists, and an attempt was made to discover a solvent that could turn the crude rubber into a substance for commercial exploitation.

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In 1818 a British medical student named James Syme first used rubber to make waterproof cloth. Another early use of the substance was as an eraser of pencil marks, hence the name 'rubber'. This was complemented by balloons, rubber bands, braces, boots for the army and other ideas that met with varying degrees of success. In 1820 Thomas Hancock, an English manufacturer of rubber goods such as driving belts, industrial rollers and rubber hoses, invented a machine he called the 'pickle', which chewed up waste strips for re-use. It was discovered that the masticated rubber was more malleable, while maintaining much of its elasticity. In Scotland at the same time Charles Mackintosh had discovered a way of using rubber as waterproofing material, by a process he patented in 1823. Hancock and Mackintosh joined forces in 1834, and three years later Hancock invented a machine for spreading rubber onto material.

Despite their beneficial qualities, such as waterproofing, rubber goods were still not particularly popular as they had some major flaws, including the fact that they dissolved malodorously. They also became pliant when warm and rigid when cold. Then in 1839 the American Charles Goodyear discovered that it was possible to stabilise rubber by mixing it with sulphur while exposing it to heat - a process he called vulcanisation - and the full versatility of this extraordinary substance became apparent.

Rubber goods could now be manufactured which had all the beneficial qualities of the material, such as durability, elasticity and variability, but which were not sticky, soluble or governed by the vagaries of the weather. The economic potential of rubber was now clearly evident. It played an important role in the Industrial Revolution, being employed in the steam engines found in factories, mills, mines and railways. It made a triumphant entrance as a new and innovative material at the Great Exhibition of 1851, where shoes, airbeds, furniture and clothing made out of newly improved rubber were proudly displayed.

One of the most important rubber inventions was made in 1888, when an Irishman called John Boyd Dunlop produced the first pneumatic tyre. Solid rubber tyres had been used for the previous 18 years, but Dunlop's new design, which he updated in 1890, immediately became popular. In 1895 Dunlop's tyres were first used in motor cars, and with the mass production of cars just over the horizon the rubber industry had never looked healthier. The import levels of rubber over the nineteenth century bear witness to its irrepressible rise. In 1830 Britain had imported just 211 kg of crude rubber. This had risen to 10,000kg in 1857, and by 1874 levels were just under six times as much again.

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Questions 1–6

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 Rubber plants grow only in certain regions of the world.

2 Rubber was extracted in Mexico as early as the sixth century.

3 Rubber from the Castilla elastica plant is of poorer quality than that from the Para plant.

4 A French mathematician inspired real interest in rubber amongst Europeans.

5 The process of vulcanisation was discovered by accident.

6 Imports of crude rubber into Britain fell during the nineteenth century.

Questions 7–13

Complete the summary below. Write **NO MORE THAN THREE WORDS** *from the passage for each answer.*

The Commercial Development of Rubber

Early European travellers gave accounts of various rubber objects in use in Central and South America, and these accounts created interest in the commercial exploitation of rubber. In 1818, 7 _____ was produced using rubber, and in 1820 a machine was invented for recycling 8 _____ of rubber. Over the next few years, other attempts were made to improve rubber, but some problems remained. For example, rubber products smelt bad when they were dissolved, and could turn either soft or 9 _____ depending on the temperature. However, in 1839 a new process to 10 _____ the substance greatly increased its potential. For example, rubber was used in the creation of the 11 _____ industry during the Industrial Revolution. Then in 1888 the 12 _____ was developed, and a few years later the 13 _____ of the motor car began.

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Reading Passage 2

The Myth of the Eight-hour Sleep

We often worry about lying awake in the middle of the night – but it could be good for you

Indications from both science and history suggest that the eight-hour sleep may not be a natural or inborn pattern for humans. In the early 1990s, psychiatrist Thomas Wehr conducted research which implies just that. Wehr kept a group of people in a darkened room for 14 hours a day for a month. It took some time for their sleep to regulate, but by the fourth week the subjects had settled into a very distinct sleeping pattern. Firstly, they slept four hours, then woke for one or two hours before falling into a second four-hour sleep. Though sleep scientists were impressed by the implications of the study, the idea that we must sleep for eight consecutive hours persists amongst the general public.

In 2001, historian Roger Ekirch of Virginia Tech University in the US published a paper drawn from 16 years of research, revealing historical evidence that humans used to sleep in two distinct periods of time each night. Ekirch's research found more than 500 references to a segmented sleeping pattern – in diaries, court recorders, medical books and literature, from the ancient Greeks to tribes in Nigeria. Much like Wehr's findings, these references describe a first sleep which began not long following sunset. Then there was a waking period of one or two hours and after that a second sleep. During the waking period, people could be quite active. They sometimes got up and moved around the house, although most people stayed in bed, and perhaps read or wrote if they had enough money for candles. In many historic accounts, Ekirch found that people used the time that they were awake between periods of sleep to think about and attempt to analyse their dreams.

In his book, Evening's Empire, historian Craig Koslofsky suggested an explanation for this divided sleep pattern in Europe. 'Associations with night before the 17th century were not good,' he writes. He goes on to explain that the streets of cities and towns at night were often populated by thieves or worse. The streets at night consequently scared many people. Even the wealthy, who could afford to light their way, had better things to spend their money on. There was no prestige or social value with going out or staying up late at night.

Ekirch found that references to the first and second sleep started to disappear during the late 17th century. The pattern began to alter first among the urban upper classes changing amongst the rest of Western society. By the 1920s, the idea of a first and second sleep had receded entirely from our social consciousness. Ekirch attributes the initial shift to improvements in street and home lighting. As the night became a time for all kinds of activity, the length of time people used for rest declined. In 1667, Paris became the first city in the world to light its streets, using wax candles in glass lamps. It was followed by another French city, Lille, in the same year and by Amsterdam in Holland two years later,

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where a much more efficient oil-powered lamp was developed. London didn't light its streets until 1684, but by the end of the 17th century, more than 50 of Europe's major towns and cities were lit at night. Coffee houses emerged as a fashionable phenomenon and many were open virtually around the clock. Going out at night became commonplace, and spending hours lying in bed was considered a waste of time. 'People were becoming increasingly time-conscious and sensitive to efficiency, certainly before the 19th century,' says Ekirch. 'But the Industrial Revolution intensified that attitude considerably.' Strong evidence of this shifting attitude is contained in a medical journal from 1829 which urged parents to force their children out of the pattern of first and second sleep.

Today, most people seem to have adapted quite well to the eight-hour sleep, but Ekirch believes many sleeping problems may have roots in the human body's natural preference for divided sleep, as well as in difficulties caused by extended exposure to artificial light in the modern world. This could be the cause of a condition called sleep maintenance insomnia, where people wake during the night and have trouble getting back to sleep. The condition first appears in literature at the end of the 19th century, at the same time as accounts of interrupted sleep disappear. Sleep psychologist, Gregg Jacobs, says that the idea that we must sleep for an extended period of time could be changing. If it makes people who wake up in the night anxious, as this anxiety can itself discourage sleep and is likely to affect waking life too. Jacobs suggests that the waking period between sleeping, when people simply rested and relaxed, could have played an important part in the human capacity to regulate stress naturally. Russell Foster, a professor of circadian (body clock) neuroscience at Oxford University in the UK, shares this point of view. 'Over 30% of the medical problems that doctors are faced with stem directly or indirectly from sleep. But sleep has been ignored in medical training, and there are very few countries where sleep is studied,' Foster says. He feels this needs to change.

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Questions 14–18

Reading Passage 2 has five sections. Choose the correct heading for each section from the list of headings below.

LIST OF HEADINGS

i historical reasons why interrupted sleep became uncommon

ii Brain structures involved in sleep patterns

iii Potential health issues related to sleep

iv Famous cases in literature of sleep

v An analysis of old documents to discover sleep patterns

vi Biological and environmental factors preventing people from falling asleep again

vii Scientific evidence that divided sleep is a natural phenomenon

Q	Section	Answer
14	Section 1	
15	Section 2	
16	Section 3	
17	Section 4	
18	Section 5	

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Questions 19–23

Match each statement with the correct researcher, **A–E**. *You may use any letter more than once.*

LIST OF RESEARCHERS

A Thomas Wehr
B Roger Ekirch
C Craig Koslofsky
D Gregg Jacobs
E Russell Foster

Q	Statement	Answer
19	In certain historical periods, the threat of criminal danger led to segmented sleep.	
20	Physicians should learn more about treating people with sleeping difficulties.	
21	Historically, when people experienced interrupted sleep, they used the waking period at night for different activities.	
22	Technological changes in Europe made people more likely to sleep throughout the night.	
23	The belief that humans should have a long continuous sleep can be psychologically harmful.	

Questions 24–26

Complete the summary below. Write **ONE WORD ONLY** *from the passage for each answer.*

Historical patterns of interrupted sleep

Historical evidence seems to show that humans had common sleep patterns which were quite unlike today's eight hours of unbroken sleep each night. People habitually went to bed early, soon after (24) _____, slept for a while, then woke up for a few hours of activity or perhaps to consider, for example, the meaning of (25) _____. There was little else to do at home in the dark, since what little means of lighting they had, such as using candles, was costly. In 16th-century Europe, people tended to stay at home at night because they feared (26) _____ on large unlit city streets.

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Reading Passage 3

Living Dunes

Things don't come much stranger than heaps of sand that can move and sing of their own accord. Sally Palmer investigates

1 Armies of giant sand dunes are advancing across the world's deserts, engulfing anything that crosses their path. They are tens of metres tall and hundreds of metres long. Fortunately, they aren't going very fast. Even the smallest, speediest dunes only travel about 100 metres over the course of a year, while the bigger ones, which weigh something in the order of 10,000 tonnes, barely move one metre in that time. However, their insidious creep can have serious consequences if there is an oil installation or a railway line in their path.

2 About 47% of the world's land mass, including Antarctica, most of Australia and large areas of Africa, is classified as arid or semi-arid desert. Only around 20% of that is sand-covered, however, and over half of that is classified as 'linear' sand dunes. These form in a long curving wave, as a result of wind blowing strongly from several quarters, flipping them from side to side. Although linear dunes are static, sand blowing off them can cause problems for desert villages, burying crops and buildings.

3 Moving dunes make up just a small percentage of the rest, but they are of the most interest to scientists. They are known as 'barchans': heavy, crescent-shaped sand piles with a ridged crest and two elongated arms, one curving away to either side. 'Barchan dunes only tend to form where you have one-directional winds on the edge of sandy deserts near coastal areas,' says Giles Wiggs, a geomorphologist at Oxford University, who has been studying the formation and movement of sand dunes for more than a decade.

4 But even with strong winds, how can entire barchans move while retaining their form? That question was first answered in the mid-20th century by British explorer Ralph Alger Bagnold, and his answer hinges on the fact that dunes aren't solid, but granular. Bagnold figured out how barchan dunes are able to move grain by grain. Imagine a single grain of sand being blown up the back of a dune by the wind and deposited on the top. More grains follow the same pattern, until the accumulated weight of piled-up sand finally pushes the top down the dune face. The grain tumbles, then stops on the face until subsequent mini-avalanches bury it. Eventually, it reappears at the back of the dune, ready to repeat the process. As this happens to every grain of sand in the dune, the whole thing creeps in the direction of the prevailing wind.

5 The relationship between the wind and barchan dunes is complex. As a dune grows, it modifies the speed and course of the wind, which in turn alters how that dune and its neighbours evolve. 'Interestingly, the dune can regulate its own shape, and maintain it as it

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moves,' says Dr Stéphane Douady, a physicist at Ecole Normale Supérieure (ENS) in France. 'Even when two dunes collide, they quickly take on their distinctive shapes again. It's like a living organism.'

6 Douady and his colleagues have also been studying an even odder phenomenon than moving dunes: some barchans actually sing. Local legends attributed the sounds to dangerous spirits which were trying to trap unwary travellers. Douady is more pragmatic. 'It's a strong booming noise with a low frequency,' he explains, making a noise like a foghorn to demonstrate. 'It can last for a long time - up to several minutes. It's a very loud sound and you don't understand where it's coming from when you first hear it.' There are about 50 dunes distributed across 35 deserts round the world that are known to sing. Douady says the sound is caused by the way sand avalanches down the faces of particular dunes. Rather than tumbling randomly, the sand grains flow in synchrony and set each other vibrating like the membrane on a gigantic loudspeaker. The synchronisation causes the air to move in and out between the grains, creating a powerful sound wave.

7 What really surprised the scientists, however, was that they were able to take samples of the singing sand back to France and replicate the sound at ENS, proving that it's the sand, not the dune shape, that causes the sound. Their studies show the grains are a uniform shape, well-rounded from years of striking each other, and that the variations in size affect the tone. Crucially, the grains are coated with a special veneer, which Douady calls 'desert glaze', made from a precise combination of minerals from surrounding rocks including iron, aluminium, manganese, silicon and calcium. The team found that after a month or so, the veneer wore off and the grains lost their 'voice'. 'We managed to reproduce the desert glaze and then the grains started to sing again,' says Douady. 'We tried putting the coating onto different grains, but they weren't round enough and it didn't work. But some American colleagues made some artificial grains and managed to make them sing, after covering them in desert glaze.' Douady has now made recordings of dunesong from all over the world which are to be made into a CD.

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Questions 27–33

Reading Passage 3 has seven paragraphs. Choose the correct heading for each paragraph from the list of headings below.

LIST OF HEADINGS

i Moving rapidly across the desert
ii Recreating the process in a laboratory
iii Strange music created by human movement
iv A potential threat to industry and communications
v Dunes coming together and re-forming
vi Needing a specific combination of conditions
vii A continuous cycle
viii The commonest type of dune
ix Old superstitions demystified

Q	Paragraph	Answer
27	Paragraph 1	
28	Paragraph 2	
29	Paragraph 3	
30	Paragraph 4	
31	Paragraph 5	
32	Paragraph 6	
33	Paragraph 7	

Questions 34–36

Choose the correct answer.

34 What are we told about linear dunes?

A They are formed by strong winds blowing from one direction.
B They can move up to 100 metres in a twelve-month period.
C They develop in a recognisable shape.
D They make up about 20% of the world's deserts.

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35 Bagnold discovered that movement in barchans was caused by

A the long straight shape of the dunes.
B the particular composition of the dunes.
C the exceptionally heavy nature of the sand grains.
D the unusual strength of the wind in certain seasons.

36 Why does Dr Douady compare a barchan dune to a living organism?

A It starts small and then increases in size.
B It has an effect on its immediate surroundings.
C Its relations with desert organisms are quite developed.
D Its outline stays the same even after a period of movement.

Questions 37–40

Complete the summary below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

Singing Dunes

Singing dunes, which belong to the type of dunes known as 37 _____, produce a very loud sound which is transmitted at a low frequency. Researchers have worked out that sand grains fall down the dune and start vibrating against other grains, forming a sound wave. Research proves that the individual grains have a similar 38 _____, but the differences in dimensions alter the 39 _____ of the 'song'. Each grain is covered with a mixture of different 40 _____ and this is vital to the sound production.

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IELTS Academic Reading — Test 6

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Reef Fish Study

Tom Holmes examines the relationship between size and survival in fish on Australia's Great Barrier Reef

It is a widely held belief among those who study animals that larger size conveys some form of advantage throughout life. The idea that certain body characteristics may influence a prey's ability to survive when being attacked by a predator has received considerable attention in the past, and has been applied to a wide range of both animals and habitats. Body size, weight, growth rate, overall condition, sensory development and escape speed have all been linked to survival. Of these, body size has received the most attention from researchers, which has subsequently led to the development of a number of theories on how this characteristic may influence the predator-prey relationship. One of the most popular of these is known as the 'bigger is better' hypothesis. This theory predicts that larger body size should increase an individual's ability to escape from a predator due to the characteristics associated with this, such as overall strength and visual development. Bigger is thought to be better, but is this always the case?

Compared with other habitats and ecosystems, not a great deal is known about predator-prey relationships within tropical reef communities. What we do know is largely limited to predator identity and, to some extent, to what these predators eat. But what drives these interactions and how does body size influence the outcome? How would fish populations respond to the removal of certain predators from a reef fish community? It is precisely these questions that have led our research team to Lizard Island, in the northern sector of the Great Barrier Reef.

We want to examine the predator-prey relationship during the 'early juvenile' or larval phase of tropical fish, and determine whether a predator's choice of prey is influenced by factors such as the prey's size, weight and swimming speed, as these characteristics vary from individual to individual at the beginning of the juvenile stage.

Most coral reef fish have a life cycle consisting of an open ocean larval phase followed by a juvenile to adult phase near the reef. The transition between the two stages is marked by rapid and dramatic changes in the body of the fish, and a sudden move from the open ocean to coral reef habitats. This point is commonly known as 'settlement'. Not

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surprisingly, it is also characterised by increased levels of mortality in the first 48 hours following settlement, as individuals are forced to adapt to life not only on the reef but also with a previously unknown set of predators.

However, studying the interactions between predators and prey during this phase is no easy task. On the northern Great Barrier Reef, coral reef fish settlement usually occurs within 2-3 days of a new moon during the warmer summer months. These short windows of opportunity, combined with the difficulties associated with working underwater, transporting fish and adapting them to the laboratory, often make research a daunting task. These obstacles are overcome using a combination of innovative techniques, long working hours and a measure of sheer persistence. We collect the larval fish immediately prior to settlement using purpose-built traps that use artificial light to attract and contain fish of this life stage only. These fish are subsequently transported to an aquarium, along with predatory fish, before being used in a series of aquarium and field-based trials.

According to the 'bigger is better' hypothesis, being larger at the time of settlement should result in an increased chance of surviving an attack by a predator during the period immediately following settlement. However, our initial research shows that this is not always true. There may be several reasons for this. Communities of reef-based predators can differ considerably over quite small distances. Species that are abundant in one location may be particularly rare in a similar habitat just 100 metres away. In the same way that different predators prefer different types of prey, so too do they prefer different sizes of prey. This relationship is driven by physical characteristics of predators, such as mouth size, and behavioural differences between different predator species.

As a result, the predator community into which a reef fish settles may very well determine which individuals survive. So why don't larger individuals survive more often than smaller fish over this period? The answer may well lie in an alternative hypothesis known as 'optimal foraging theory'. According to this theory, being either very large or very small in size should assist prey to escape during a predatory encounter. The theory predicts that a predator should prefer those individuals that provide the greatest energy return. This is generally a trade-off between the potential energy intake from eating a fish and the amount of energy required to catch it. Larger prey provide a higher energy return but smaller prey generally require less energy to catch. This usually results in a preference for medium-sized prey.

At the time of settlement, juvenile reef fish vary in the characteristics that can help them survive a predatory attack. However, these characteristics are not strongly associated with body size. For example, larger individuals are not necessarily faster. This means that they may be easier to catch than smaller prey. If this is the case then, according to the optimal foraging theory, predators should prefer larger prey because of their greater energy return.

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Consequently, larger prey individuals may be targeted by some predators at the time of settlement. Our research hopes to show that larger size may, in fact, be a distinct disadvantage during certain life stages.

Questions 1–4

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 Research to date has concentrated on the importance of escape speed in surviving an attack by a predator.

2 According to the 'bigger is better' hypothesis, larger animals have better eyesight.

3 Early-juvenile reef fish share similar characteristics.

4 Fully developed reef fish have a slower swimming speed than juveniles.

Questions 5–7

Complete the flow-chart below. Write **ONE WORD ONLY** *from the passage for each answer.*

Life cycle of fish in the Lizard Island study

Phase 1: Larval stage — takes place in the 5 ———

Phase 2: Juvenile stage — takes place close to the 6 ———

the move from one habitat to another is called settlement — 7 ——— rates are high in the early part of this stage

Phase 3: Adult stage

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Questions 8–13

Complete the notes below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

Lizard Island study

AIM

- to study the relationship between predator and prey in juvenile reef fish

PROBLEMS

- i) time-change of habitat takes place in summer and is linked to the (8) _____
- ii) working conditions
- iii) adaptation - difficult to move and settle fish into the (9) _____

METHOD

- larval fish are drawn into a special trap with the aid of (10) _____
- (11) _____ are also collected
- both undergo trials

POSSIBLE REASONS WHY 'BIGGER' IS NOT ALWAYS 'BETTER'

- the number and type of predatory fish on the reef can vary from place to place
- physical features (e.g. (12) _____) affect which prey a predator can eat
- optimal foraging theory suggests that the most common choice of prey should be (13) _____ in size
- large juvenile fish may swim slowly and so may be targeted more often

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Reading Passage 2

Will Eating Less Make You Live Longer?

The latest in a conflicting series of studies suggests calorie restriction could potentially slow ageing in humans

A Calorie restriction, or 'semi-starvation' as some refer to it, has been proven to extend lifespan in many living organisms from yeast to mice, but the picture for primates, including humans, is not so clear. Research published by a team at the University of Wisconsin in the United States shows that rhesus monkeys also live longer on a calorie-restricted diet. But those findings disagree with research by the National Institute on Aging (NIA) in Maryland, also in the United States. Rozalyn Anderson, of the Wisconsin team, says the research is not intended as a recommendation of calorie restriction. 'I find the idea monumentally unattractive!' she says. 'We study it because it is so effective at delaying ageing and the onset of age-related disease. It's a way to tease out what it is that creates increased disease vulnerability as a function of age.'

B Both groups started long-term trials on rhesus monkeys in the late 1980s to determine whether calorie restriction would extend the lifespan of primates. In mice, many experiments had come to the same finding: feed them a diet with 30% fewer calories and see a lifespan extension of 40%. The monkey trials were set up in a similar way: researchers took the calorie content of a standard monkey diet, cut it by 30% (while continuing to supply all essential nutrients) and monitored whether those monkeys lived longer, healthier lives than those on the standard diet.

C In 2009, with the monkeys approaching old age, preliminary findings of the two trials started coming in. For the Wisconsin monkeys, calorie restriction seemed to be working. Compared to well-fed control animals, the lean monkeys were living longer and suffering less from the diseases of ageing: diabetes, heart disease and brain diseases. However, in 2012 the NIA results emerged with a dramatically different conclusion: their monkeys were not living any longer than the controls, although they were healthier.

D The Wisconsin group's latest results confirm that their calorie-restricted monkeys are living longer than the controls. They also offer a possible explanation of why the two groups' findings don't agree, and that lies in the treatment of the control group. The Wisconsin study began with monkeys in early adulthood. Initially, all the monkeys were allowed to eat as much as they liked. A few months into the trial, the monkeys were placed into one of two groups: the controls (who continued to be fed as much as they wished) and the calorie-restricted monkeys (who were given an individualised diet of 30% less than they were previously eating).

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E The NIA study differed in two ways. First, the control group of monkeys were not allowed to eat as much as they wished. They were given a diet considered to represent a normal calorie count, while the calorie-restricted monkeys were fed 30% less than that. Second, whereas the Wisconsin monkeys were given highly processed food high in sucrose, making it easy to standardise, the NIA diet was based on whole grains, fish oils, and was very low in sugar. These different settings for the normal control diet may provide an explanation of why the two groups showed different results.

F The Wisconsin group may in effect have studied the effects of 'overeating'. Their control animals weighed up to 10% more than average for their age and gender. Compared to them, the calorie-restricted animals not only suffered fewer diseases, they lived longer. Julie Mattison, head of the NIA study, observes that an overweight person who goes to a fast food restaurant every day will obviously benefit if their calories are cut back by 30%.

G The NIA study fed their control monkeys what they considered a 'standard' caloric intake and saw that longevity for monkeys that were fed 30% less was the same. But in Anderson's view, the NIA control monkeys were not fed enough; the NIA was calorie restricting both groups of animals. Their older female control monkeys, for example, weighed nearly 20% less than the national average. Indeed, Anderson points out that several of the NIA control monkeys have lived past the age of 40, far exceeding the 27-year average lifespan for captive rhesus monkeys. 'That's the maximum lifespan ever detected for the species, so the idea that their intervention is doing nothing is really at odds with the data,' says Anderson.

H However, neither group had the right notion of a standard diet according to Leonie Heilbronn, who researches calorie restriction and healthy ageing at the University of Adelaide in Australia. Heilbronn points out that the NIA controls were a very healthy set of monkeys and leaner than a control monkey should be. On the other hand, she notes, the Wisconsin controls were a little bigger than they had to be. On balance, Heilbronn agrees with the Wisconsin researcher's argument. 'I think these studies suggest calorie restriction definitely will increase lifespan,' she says.

I Both studies agree that cutting calories is beneficial to health in both cases the calorie-restricted monkeys had fewer diseases related to ageing. As to the question of whether calorie restriction extends lifespan in primates, Anderson and Mattison are currently working together to compare the two studies' raw data. They plan to co-publish a joint analysis. The current data conflict will ultimately provide deeper insights into ageing, Anderson predicts. 'The fact the two studies were set up differently, asking the same question in different ways I think we will gain maximally from that.'

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Questions 14–19

Reading Passage 2 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
14	a reference to studies based on the same level of calorie reduction with different animals	
15	a comment that some control monkeys in one study reached an older age than normal for animals of their kind	
16	distinctions between the types of food in each study that may have led to a contrast in findings	
17	examples of health problems which monkeys on calorie-restricted diets were less likely to get	
18	a researcher's negative opinion of a calorie-restricted diet	
19	a reference to the stage in the monkeys' lives at which research commenced in one study	

Questions 20–23

Match each statement with the correct researcher, **A, B or C**.

LIST OF RESEARCHERS

A Rozalyn Anderson

B Julie Mattison

C Leonie Heilbronn

Q	Statement	Answer
20	It is good that the two studies took dissimilar approaches.	
21	Neither study used diets that are typical for monkeys.	
22	Calorie restriction is a method of finding out about health issues connected with ageing.	
23	Calorie reduction will have a positive effect on people who have unhealthy diets.	

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Questions 24–26

Complete the summary below. Write **ONE WORD ONLY** *from the passage for each answer.*

Agreement between the two studies on calorie restriction in rhesus monkeys

The two studies were in agreement that it is beneficial to cut calories, in terms of health. Monkeys whose calories were restricted suffered fewer diseases that are associated with the (24) ———— process. Anderson and Mattison have undertaken research to determine if a calorie-restricted diet increases primates' (25) ————. Although there is a (26) ———— between the two sets of data, Anderson believes they will help researchers reach a better understanding.

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Reading Passage 3

Saving the Skink

What can be done to save the skink, one of New Zealand's most endangered lizards?

A New Zealand's reptiles come in three forms: the tuatara, the geckos and the skinks. With their gleaming scales, long bodies and often surprisingly short limbs, skinks are instantly recognisable. There are more than 90 species, half of which still await formal description. Identification depends on where specimens are found, and such minutiae as the number of scales on their toes. Two of the largest species of skink - the grand and the Otago - are found only in Otago, in New Zealand's South Island. Skinks, being reptiles, need to regulate their temperature, and lie on sunny rocks to absorb heat. However, the grand and Otago are the only kinds that do not hibernate through winter; when snow can fall even in summer in Otago, there is no point in them lying dormant for extended periods. Like all other New Zealand reptiles except tuatara and the egg-laying Suter's skink, they give birth to tiny baby lizards. Today, the Otago and grand skinks are lucky to reach five years of age in the wild, although in zoos or research centres they can live beyond thirty.

B Unfortunately, extinction may loom for these reptiles. With any decline in New Zealand fauna, predation is always suspected, so the efforts of the Department of Conservation (DOC) have focused on controlling specific pests. An example was a six-year project to control the feral cats on the Otago Skink Reserve, launched in the 1990s. Despite the capture of hundreds of cats, skinks showed no signs of recovery; indeed, there were further declines. The trapping programme also revealed rats and hedgehogs to be predators.

C Parasite infections are also thought to be on the rise. Although no direct health implications have been detected the trend is a concern, as it may signal a subtle deterioration in the skinks' living environment. Infection with mites is suspected of inhibiting reproduction among captive animals.

D Another potential threat is habitat loss. Eight hundred years ago, Otago was covered in forest. However, extensive burning produced wide open tussock country. The skinks' response to this change isn't known, nor what their habitat was in the earlier environment, with its predominance of shrubs and trees. Furthermore, recent intensive farming has led to poor soil. Even if we had the answer to the eradication of predators, we haven't a clue to the habitat we should be recreating for the skinks, or if where they persist is best for conservation.

E It is also puzzling that grands and Otagos are disappearing more rapidly than other species in the same region. The explanation likely lies in their delayed sexual maturity. Taking five years before you can breed is a handicap, since there is little likelihood of

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surviving that long. If survivors have only two offspring annually, they do little to remedy the situation. The commoner skinks mature earlier and produce more offspring, thereby doing more to counter predator pressure. Not that there is room for complacency concerning these species. As more is learned about the size of skink populations, it becomes clear that even common species may be facing a doubtful future.

F So far, DOC's efforts to eliminate specific predators have not resulted in a recovery of skink numbers. The prediction that less than six years is all the Otagos and grands have left is supported by data from surveys conducted outside the protected area between 1984 and 2002. These show that more than 50 per cent of sub-populations of both skinks have died out over the past years. Populations in other regions are believed to be smaller, although access issues on some properties limit knowledge. No more than five years remain in which to turn things around - a close call considering it will be at least three before detectable response in the populations can be expected owing to their low breeding rate.

G DOC's recovery-programme team has recently embarked upon an experiment to extend nearly two kilometres of mammal-proof fencing around a population of Otagos and grands in the Macraes Flat reserve. Choosing a site was difficult, as with a limited budget and the high cost of the fencing it had to be small. It also had to include vacant space and resources for a growing population. An added complication was that grands and Otagos have subtly different environmental preferences, both of which need catering for.

H Because of the danger of imminent extinction, the team has also had to develop emergency captive-management techniques. Specially permitted breeders, who have had some success in breeding the Otago in captivity, are now developing techniques to breed grands. Captive management, however, is a last resort. The priority is to manage and protect skinks in the wild.

I Getting the public to understand the plight of these creatures is essential. Without public support, their chances are slim. Conservation is as much driven by public opinion as it is by political decisions. In New Zealand, the situation is more pressing than just about anywhere else on the planet because many of its native birds and lizards have simply not evolved to defend themselves against species introduced in the last few hundred years. In the once-isolated islands that form New Zealand, ecosystem collapse didn't happen 5000 years ago or even 500 years ago; it's happening now.

J Fortunately, DOC no longer faces these problems alone. The challenge has been taken up by a group of conservationists representing business, science and agriculture. Central Otago Ecological Trust (COET) aims to establish a refuge for the reintroduction of native species. It has identified suitable habitat and, with the generous support of a local farmer

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with grazing land there, has begun the construction of a mammal-proof fence to protect a 24-hectare area. The recovery programme and COET provide hope that skinks might once again flourish.

Questions 27–31

Complete the summary below using words from the box.

Characteristics of New Zealand skinks

WORD LIST

A eggs **B** location **C** hooded eyes **D** the wild **E** body heat **F** winter sleep
G captivity **H** live young **I** digestive process **J** small legs **K** warmer areas

There are three kinds of reptiles in New Zealand: the tuatara, the gecko and the skink. Skinks are easily distinguished from these other natives by their shiny scales and (27) _____. In common with other reptiles, the grand and Otago species of skink have cold blood and therefore must have their (28) _____ externally controlled. They are unique in that they do not require a long (29) _____. Many species are yet to be identified, and identification depends primarily on their (30) _____ and extremely small physical differences. In (31) _____, skinks may potentially live for about 30 years.

Questions 32–36

Do the following statements agree with the claims of the writer?

YES *if the statement agrees with the claims of the writer* • **NO** *if the statement contradicts the claims of the writer* • **NOT GIVEN** *if it is impossible to say what the writer thinks about this*

(32) The DOC project to remove feral cats helped stop the decline in Otago skink numbers.

(33) Parasite infections are having a serious effect on the physical condition of wild skinks.

(34) The best possible habitat for the conservation of skinks is unknown.

(35) It is possible that skinks used to have larger numbers of young than they do now.

(36) The survival of the common species of skink may soon be at risk.

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Questions 37–40

Choose the correct answer.

37 According to the sixth paragraph, what is being said about the future of the grand and Otago skinks?

A More data is necessary in order for DOC to target the right areas.
B DOC is uncertain how long a new strategy will take to be effective.
C There is a considerable risk that DOC will not succeed in saving them.
D The first step in DOC's strategy is to identify their main predator.

38 According to the writer, a problem with the DOC's fencing project was that

A the fence needed to surround various kinds of habitat.
B the fence was not the preferred choice of some conservationists.
C the fence had to be constructed near a heavily populated urban area.
D there was no way of knowing how many skinks were inside the fence.

39 The writer uses the phrase 'a last resort' in the eighth paragraph in order to

A introduce a recent alternative approach to skink conservation.
B question whether captive breeding has worked in the past.
C imply that skinks are unlikely to reproduce outside their usual habitat.
D express a preference for conserving skinks in their natural environment.

40 What is the writer doing in the final paragraph?

A referring to obstacles in the way of skink conservation
B appealing to landowners to donate land for conservation purposes
C describing a conservation scheme that is being set up on farmland
D showing how commercial profit can be made from ecological ventures

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IELTS Academic Reading — Test 7

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Deep Sea Discovery

Recent research has provided new insights into how fish communicate

A Nico Michiels is an ecologist from the University of Tubingen in Germany who spends part of each year in Egypt, where he dives in the Red Sea, observing fish life and gathering data on its coral reefs. In September 2007 he decided to find out how far red light could penetrate the ocean depths. Seawater absorbs different colours at different depths, and as an experienced diver, Michiels was aware that red light is extinguished not far below the surface whereas blue-green light penetrates deeper. To find out the depth at which red disappeared in this particular ocean, however, he attached a special plastic filter to his dive mask which was designed to block out all colours except red. Then he began to descend. In theory, once he reached about 15 metres, he should have been plunged into darkness. Instead, something totally unexpected happened. Sure enough, 20 metres down it was as dark as night. 'All the fish disappeared. With no light from the surface, they were effectively black and had become invisible,' he says. But it didn't stay black for long. 'Then I saw a group of goby fish with bright red eyes lit up against the background. After that red spots began to show up all over the reef.'

B Even with the red filter removed, Michiels could pick them out without much trouble once his eyes grew accustomed to the gloom. It seems strange that no diver or researcher had spotted all this red before, but as Michiels points out, no one saw it because no one expected to see it. On that one dive Michiels discovered three fish species with prominent red markings, and has found many others since.

C But how can fish appear red where there's no red light? Ordinary red pigments look red because they reflect red light while absorbing all other wavelengths. At 20 metres down, there had to be some other explanation for the red Michiels was seeing. He suspected fluorescence. Fluorescent pigments behave differently from ordinary ones: they receive incoming light of one wavelength, for example blue, and emit light of a longer wavelength, in this case red. On the reef in the Red Sea during daytime, the most likely explanation was that the predominantly blue and green wavelengths at depth triggered the emission of fluorescent red in the fish.

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D With only a week left in Egypt, and lacking the equipment to confirm that the fish were fluorescent, Michiels photographed as many of them as he could. Then once back in Germany, he bought an assortment of tropical fish and installed them in his lab. Here he confirmed that the fish did indeed fluoresce. In most of the fish he looked at, the fluorescence could be traced to specialised pigment cells that lie in the skin beneath the scales. These cells contain 'guanine crystals', which scatter light to give fish their silvery sheen. However, Michiels says they are still not sure exactly what is fluorescing. 'It's not the crystals themselves. It's probably a fluorescent protein built into the crystals, and we have a suspicion that it might be made by bacteria.'

E Intrigued, Michiels began a systematic search for red fluorescence in reef fish. He and his colleagues, Nils Anthes and Dennis Sprenger, have identified some 50 species with red fluorescence. The most common markings tend to be on the body towards the head and to a lesser extent around the eyes, and then the fins. To Michiels, the distribution of these markings is one of the strongest indications that red fluorescence has a very particular function: communication with other members of the species. According to several recent studies, a whole range of animals employ fluorescence as a natural highlighter to boost the visibility of body parts they use to signal, for example to ward off enemies. In reef fish, the red tends to be confined to parts of the body used to signal, suggesting these markings serve a similar function. But instead of highlighting an existing colour, the fluorescence gives the fish a colour that otherwise wouldn't exist. For example, fish commonly use eye rings to signal that they are present and their direction of gaze, and Michiels suspects that red-eyed gobies use signals to indicate their location and keep their group together.

F Red light, whatever its source, doesn't travel far through water, which suggests signals are intended to be private, seen only by nearby fish of the right species. There are several lines of evidence to support this, says Michiels. And closely related species do not have completely identical markings, which suggests they might be important in species recognition.

G Michiels suspects red fluorescence has another important role for some reef fish: helping them blend in. During his first dive with the red filter, he noticed corals glow a dark but faint red too. Against this irregular red background, a fish that glows red all over would be hard to distinguish. More compelling for Michiels is the case of the scorpionfish, which lies perfectly still until food swims past which it then sucks in.

H Yet if red plays any part in a fish's life then it must be able to see it. Fish that live in a world dominated by blue-green light are assumed to have eyes tuned to those wavelengths, and most marine fish that have been studied are thought incapable of seeing red. One exception is the seahorse, whose eyes are sensitive to red. As for the other fish, it remains to be seen.

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Questions 1–6

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 During his 2007 dive, Michiels expected to encounter total darkness at about 15 metres.

2 Michiels could see the red markings on fish without the aid of the red filter.

3 Other divers had assumed they would see fish with red markings.

4 All the fish with red markings that Michiels found during his diving expeditions came from the Red Sea.

5 Michiels first thought of the possibility that fish could fluoresce while he was in Germany.

6 Michiels remains uncertain as to what creates fluorescence in fish.

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Questions 7–13

Complete the notes below. Write **ONE WORD ONLY** *from the passage for each answer.*

Michiels's findings

MICHIELS HAS OBSERVED

- 50 types of fish with red fluorescence in total
- markings mainly near the (7) _____

SOME OF MICHIELS'S BELIEFS ARE THAT

- red fluorescence is used specifically for (8) _____ purposes
- fish, like some animals, use fluorescence to keep (9) _____ away
- gobies depend on red fluorescence to show their (10) _____
- there are variations in the markings of fish among those (11) _____ which are very similar

OTHER BENEFITS OF RED FLUORESCENCE

- fish cannot easily be seen near backgrounds of (12) _____ which give off a red light
- helps some fish catch their prey

THE ABILITY TO SEE RED AMONGST FISH

- the only fish proven to have this ability is the (13) _____

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Reading Passage 2

Jellyfish – The Dominant Species

A Jellyfish have become the curse of beach holidays, permeating every ocean on the globe, thriving in the Arctic and the tropics. In an ever-changing world where other species struggle to endure, jellyfish populations are on the rise. To the untrained eye, these creatures drift aimlessly on the oceans' currents and appear benign. In addition, they lack sharp claws, piercing teeth or even a brain. Despite this, they are armed with an amazing arsenal of weapons, especially the stinging power of their tentacles. As a result, jellyfish are among the most-feared, least-understood creatures in the seas.

B According to Dr Monty Graham, a jellyfish scientist at the University of South Alabama, US, 'Jellyfish are a pretty good group of animals to track coastal ecosystems. When you start to see jellyfish numbers grow, that usually indicates a stressed system.' While populations appear to be down this year, Dr Graham sees 'a statistically solid increase' over the longer term. This increase first gained attention in the 1980s when a huge number of jellyfish, Atlantic Ocean natives named Mnemiopsis leidyi, devastated the Black Sea, an ecosystem already weakened by overfishing of anchovies. Scientists believe that this species of jellyfish came in on the bottom of a ship and then rapidly multiplied, feeding on anchovy eggs and the plankton that young fish rely on.

C Dense jellyfish aggregations can be a natural feature of healthy ocean ecosystems, but a clear picture is now emerging of more severe and frequent jellyfish outbreaks worldwide. Dr Anthony Richardson, from the University of Queensland, Australia, explains that once jellyfish gain a foothold, if conditions are right they can establish a massive population at the expense of other ocean life. The problem is that parts of the ocean might switch from being dominated by fish to being dominated by jellyfish.

D A study done by Richardson and his colleagues explores the causes behind jellyfish infestation, and the need for swift, decisive action to stem jellyfish takeover. Jellyfish outbreaks are linked directly to human actions, including overfishing, the input of fertilizer and sewage into the ocean, and climate change. Overfishing has removed fish from marine ecosystems at astounding rates. According to Richardson, this has made it possible for jellyfish to take their place. 'This is because small fish appear to keep jellyfish in check by predation (on jellyfish when they are very small) and competition (when feeding). So, once we remove fish, jellyfish can proliferate.' Eutrophication is another human-caused change in the ocean that has likely contributed to jellyfish explosions. Eutrophication is an increase of nitrogen and phosphorus in the ocean, largely caused by fertilizer and waste run-off. This leads to algae blooms, which lower oxygen in the marine ecosystem, creating so-called 'dead zones', which have been increasing dramatically around the world.

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According to Richardson, these low-oxygen waters give jellyfish the advantage. 'Fish avoid low-oxygen waters but jellyfish, having lower oxygen demands, not only survive but can thrive in these conditions as there is less predation and competition from fish.'

E Furthermore, Richardson and his colleagues speculate that climate change may expand the traditional geographical range of jellyfish. 'As water warms, tropical species are moving towards the poles. Many venomous jellyfish species are tropical and could move into more densely populated subtropical and temperate regions.'

F Once jellyfish appear en masse in an ecosystem they can make it very difficult for fish to stage a comeback. By feeding on fish, the jellyfish successfully prevent fish from returning to their normal population numbers, says Richardson. 'One can thus think of two alternate states with each being stable: one dominated by fish and the other by jellyfish. Unfortunately, where there is a jellyfish-dominated state then this does not support the nutritional needs of other fish, marine mammals, and seabirds.' In other words an ecosystem that loses fish also loses the species that depend on fish for survival. This state has been defined as a 'monoculture of jellyfish', an apt analogy since the situation shares similarities with other monocultures. When the rich biodiversity of tropical forests is replaced by plantations growing a single species of tree, an area of rich variety becomes a desert in terms of biodiversity, as do ocean ecosystems when jellyfish become the dominant species. One result of large jellyfish populations is the economic effect it has had on the fishing industry. In the Gulf of Mexico, shrimp fishermen are struggling with a jellyfish boom that fills nets, causing them to break and resulting in millions of dollars in losses.

G Experts say that a greater understanding of jellyfish, including their ideal water temperature and feeding habits, is necessary to determine with certainty what is causing the recent massive invasion, and to come up with ways to combat it. Due to the difficulty of turning ecosystems around once jellyfish have become dominant, Richardson and his colleagues propose focusing on 'prevention rather than cure'. They recommend a halt to overfishing small fish that are vital to keeping jellyfish in check, reducing the amount of fertilizer and sewage running off into the oceans, and finally, if possible, confronting climate change.

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Questions 14–17

Reading Passage 2 has seven sections, **A–G**. *Which section contains the following information?*

Q	Statement	Answer
14	a prediction as to the direction in which the jellyfish population may spread	
15	a description of some physical characteristics of jellyfish	
16	an account of the consequences of jellyfish as lone survivors	
17	suggestions on how to avoid further jellyfish invasions	

Questions 18–19

Choose **TWO** *letters, A–E. The list below gives some effects that jellyfish have had on the world. Which TWO of these effects are mentioned by the writer?*

A They have damaged the tourism industry in some areas.
B They have led to a reduction in the oceans' oxygen levels.
C They have contributed to the decline in the Black Sea anchovy population.
D They have caused the shrimp business in the Gulf of Mexico to shut down.
E They have created financial hardship in the fishing industry.

Questions 20–21

Choose **TWO** *letters, A–E. Which TWO of the following are possible causes of an increase in jellyfish numbers?*

A a shortage of small fish in the oceans
B the dumping of chemicals into the oceans
C a decline in biodiversity in the oceans
D more competition among other fish in the oceans
E a decrease in seabird populations

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Questions 22–26

Complete the sentences below. Write **ONE WORD ONLY** *from the passage for each answer.*

Some fish in the oceans may be unable to sustain their population as the jellyfish eat their (22) _____.

The state of jellyfish becoming the main ocean species has been named (23) _____.

Increasing numbers of jellyfish can damage (24) _____ used for commercial fishing.

Understanding basic facts about jellyfish, such as the (25) _____ of the ocean which suits them best, may help control their numbers.

Richardson believes it is better to direct attention to (26) _____, instead of just trying to solve existing problems.

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Reading Passage 3

The Pirahã People of Brazil

The Pirahã language has stirred up debate among experts

A The Pirahã tribe live deep in Brazil's Amazon forest, and their language is hotly debated by linguists. Since 1977, the ethnologist Daniel Everett has spent a total of seven years living with the Pirahã and has committed his career to studying their puzzling speech. Indeed, he was uncertain for so long about what he was actually hearing while living among the Pirahã that he waited nearly three decades before publishing his findings. The debate over the Pirahã language goes right to the core of the riddle regarding how Homo sapiens managed to develop vocal communication. Although bees dance, birds sing and whales even sing with syntax, human language is unique, if only for the reason that it enables humans to piece together never-before-constructed thoughts, and be infinitely imaginative - think of Shakespeare's plays or Einstein's theory of relativity.

B Linguistics generally focuses on what features all human languages have in common, but the Pirahã language departs from what some academics have long maintained are essential and inalienable features of all human languages. Most of all, it may be unique for not employing subordinate clauses. Instead of saying, 'When I have finished eating, I will speak to you,' the Pirahã say, 'I finish eating, I speak with you.' Equally perplexing, the Pirahã appear not to use numbers. During the time he spent with them, Everett never heard words like 'all', 'every' and 'more'. There is one word, 'hoi', that comes close to the numeral 1, but it can also mean 'small'. And they were never observed to count without language, on their fingers for example, in order to determine important tasks in village life like how many pieces of meat to grill.

C Everett's findings among the Pirahã have brought new life to a controversial theory by the linguist Benjamin Whorf, who suggested that people are only capable of constructing thoughts for which they possess actual words. Or to put it another way, because they have no words for numbers, they cannot even begin to understand the concept of numbers or arithmetic.

D The Warlpiri language - spoken by a group of Australian Aborigines - like that of the Pirahã, features only the most rudimentary system of counting. However, the Warlpiri people had no difficulty counting farther than three in a foreign language, in this case English, but when Everett attempted to teach the Pirahã how to count in Portuguese like other Brazilians, not a single person could count to 10.

E Everett is at pains to point out that the Pirahã are not unintelligent, for their thinking is not any slower than that of the average university student. And although they reside in a remote part of the forest, they do not live in complete genetic isolation, but mix with

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people from the surrounding populations and share similar intellectual capacities with their neighbours whose languages do contain numbers.

F Eventually, after some 30 years of research, Everett has come up with a surprising explanation for the peculiarities of the Pirahã language. Language, he believes, is created by a people's way of life, their belief system and values. In this way, variety in human language is almost limitless, a function of the human capacity to live in different ways, such as in the forest. What Everett's research has revealed is that the central tenet of the Pirahã culture is to live in the here and now. The only thing of importance that is worth communicating to others is what is being experienced at that very moment, though this can often be described with great care and detail. In consequence, the language has no means to conjugate verbs in order to describe 'yesterday' or 'last week' or 'when I was a child'. Their very literal view of the world curtails abstract thought, and many features taken for granted among other peoples are absent among the Pirahã, such as a creation myth, story telling and painting. One manifestation of their beliefs is that by tradition the names they give their children are not particularly imaginative. Often they are named after other members of the tribe with whom they share similar character traits. Standing out or being different is not encouraged by the Pirahã, and this is reflected in their perhaps colourless choice of names.

G Everett anticipated that these findings would be controversial and the reaction came as expected. Until this point, many linguists had defended the theories of Noam Chomsky, according to which all human languages have a universal grammar. What exactly makes up this universal grammar is the subject of debate, but at its heart is the concept of 'recursion', which is defined as replication of a structure within its single parts. Without it, humans would not be able to view separate thoughts as subordinate parts of a complex whole. And, most pertinent to Everett's work, there would not be subordinate clauses, which are responsible for translating the concept of recursion into grammar. But if the Pirahã do not form subordinate clauses, then recursion cannot explain the uniqueness of human language, and this would negate Chomsky's theories.

H The logical way forward now would be to try to prove that the Pirahã can think in a recursive fashion. The only problem is, nobody can confirm or deny Everett's observations since no other researcher can speak Pirahã as well as he does. Despite this, several researchers - including two of Chomsky's colleagues - will soon travel to Brazil to check his claims. My concern is that soon the Pirahã will simply become one more scientific oddity with every aspect of their lives being exploited and analysed.

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Questions 27–32

Choose the correct answer.

27 What are we told about Daniel Everett in the first paragraph?

A He has lived among the Pirahã since 1977.
B It took him seven years to learn the Pirahã language.
C No one would publish his research for three decades.
D Studying the Pirahã language is the focus of his work.

28 Which of the following is the best summary of the second paragraph?

A Humans are the only species to be linguistically creative.
B Humans, bees, birds and whales share a characteristic.
C Human language is not fully understood by scientists.
D Humans are the only species to use syntax.

29 Why does the writer refer to subordinate clauses?

A to criticise the general approach of linguistics
B to compare two features of the Pirahã language
C to explain why the Pirahã language is difficult to learn
D to exemplify an unusual feature of the Pirahã language

30 What point does the writer make about the work of Whorf?

A He thought that numbers were common to all human languages.
B His theory might be supported by Everett's research.
C His research enabled him to find a new life among the Pirahã.
D He predicted that people like the Pirahã would never be found.

31 The writer refers to the Warlpiri people in order to

A suggest that the Pirahã be taught to count in English.
B show how tribal peoples learn a foreign language.
C compare counting in English and Portuguese.
D illustrate the uniqueness of the Pirahã.

32 What is Everett's point about the Pirahã's intellectual capacities?

A Pirahã students have not graduated from universities.
B He does not want people to criticise their intelligence.
C Their isolation makes it difficult to evaluate their intelligence.
D He believes their language is more complex than their neighbours'.

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Questions 33–36

Choose **FOUR** *answers from the box.*

WORD LIST

A present **B** past **C** time **D** future **E** culture
F grammar **G** art **H** individuality **I** children

Everett believes that a group's language is a product of their (33) _____ and thus language is infinitely varied. During the time he spent living among them, he observed that the Pirahã only place value on the (34) _____. The language has no (35) _____ to describe completed events. The types of names they use reflect the fact that they do not celebrate (36) _____.

Questions 37–40

Do the following statements agree with the claims of the writer?

YES *if the statement agrees with the claims of the writer* • **NO** *if the statement contradicts the claims of the writer* • **NOT GIVEN** *if it is impossible to say what the writer thinks about this*

(37) Everett was surprised by the way his research was greeted.

(38) Chomsky has been critical of Everett's research methodology.

(39) If 'recursion' as a universal concept is disproved, Chomsky's ideas about language would be wrong.

(40) The Pirahã will benefit from their new-found status among academics.

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IELTS Academic Reading — Test 8

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

The History of the Guitar

An overview of the origins of the modern guitar

A The earliest stringed instruments currently known to archaeologists are bowl harps. For millennia, people made bowl harps using, for example, tortoise shells as resonators, with a bent stick for a neck and one or more gut or silk strings. The world's museums contain many such harps from the ancient Sumerian, Babylonian, and Egyptian civilisations. Around 2500-2000 BC more advanced harps, such as the beautifully carved 11-stringed instrument found in the tomb of Queen Shub-Ad in ancient Mesopotamia, now modern day Iraq, started to appear.

B The tanbur probably developed from the bowl harp. It was different from the bowl harp in that its neck was straightened out to allow the strings to be pressed down to create more notes. Tomb paintings and stone carvings in Egypt indicate that harps and tanburs – plus flutes and percussion instruments – were being played together 3,500-4,000 years ago. Archaeologists have also found many similar relics amongst the ruins of the ancient Mesopotamian civilisation. Many of these instruments have survived into modern times in almost unchanged form, for example folk instruments of the region such as the Turkish saz and Afghan panchtar.

C At 3,500 years old, the tanbur which belonged to the Egyptian singer Har-Mose is the earliest known example of this instrument. Har-Mose's tanbur had three strings and a plectrum suspended from the neck by a cord. The soundbox, which increased the volume, was made of beautifully polished cedarwood and covered in rawhide. It can be seen today at the Archaeological Museum in Cairo.

D In order to distinguish guitars from other stringed instruments, it is helpful to have a broad definition of the guitar. Music expert Dr Michael Kasha defines a guitar as having 'a long, fretted neck, flat wooden soundboard, ribs, and a flat back, most often with sides that curve inwards'. The oldest known visual representation of such an instrument is a stone carving at Alaca Huyuk in Turkey, which shows a 3,300-year-old instrument with a long neck and sides that clearly curve inwards.

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E The name 'guitar' comes from the ancient Sanskrit word for 'string' – 'tar'. Many popular stringed instruments used in central Asia today have existed in an unchanged form for several thousand years, as shown by archaeological finds in the area. Many have names that end in 'tar', with a prefix indicating the number of strings, such as the doter, a two-stringed instrument found in Turkestan, and the Persian three-stringed setar and four-stringed chartar. The Indian sitar almost certainly took its name from the setar, but over the centuries it evolved radically, following the Indians' own aesthetic and cultural ideals.

F Tanburs and harps spread around the ancient world with travellers, merchants and seamen. The earliest guitar-like instruments to arrive in Europe had, most often, four strings. Many such instruments, and variations with from three to five strings, can be seen in mediaeval illustrated manuscripts. They were also carved in stone in European churches and cathedrals, from the first century AD through until the 13th century.

G When the four-stringed Persian chartar arrived in Spain, however, it changed in form and construction, acquiring pairs of strings tuned to the same note instead of single strings. It became known as the chitarra. By the middle of the 14th century, the chitarra had become dominant, at least in most of Europe. The earliest known music for the eight-stringed chitarra was written in 16th-century Spain. The ten-string version first appeared in Italy at the same time, and gradually replaced the eight-stringed instrument. A further two strings first appeared in the 17th century, an innovation which guitar makers all over Europe quickly took up. However, this twelve-string arrangement gradually gave way to six single strings across the continent. The six-stringed guitar can thus be said to be a development of the twelve-string, rather than vice versa, as was thought previously.

H At the beginning of the 19th century, the present-day guitar began to take shape, although bodies were still fairly small and narrow-waisted. The modern classical guitar first appeared in its current form in the mid-19th century, when the Spanish guitar maker Antonio Torres increased the size of the body, altered its proportions, and introduced the revolutionary fan-braced top. His design radically increased the volume and improved the tone of the instrument, and very soon became the norm. This design has remained essentially unchanged to this day.

I At the time when Torres made his breakthrough, German immigrants to America – among them Christian Fredrich Martin – began making guitars with X-braced tops. Steel strings, which became widely available several decades later in the early 1900s, offered the promise of much louder guitar, but the increased tension was too much for the fan-braced top. The stronger X-braced top proved equal to the job, and quickly became the industry standard.

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J At the end of the 19th century, guitar manufacturer Orville Gibson added steel strings to a body constructed like a cello, a combination which produced more volume. The electric guitar was born when pickups were fitted to Hawaiian and jazz guitars in the late 1920s, but met with little success until 1936, when Gibson introduced its famous ES150 model.

Questions 1–6

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 The instrument found in Queen Shub-Ad's tomb is the world's oldest known version of a harp.

2 Today's Afghan panchtar is very similar to an ancient Mesopotamian instrument.

3 The Egyptian singer Har-Mose was an excellent tanbur player.

4 Cairo Archaeological Museum contains many historic musical instruments.

5 The instrument carved in stone at Alaca Huyuk is consistent with Dr Michael Kasha's definition of a guitar.

6 The different instruments that appeared in medieval literature had the same number of strings.

Questions 7–13

Complete the table below. Write **ONE WORD ONLY** *from the passage for each answer.*

Date	Type of instrument	Notes
13th–19th century	Chitarra	was a development of an earlier instrument called the 7 ——— extra strings added in 16th century in Italy
from around the 1850s	Classical guitar	its shape a result of modifications including a larger 8 ——— introduced by Antonio Torres changes produced better tone and greater 9 ———
mid-19th century	X-braced top guitar	first made in 10 ——— in mid-19th century strings made of 11 ——— became available around 1900
1920s onwards	Electric guitar	in the 1920s, 12 ——— added to guitars a well-known version brought out by 13 ———

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Reading Passage 2

Playing Soccer

There are many differences between playing soccer in the street and joining a youth team in an organized league in the USA

A Street soccer, as its name implies, is an informal variation of the sport, often played on the street, particularly in urban areas. There are many reasons for the widespread popularity of street soccer. Unlike youth soccer, its more formally organized counterpart, no large space is needed, and goal posts, corner markers, and marked lines, associated with the formal game, are typically absent, as are game officials or referees. Another attraction of street soccer is that it is played frequently and competitively, but does not necessarily require standard 11-a-side teams or fixed playing positions. Unlike in youth soccer, inexperienced street soccer players rarely learn from repetitive technical and tactical drills. Instead, they learn from their poor performance in competition, unconscious of the skills they are nonetheless developing, and without older adults or coaches present. Players learn without effort through playing the game, and soon attain an almost natural feeling for the sport.

B However, there are lots of cities in the world today where conditions are such that street soccer is no longer possible. Congested traffic now dominates where games were once played. Parks and open fields are used as hangouts for older teenagers with other interests. Add to this the requirement in many localities for official permits to use public spaces and the managed schedules that many young people have today, and spontaneous play of any kind is hard to imagine.

C In spite of all these obstacles, which are probably solvable in most instances, there is another sociological explanation of why in many places street soccer doesn't enjoy the same popularity it once did. In his book How Soccer Explains the World, US writer Franklin Foer observes: But for all the talk of freedom, the 1960s parenting style had a far less relaxed side too. Like the 1960s consumer movement which brought seat belts and airbags to cars, the youth soccer movement felt like it could create a set of regulations that would protect both the child's body and mind from damage. Soccer leagues like the one I played in as a child handed out 'participation' prizes to every player, no matter how few games his or her team won. Where most of the world accepts the practice of using your head to hit the ball as an essential element of the game, some youth soccer parents have worried over the potential for injury to the brain. An entire industry grew up to manufacture protective headgear. Even though very little medical evidence supports this fear, some youth leagues prohibited heading the ball altogether.

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D A growing body of people don't believe street soccer involves a legitimate educational method. They argue that children need to be taught by experts. Youth soccer instruction now begins with four-year-olds, so that they will have an advantage as six-year-olds. This need to get ahead brings with it a fear of falling behind that only expert instruction can prevent. This type of instruction leaves no room for the trial and error approach of street soccer.

E One of the basic ideas of street soccer is that young players are assigned a particular role by a better player and are expected to play for the good of the team. Such an assignment runs counter to the idea of youth soccer that every child needs to learn every position and will benefit from doing so. In street soccer, you fill the role that you are best suited to at a particular time. While this role assignment can change from game to game, the purpose is always the same: to get the best out of each individual at any given moment.

F In street soccer, children have to learn patience, to wait their turn, to realize that they are not entitled to make decisions, or even be listened to simply because they show up. Positions of responsibility are earned through competition within the team. Younger players in street soccer must wait to attain those positions. In youth soccer, however, with its overly democratic values, youngsters are guaranteed their time in the spotlight. Whether it's their turn to be captain, to play a central position or to take a crucial shot, youth soccer players come to believe that hard work and patience aren't really necessary.

G Not only does every youth soccer player get a chance, it is assumed that each individual has played well. 'Everyone's a winner; no one's a loser' is a guiding principle of youth soccer. This ensures each individual goes away positive about themselves. No one can leave a game or a practice feeling bad. But, if there really are no losers, then why try at all? Since giving less than your best receives the same reward as giving your best, why go to any extra effort? In street soccer, every game results in a winner and a loser and everyone knows who is who. Losing a game is a common experience and players learn early on how to handle this. As a result, unlike most youth soccer players, they acquire resilience. A further difference between these two strands of soccer is that in street soccer a formal record is not kept. You can lose one day and win the next. The results are only temporary and are forgotten within minutes of the end of the match. But in organized youth soccer, the position each person plays and the results are formally noted and maintained throughout a season.

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Questions 14–19

Reading Passage 2 has seven sections, **A–G**. *Which section contains the following information?*

Q	Statement	Answer
14	a contrast between the ways young players gain experience of playing different positions	
15	examples outside sport of greater emphasis on individual safety	
16	a description of methods of selection for leadership on soccer teams	
17	details of urban changes that discourage street soccer	
18	a mention of the lesson that failure teaches street soccer players	
19	an explanation of why youth soccer emphasises the need for coaches	

Questions 20–21

Choose **TWO** *letters, A–E. The list below gives some possible reasons for the popularity of street soccer. Which TWO of these reasons are mentioned by the writer?*

A Many famous soccer players got their start in street soccer.
B Young people can begin playing street soccer at a very early age.
C You do not need elaborate facilities to play street soccer.
D Inexperienced street soccer players are not criticised for mistakes.
E Street soccer teams can have varying numbers of players.

Questions 22–23

Choose **TWO** *letters, A–E. The list below gives some possible results of the 1960s parenting style. Which TWO of these results are mentioned by Franklin Foer in the excerpt from How Soccer Explains the World?*

A Participation in youth soccer became much more expensive.
B Some youth soccer leagues adopted more restrictive rules of play.
C Fewer young people joined youth soccer teams.
D Youth soccer players were sometimes rewarded for simply playing in games.
E Soccer equipment manufacturers directed advertising towards parents.

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Questions 24–26

Complete the summary below. Write **ONE WORD ONLY** *from the passage for each answer.*

Winners and losers

For youth soccer players, a key (24) _____ is that they should always come away from the game with a positive attitude. In this respect, regardless of the effort the players make, they get some kind of (25) _____ at the end of a game. In street soccer, however, players gain resilience because they have to learn to cope with failure. But the outcome of a match isn't remembered for long. In fact, no-one ever keeps a (26) _____ of the results of games.

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Reading Passage 3

The History of Chocolate

A Chocolate is now perceived as a ubiquitous indulgence, whether consumed as confectionery or a beverage, readily accessible in global markets. However, it may come as a surprise that chocolate was once regarded as a commodity of extraordinary value. The discovery of the cocoa tree, the source of chocolate, occurred approximately 2,000 years ago in the lush tropical rain forests of the Americas. The evolution of chocolate — from a regionally significant Mesoamerican drink to globally consumed confectionery — has been shaped through cross-continental cultural exchanges and the successive influences of multiple civilisations.

B Historians suggest that the ancient Maya pioneered cocoa tree cultivation in Mesoamerica, with evidence dating back approximately 2,000 years. They transplanted cocoa trees from the dense rain forest into their domesticated gardens, where cocoa beans were initially roasted and ground into a smooth paste. The paste was combined with spices and water to produce unsweetened chocolate beverages. The Maya also employed a distinctive technique of transferring the liquid back and forth between containers, generating a foamy layer on the drink's surface.

C Cocoa and chocolate were integral to Maya culture, commonly depicted in their architectural structures and artistic works. While the aristocracy, particularly the royal families, consumed chocolate during elaborate ceremonies, even less privileged groups occasionally savoured the drink. Furthermore, cocoa beans served as a powerful emblem of marital union and were ritually exchanged during wedding ceremonies.

D The Aztec civilisation in Mexico regarded chocolate as a prestigious commodity, despite being geographically unable to cultivate cocoa trees within their heartland. Accordingly, they sourced cocoa beans through extensive trade networks, even utilising them as currency for tax payments. Chocolate held an exclusive function within Aztec imperial and religious ceremonies, where priests offered cocoa beans as divine offerings and prepared sacred chocolate drinks. Due to the scarcity of cocoa beans, consumption remained restricted to the societal elite, with historical accounts suggesting Emperor Montezuma consumed up to 50 cups daily. Linguists trace the term 'chocolate' to the Aztec Nahuatl word xocolatl, meaning 'bitter water' in the Nahuatl language, though some scholars argue that it has hybrid origins combining Maya and Nahuatl linguistic elements.

E In 1502, explorer Christopher Columbus introduced cocoa beans to Spain after voyaging to Mesoamerica, and later Spanish explorer Hernán Cortés quickly recognised the potential economic value of chocolate. Upon arriving in present-day Mexico in 1519, Cortés observed that chocolate beverages would appeal to the Spanish. After conquering the Aztec

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Empire, Spanish forces exported confiscated cocoa supplies to Spain. In response to the growing demand, Spain began cultivating cocoa trees in its American colonies. Spanish elite first developed a taste for sweetened chocolate drinks, but the beverage rapidly spread across Europe. England, the Netherlands, and France followed suit by establishing cocoa plantations in their own colonies. By the 18th century, chocolate remained primarily a luxury restricted to the affluent. However, the Industrial Revolution's technologies enabled mechanised production, dramatically reducing costs and making chocolate available to a broader population.

F Cocoa trees are cultivated across multiple countries in Africa, Mesoamerica, and South America, thriving in cool, shaded environments within equatorial zones. Despite widespread cultivation, cocoa farming remains labour-intensive, due to the trees' requirement for specific water thresholds, temperature ranges, soil compositions, and protection from environmental threats. Approximately five years after planting, cocoa trees begin to yield large, pod-shaped fruits that grow on the trunk. These pods contain the beans used in chocolate production. While several types of cocoa trees exist, the bulk of the chocolate is produced from the seeds of the most commonly cultivated varieties. However, farmers opt to grow more specialised types, such as Criollo or Trinitario, valued for distinctive flavour profiles. Unexpectedly, cocoa trees grown on farms are more vulnerable to disease and pests than their wild counterparts. Cocoa cultivation presents a strenuous endeavour for farmers, who sell their harvests on futures markets, exposing their incomes to external market fluctuations.

G The process of chocolate production involves a series of steps, carried out in specialised factories where cocoa beans are meticulously processed. Workers first sort, clean, and roast the beans before extracting the outer shells, leaving only the inner nibs. These nibs are then ground into a dense paste known as chocolate liquor, which is subsequently separated into two components: cocoa solids and cocoa butter. Following standardised formulations, chocolate manufacturers blend the chocolate liquor with precise ratios of sugar, milk, and cocoa butter. The resulting mixture, often referred to as the 'crumb', undergoes fine grinding to create a smooth consistency. Finally, the mixture is then subjected to two additional processes — conching and tempering — before being placed into moulds to create various forms, such as bars or other chocolate products.

H Chocolate holds significant global economic value, with the manufacturing industry generating over $5 billion annually from cocoa crops, and its reputation remains most prominent in Europe and the United States. For instance, US cocoa product imports reached $1.4 billion in 2005. On average, Americans consume over five kilogrammes of chocolate per person each year. In addition, upscale chocolate boutiques, which offer premium chocolates, have gained significant popularity. Many of these shops specialise in artisanal chocolates, crafted with unique flavour profiles. Renowned chocolate brands such

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as Cadbury, Godiva, Lindt, and Ferrero Rocher dominate the market. However, the cocoa and chocolate production sectors often face challenges related to fair trade, sustainable development, and environmental preservation. These issues are now being actively addressed by industry stakeholders, including officials, activists, and researchers, who are working closely with farmers to promote farming practices that foster sustainability and protect the environment.

Questions 27–31

Reading Passage 3 has eight paragraphs, **A–H**. *Which paragraph contains the following information? You may use any letter more than once.*

Q	Statement	Answer
27	the part of cocoa trees used for chocolate production	
28	annual chocolate consumption per capita in the United States	
29	the higher likelihood of cultivated cocoa trees being attacked by insects	
30	a community which developed a preference for sweetened chocolate beverages	
31	an approach by which ancient Mexicans could acquire cocoa beans	

Questions 32–36

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

32 The Mayan ruling class forbade the use of cocoa and chocolate in rituals.

33 The Spanish explorer Hernán Cortés had made an investment in chocolate and chocolate beverages.

34 Wild cocoa trees are more susceptible to infectious diseases than cultivated ones.

35 During chocolate production, some of the seeds are discarded.

36 Chocolate enjoys more prestige in Europe and the United States than in any other country.

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Questions 37–40

Complete the flow-chart below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

From Cocoa Beans to Chocolate

Seeds → classifying, cleaning, and roasting → stripping their (37) _____ → Nibs → Crushing → producing (38) _____ → mixing with measured quantities of substances such as sweeteners, liquid dairy, and (39) _____ → Crumb – crushing finely to obtain a smooth texture; manufacturing diverse shapes of chocolate in (40) _____ → Chocolate

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IELTS Academic Reading — Test 9

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Dust and the American West

A Images of deserts in the United States show dusty, barren landscapes, but the land may not always have been this way. Ever since settlers moved west across the US there has been dust-clouds of it everywhere. It was part of the landscape, or so it seemed to them. But there were no records of the landscape of the West until the settlers arrived. Now evidence is starting to emerge which suggests that before the settlers, there was very little dust.

B The evidence comes from the San Juan mountains of southwest Colorado, downwind of Arizona and New Mexico. There Jason Neff, a geochemist from the University of Colorado, has been analyzing sediments, the sand, stones and mud, laid down over the past 5000 years. Atmospheric dust was minimal throughout those five millennia until the mid-19th century, he says, but then, 'from about 1860 to 1900, dust deposition rates shot up.'

C This is surprising because usually dry means dusty, and the American West has almost always been dry-often drier than today. There was a near-permanent drought between 900 and 1300 which was so intense that it destroyed a series of native American civilizations, including the Anasazi, whose cliff homes are now US national treasures. Yet the evidence from the San Juan lakes is that it was not dusty. Even as their civilization was collapsing, the Anasazi seem to have protected their soils from erosion.

D This was not the case with the European settlers once they brought their cows. The landscape the cattle were introduced to was remarkably ill-equipped to cope with grazing animals, says Neff. 'Unlike most other parts of the US, there were few grazers in the American Southwest until the Europeans came. No bison and few antelope or deer.'

E In the Great Plains to the east and north, bison roamed in vast herds. Their regular grazing had created tough grass, while the herds manured the soil. In the Southwest, the land had few defenses against a sudden invasion of millions of livestock, whose teeth stripped the grass and whose hooves punctured the hard crust of desert soils that protected them from the wind. The invasion was sudden, funded by a bubble of speculative investment, much of it from Britain. The money went into railroads, and herds of cattle and sheep that rode the rails to the wide open pastures. By 1900, when sedimentation rates peaked, there were 20 million cattle and 25 million sheep in the West.

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F One of the biggest ranches was owned by the Aztec Land and Cattle Company, which owned a million acres of land by 1884. Each acre had cost the company a mere 50 cents, and like many other speculators, it was only interested in quick profits and had little incentive to protect the soils from overgrazing. By the time Aztec sold the ranch in 1901, it was barren, with cattle carcasses scattered across the exhausted land. Such was the damage to the grasslands that even now few of the pastures have recovered. The parched and exposed soil simply blew away. The 1862 Homestead Act was passed in order to encourage the populating of the West. Any family willing to make the journey was entitled to claim 160 acres and farm it. Yet by the time people moved west in large numbers, the cattle companies had taken most of the land and surrounded it with barbed wire. The water sources were therefore cut off, and were aggressively guarded. It was only in the 1930s, with the passage of the Taylor Grazing Act, that federal authorities finally sought to limit cattle herds.

G Soil scientists have known for a while about the importance of the hard crust that forms on arid soils. 'These crusts can survive winds of up to a hundred miles an hour, but cattle hooves break the crust,' says Jayne Belnap, a soil ecologist at the US Geographical Survey, Utah. The scale of the dust clouds created by the livestock invasion has until now been largely unknown. When Neff first discovered dust in Colorado lake sediments laid down in the 19th century, he was initially unsure where it came from. Maybe it had crossed the Pacific from China's Gobi Desert.

H But after investigating the size and chemical composition of the dust, Neff was clear that it mostly came from the American Southwest, mainly Arizona and New Mexico. Now, with the soil crusts gone, dust clouds still head north and are having significant ecological effects in the Colorado mountains. They carry nutrients with them into areas which previously evolved and survived without them. But perhaps the most dramatic impact of the dust has been on snowfields in the Rocky Mountain Range. Even a thin sprinkling of dark material means snow absorbs more solar radiation, meaning that snowmelt occurs far more rapidly during springtime. The impact on the ski tourism industry is obvious.

I The loss of snow and the shrinking of glaciers across the American West in the past century has been dramatic. Glacier National Park in Montana, for example, has lost three quarters of its snow cover since 1910. All this is frequently attributed to global warming. While this almost certainly plays a role, Neff's findings suggest that dust may also contribute.

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Questions 1–7

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

1 Jason Neff discovered there had been a dramatic rise in dust levels in the second half of the nineteenth century.

2 The Anasazi civilization disappeared due to the impact of dust in the atmosphere.

3 Before cattle were introduced to the American Southwest, large numbers of bison occupied the area.

4 The bison population in the Great Plains diminished because European settlers found it easy to hunt them.

5 The development of railroads across the US was more expensive than originally expected.

6 The Aztec Land and Cattle Company worked hard to take care of the grazing land it owned.

7 Most of the land once owned by the Aztec Land and Cattle Company remains infertile today.

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Questions 8–13

Complete the notes below. Write **NO MORE THAN TWO WORDS** *from the passage for each answer.*

American Southwest: 1860s to present day

1862–1930S

- 1862 – American Homestead Act was passed to encourage settlement in the West
- 1930s – laws were passed to control the size of (8) ———

TODAY — JAYNE BELNAP BELIEVES

- soil was not affected by high (9) ———
- soil was seriously affected by cattle hooves

JASON NEFF

- found that (10) ——— in the Colorado region contained dust
- examined the dust for its chemical content as well as its (11) ———
- concluded that the dust was from the American Southwest
- found that dust affects mountain environments by bringing in (12) ——— that are not normally found there
- argues that dust is partly to blame for the gradual disappearance of some (13) ——— and snow

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Reading Passage 2

Urban Regeneration: An Award-Winning Redevelopment Project in Berlin

A Just over a kilometer south of Berlin's Potsdamer Platz, near the left bank of the Landwehr Canal, an extensive, triangular-shaped area of waste ground once separated the neighbourhoods of Kreuzberg to the east and Schöneberg to the west. Known as Gleisdreieck, meaning 'triangle of rails', it was formed by the intersection of different railway lines built in the mid-nineteenth century that entered Berlin from the south. Tracks, sheds and warehouses belonging to three old railway stations – Dresdner Bahnhof (1875-1882), Potsdamer Bahnhof (1838-1944) and Anhalter Bahnhof (1839-1952) – are situated on a raised area some twenty hectares in size, at a height of four metres above the surrounding ground level.

B As the railway infrastructure gradually ceased to be used, the whole area of Gleisdreieck became increasingly run down and abandoned, to the point of being used as a rubbish tip after 1945. Meanwhile, vegetation took over, turning it into a surprising, natural haven in the middle of built-up areas. The closeness of the wall which divided the city of Berlin between 1961 and 1990 also contributed to the fact that Gleisdreieck was for decades clearly identified as no-man's-land. Only a station of the same name in Berlin's underground railway system testified to its existence. Shortly before the fall of the Berlin Wall, however, the German Museum of Technology was established there, attracting large numbers of visitors and giving visibility to the site. More importantly, the unification of the formerly divided city gave the zone more centrality, at which point it became imperative to identify a purpose for it.

C After 2006, the State of Berlin put forward the proposal of converting Gleisdreieck into a large urban park that would integrate the different urban zones which converged there. The decades-long isolation of the site, which had formerly presented a problem, now represented an opportunity for joining the southern area of Potsdamer Platz with Kreuzberg and Schöneberg. The creation of the park would trigger one of the biggest urban expansions inside Berlin, all within a framework of multiple uses, and social characteristics. It was necessary to stimulate the development of sixteen new hectares of local amenities that would be capable of integrating different generations and social groups in a sustainable way, and in harmony with nature. Following intense discussion with local proprietors and residents, the need to reconcile these goals with the conservation of railway heritage also emerged.

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D Once the area had been subjected to a process of undergrowth clearance and decontamination, it was then organized around a combination of existing and added elements. The project as a whole was planned around a large central meadow, intersected from east to west by a concrete footpath, and from north to south by a pair of railway lines. Once a month, a train slowly travels along these lines from its parking shed to the German Museum of Technology. The concrete footpath, which is a continuation of one of the main Kreuzberg boulevards, starts in the east, bridges the four-metre difference in ground level by means of a stairway, and suddenly ends in the west on reaching the underground lines.

E North of the meadow, there is a large concrete slab with rounded edges. Being south-facing, this functions as a big sunny terrace, full of benches complete with footrests. In the south, the meadow overlooks Yorckstrasse, an underpass crossed by more than fifteen former railway bridges. To the east of the meadow there is quite a dense forest of pre-existing maples, oaks and birches, as well as newly planted trees of the same species. In this area, a couple of large metal frames each hold two swings. The edges of the park are finished with a collection of distinctive, functional spaces, for example a nursery, sports fields, concave surfaces for skateboards, stages for dancing, community gardens, or simply areas covered in gravel obtained from the site itself.

F Although initially the regeneration of Gleisdreieck Park was the subject of disagreement between those who were in favour of safeguarding the railway heritage, and those who wanted to regenerate the adjacent neighbourhoods, it was opened to the public in September 2011. The heart of Berlin now has a new green lung, in which the atmosphere of various small, very different corners fits neatly into a large-scale, wide-ranging and robust general order. This has been possible precisely because the intervention was not limited to conserving industrial remains in order to promote railway history. And as a reminder of the six decades of human absence, during which nature took over, the park has been able to conserve to some extent the spirit of the non-place that preceded it.

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Questions 14–19

Reading Passage 2 has six paragraphs, **A–F**. *Choose the correct heading for each paragraph from the list of headings below.*

LIST OF HEADINGS

i Objectives of the regeneration project
ii Architectural details of the station buildings
iii Features and facilities of the completed development
iv The remains of a railway system
v A positive evaluation of the finished project
vi A way of controlling development costs
vii Ways of crossing the site
viii Reasons why the area had become neglected

Q	Paragraph	Answer
14	Paragraph A	
15	Paragraph B	
16	Paragraph C	
17	Paragraph D	
18	Paragraph E	
19	Paragraph F	

Questions 20–21

Choose **TWO** *letters, A–E. Which TWO reasons are given for the choice of Gleisdreieck as a site for a park?*

A It was a habitat for some rare wildlife species.
B It contained a particular tourist attraction.
C It lay at the heart of the reunited city.
D It was served by a major rail network.
E It contained buildings which could be easily converted.

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Questions 22–23

Choose **TWO** *letters, A–E. Which TWO things did the State of Berlin hope to achieve with the Gleisdreieck development?*

A to bring people of different ages together
B to encourage tourism
C to improve transport links
D to preserve industrial remains
E to generate income for the city

Questions 24–26

Complete the summary below. Write **ONE WORD ONLY** *from the passage for each answer.*

Gleisdreieck Urban Park

In the middle of the new park is a (24) _____, where there is a path for pedestrians and two railway lines. A slow train uses these lines to make a regular journey between its base and a (25) _____. Benches have been provided on the north side, so that people can relax and enjoy the sunshine, and to the east is an area with mature trees of various kinds, as well as sets of swings. A range of sports and leisure areas are situated around the (26) _____ of the site.

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Reading Passage 3

The Fluoridation Controversy

The long-standing debate about whether to fluoridate our drinking water continues

A Fluoridation is the addition of fluorine to public water supplies with the aim of reducing tooth decay. The fluorine, when mixed with water, becomes fluoride and the desired concentration of fluoride in public water is approximately one part per million, depending on the regional temperature and hence the amount of water people are likely to drink. Many studies, such as those by McClure in 1970 through to Burt in 1983, have shown that when children drink fluoridated water, their average rate of tooth decay seems greatly reduced. A typical figure claimed is 50 percent reduction. This apparently enormous benefit for children's teeth is the major argument in favor of fluoridation.

B Three main grounds for opposition to fluoridation have been expressed. First, opponents claim the benefits are exaggerated or not established. Second, there are claims of health risks to parts of the population, for example, allergic reactions. It is also accepted that high levels of fluoride can cause discoloration of otherwise healthy teeth. Proponents do not consider this to be a problem in such small concentrations, whereas opponents disagree - especially because some people drink more water and obtain much more than the standard milligram of fluoride per day. Third, fluoridation is thought to be an infringement on individual rights because it is compulsory medication of all members of a community.

C An understanding of the fluoridation issue has important implications. If, according to the experts, fluoridation is unquestionably a beneficial and non-hazardous measure, then the wisdom of allowing the public to vote on, and reject it must be questioned.

D Almost all studies that have been done have assumed that the scientific aspects of the controversy are unproblematic, and they have excluded science from sociological examination. The traditional view is that science is a special kind of knowledge, which is established through scientific methods and objectively applied by members of a scientific community. However, in recent years there has been a major challenge to this picture by a sociology of science that shows how scientific knowledge is socially negotiated, and inevitably linked to the values of the relevant parties, both scientists and non-scientists. These challengers do not see scientific knowledge as exempt from social inquiry.

E Kuhn (1970) argued that scientific knowledge does not always develop as an orderly process, but is characterized by periodic revolutions, in which the methods of study and the assessment criteria change in a fragmented way. According to Kuhn, the shift from one scientific way of thinking to another is not made solely on the basis of clear rules of formal scientific practice, but can include social factors, though Kuhn has never developed a full

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analysis of what these might be. Collins (1975) took this concept further when he asserted that the outcome of experiments was not something whose meaning could be immediately comprehended, but rather something for interpretation, discussion between scientists, and reinterpretation in the light of other experiments.

F One interpretation of this analysis of science is that traditional distinctions between facts and theories, and between scientific knowledge and values, can no longer be justified. Because social processes are involved at all stages of the creation, evaluation, and establishing of scientific knowledge, social values may also be involved.

G In the same way as many scientists who study fluoridation have overlooked social values, sociologists have also downplayed an important part of the debate by ignoring the number of eminent scientists who have questioned aspects of fluoridation. An example is the study by Sutton in 1960, which analyzed the classic North American studies of the effect of fluoridation on tooth decay, and found that each showed significant methodological shortcomings. Sutton's detailed study throws doubt as to the extent of reductions in tooth decay from fluoridation. Yet Sutton's book is not cited in a single analysis of the fluoridation issue by any sociologist. In a situation of some scientific uncertainty, differences in values are highlighted. A supporter of fluoridation might argue, 'The evidence for the benefits of fluoridation is quite substantial, while the evidence for harm is limited and dubious. I think the likely benefits outweigh the possible dangers; hence I support fluoridation because it is the cheapest and easiest way to make sure every child reaps the benefits.' An opponent might argue, 'Though the evidence for the benefits of fluoridation is substantial, there is some doubt about it. Since fluoridation is not necessary for good teeth, we should forego the benefits if there is some slight chance of harm. Some scientists claim that a small percentage of the population could be harmed by fluoride. Therefore I oppose fluoridation of water supplies and favor the voluntary use of fluoride tablets by those who want to take them.' Both arguments consider the scientific evidence concerning fluoridation, but differ in their assessments of the social benefits and costs. This difference is not between rationality and irrationality but is a legitimate difference in values, for example, the positive value placed on good teeth, the negative value placed on possible health risks, and the social benefits or costs of compulsory or voluntary intake of fluorides.

H From the sociological point of view, opposition to fluoridation is not necessarily irrational. Rather, claims to rationality and to scientific authority are better seen as part of a strategy to promote fluoridation than as incontrovertible statements of fact. Second, social values are likely to be bound up in any decision about fluoridation, so this is not an issue on which declarations by scientific experts ought to be considered the final word.

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Questions 27–31

Choose the correct answer.

27 The optimum amount of fluorine in fluoridated water is calculated partly according to

A how hot the area is.
B how warm the water is.
C how many dental problems there are in the community.
D how much fluorine the community chooses to have in its water.

28 One reason given by the writer for opposing fluoridation is that

A it may contribute to tooth decay
B it will be unacceptably expensive for the public.
C obligatory fluoridation takes away personal freedom.
D excessive fluoride could be added to the water by mistake.

29 The writer mentions Kuhn in order to

A provide a contrast with the view of Collins.
B support the rational nature of scientific inquiry.
C demonstrate that Kuhn did not argue his case adequately.
D show that science can be influenced by non-scientific considerations

30 What did Sutton's research discover about earlier studies in North America?

A There were failings in the way they were carried out.
B The scientists involved had achieved unique results.
C Proponents of fluoridation had not understood its long-term effects.
D Fluoridation had a greater effect on tooth decay than previously believed.

31 In the last paragraph, what does the writer say about scientists?

A They should reveal their true motivations.
B They should not decide the fluoridation policy.
C They are solely concerned with scientific truths.
D They cannot reach agreement on the fluoridation issue.

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Questions 32–35

Do the following statements agree with the claims of the writer?

YES *if the statement agrees with the claims of the writer* • **NO** *if the statement contradicts the claims of the writer* • **NOT GIVEN** *if it is impossible to say what the writer thinks about this*

32 Scientific knowledge should be kept separate from social values.

33 Many sociologists have disregarded the doubts that some scientists have concerning fluoridation.

34 Sutton's findings have been given insufficient attention by scientists outside of North America.

35 There are valid arguments on both sides of the fluoridation debate.

Questions 36–40

Complete each sentence with the correct ending, **A–G**, *below.*

LIST OF ENDINGS

A the results of scientific research are not always understood at first.

B it is established through scientific methods and objectively applied by the scientific community.

C people should be able to choose whether they want fluoride.

D there is insufficient evidence to support a cautious approach.

E the serious damage fluoride causes far outweighs any positive effects.

F children are not the only ones who benefit from fluoridation.

G scientific knowledge is affected by the beliefs of everyone concerned.

36 The traditional view of science is that ____

37 A sociological view of science argues that ____

38 Collins is of the opinion that ____

39 The writer suggests that a supporter of fluoridation may conclude that ____

40 The writer suggests that an opponent of fluoridation may conclude that ____

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IELTS Academic Reading — Test 10

Volume 7 · 3 Passages · 40 Questions

Reading Passage 1

Dolls Through the Ages

A What is today a simple children's toy has a surprisingly rich history. Dolls have been a part of humankind for thousands of years. Often depicting religious figures, or used as playthings, early dolls were probably made from primitive materials such as clay, fur, or wood.

B Dolls constructed of flat pieces of wood, painted with various designs, and with 'hair' made of clay, have often been found in Egyptian graves dating back to 2000 BC. Egyptian tombs of wealthy families have included pottery dolls. Dolls being placed in these graves leads some to believe that they were cherished possessions.

C Girls from ancient Greece and Rome offered their wooden dolls to goddesses after they were too 'grown-up' to play with dolls. Most ancient dolls that were found in tombs were very simple creations, often made from such materials as clay, rags, wood, or bone. Some of the more unique dolls were made with ivory or wax. The main goal was to make the doll as lifelike as possible. That ideal led to the creation of dolls with movable limbs and removable garments, dating back to 600 BC.

D Following the era of the ancient dolls, Europe became a major hub for doll production. These dolls were primarily made of wood. Fewer than 30 examples of primitive wooden stump dolls from England survive today. The Grodnertal area of Germany produced many peg wooden dolls, a type of doll that has very simple peg joints and resembles a clothespin (a device for hanging washing on a clothesline). An alternative to wood was developed in the 1800s.

E 'Composition' is a collective term for mixtures of pulped wood or paper that were used to make doll heads and bodies. These mixtures were moulded under pressure, creating a durable doll that could be mass-produced. Manufacturers closely guarded the recipes for their mixtures, sometimes using strange ingredients like ash or eggshells. Papier-mâché, a type of composition, was one of the most popular mixtures.

F In addition to wooden dolls, wax dolls grew in popularity in the 17th and 18th centuries. Munich in Germany was a major manufacturing centre for wax dolls. Wax dollmakers would model a doll's head in wax or clay, and then cover it with plaster to create a mould. Then they would pour melted wax into the cast. The wax for the head would be very thin,

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no more than 3 mm. Some of the most distinctive wax dolls were created in England between 1850 and 1930. One of the first dolls that portrayed a baby was made in England from wax at the beginning of the 19th century.

G Around the same time, porcelain became popular. It is made by firing special clays in a kiln at more than 2,372 degrees Fahrenheit (1300°C), and only a few clays can withstand firing at such high temperatures. Porcelain is used generically to refer to both china and bisque dolls; china is glazed, whereas bisque is unglazed. Germany, France, and Denmark started creating china heads for dolls in the 1840s. These china heads were replaced in the 1860s by ones made of bisque. Bisque, which is porcelain fired twice with colour added to it after the first firing, looked more like skin than china did.

H In France, the bébé was popular in the 1880s, and it has become a highly sought-after doll today. The bébé, first made in the 1850s, was different from its predecessors because it depicted a younger girl. Until then, most French dolls were representations of adults. Although the French dolls were unrivalled in their artistry, German bisque dolls became quite popular because they were not as expensive. Kammer & Reinhardt introduced a bisque character doll in the 1900s, starting a trend of creating realistic dolls.

I For many centuries, rag dolls were made by mothers for their children. The term 'rag doll' refers generically to dolls made of any fabric. 'Cloth doll' refers to a subset of rag dolls made of linen or cotton. Commercially produced rag dolls were first introduced in the 1850s by English and American manufacturers. Although not as sophisticated as dolls made from other materials, rag dolls were well loved, often as a child's first toy.

J Dollmaking did not become an industry in the United States until after the Civil War in the 1860s. Doll production was concentrated in the New England region of the United States, with dolls made from a variety of materials such as leather, rubber, papier-mâché, and cloth. Celluloid was developed in the state of New Jersey in the late 1860s and was used to manufacture dolls until the mid-1950s. German, French, American, and Japanese factories churned out cheaply produced celluloid dolls in mass quantities. However, celluloid fell out of favour because of its extreme flammability and propensity to fade in bright light.

K After World War I, dollmakers experimented with plastics. Hard plastic dolls were manufactured in the 1940s. They resembled composition dolls, but they were much more durable. Other materials used in doll manufacturing included rubber, foam rubber, and vinyl in the 1950s and 1960s. Vinyl changed dollmaking, allowing dollmakers to root hair into the head, rather than using wigs or painting the hair. Although most dolls are now mass-manufactured using these modern materials, many modern dollmakers are still using the traditional materials of the past to make collectible dolls.

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Questions 1–6

Complete the notes below. Write **ONE WORD ONLY** *from the passage for each answer.*

Dolls

EARLIEST KNOWN DOLLS

- represented religious figures
- used as toys

EGYPT, 2000 BC

- bodies were made of wood
- (1) ——— was used for the hair

ANCIENT GREECE AND ROME

- dolls were given to (2) ——— by older girls

600 BC

- realistic dolls had separate clothes and (3) ——— that could be put in different positions

17TH AND 18TH CENTURIES

- dolls made of (4) ——— became more common
- moulds made of (5) ———

1800S

- new manufacturing process developed
- new group of mixtures known as (6) ———
- recipes for these mixtures kept secret

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Questions 7–13

Do the following statements agree with the information given in the passage?

TRUE *if the statement agrees with the information* • **FALSE** *if the statement contradicts the information* • **NOT GIVEN** *if there is no information on this*

7 Bisque dolls appear less realistic than dolls made of china.

8 French dolls tended to cost more than German bisque dolls.

9 The first rag dolls were made in the 1850s.

10 Only dolls made of cotton or linen are classified as cloth dolls.

11 Dolls made of celluloid tended to lose their colour.

12 Composition dolls lasted longer than the plastic dolls that were made in the 1940s.

13 Doll collectors prefer a doll to be dressed in its original clothing.

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Reading Passage 2

Biophilic Design

A Biophilic design, a movement related to green architecture, has gained much momentum within the building community in recent years. The premise of biophilic design seeks both to avoid or minimize harmful impacts on the natural environment, and equally importantly, to provide and restore beneficial contacts between people and nature in the built environment.

B People have a psychologically developed need to commune with nature but this has frequently been neglected. All too often, architects have put creative originality before the needs of the people who have to live and work in their futuristic constructions, with scant regard for emotional, as well as practical, considerations. Biophilic design does not advocate tree houses or cave-dwelling, but it does provide the nature-based features that maximize human functioning and health. Though not technically a biophilic design, Fallingwater – the stunning house in rural Pennsylvania designed in the 1930s by Frank Lloyd Wright – arguably speaks to the human soul much more than the box-like blocks of flats designed by his contemporary Le Corbusier.

C In the modern world, much of our built environment is obstructing the age-old connection between humans and nature. The rates of technological progress are far exceeding rates of psychological evolution, leaving us ill-equipped to cope with our lifestyle, which in turn leads to increased stress. Because biophilia attempts to integrate basic and current needs, it can alleviate the difficulties caused by the brain's constant attempts to function in a modern environment it has not yet evolved to handle. For example, one crucial element of the natural landscape to human health is sunlight. We are evolutionarily programmed to respond positively to sunny areas over dark or overcast ones, and sense that they will foster restoration, improve emotional well-being and promote health.

D Because of its tremendous impact, biophilic design plays a vital role in healthcare and healthcare delivery. The current healthcare system contains many flaws, especially in its physical clinics, and offices are high-stress environments for patients, visitors and healthcare professionals alike. One well-known study looked at the impact of nature on patients after surgery. One group had a view of tree; the others had a window looking onto a brick wall. The first group had shorter hospital stays, received fewer negative comments from the nurses, required fewer analgesics, and had slightly fewer post-operative complications.

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E As a consequence of this and other studies, nature and nature-based design have been integrated into the physical design of many hospitals. For example, Dartmouth Hitchcock Medical Center (DHMC) in New Hampshire boasts an atrium design, illuminating the entire facility. Natural elements also permeate the building, including wood, stone, and numerous live plants. While DHMC was built to incorporate these qualities, other hospitals have had biophilic features added to existing structures. The application of biophilia's concepts to interior design in hospitals has increased substantially as administrators have witnessed patients' positive responses to nature.

F Many existing buildings contain biophilic elements, but only a few have been built with the specific idea of biophilic design in mind. One such building is the Adam Joseph Lewis Center for Environmental Studies at Oberlin College, in Ohio. The director of Oberlin's Environmental Studies Program, David Orr, explained the building's goals were 'to create not just a place for classes but rather a building that would help to redefine the relationship between humankind and the environment – one that would expand our sense of ecological possibilities'. The Lewis Center is sustainable in a broader sense than in the typical application of the word. It harnesses solar power, utilizes both active and passive air systems, and monitors the weather to adapt to conditions. The Center's 'Living Machine' treats wastewater by combining traditional wastewater technology with wetland ecosystems' purification processes, producing water that can be used in the toilets and for irrigation. In their design, Orr and his team of architects engineered a healthy and comfortable space for students, while ensuring the surrounding environment was undamaged.

G Another example is the University of Guelph-Humber building in Ontario, Canada. It contains a centrally located bio-wall, vertically spanning the building. The wall is covered in dense foliage which not only connects people to nature on the inside of the building, but also functions as a new filtration system prototype. The wall purifies the air and has the potential to fulfill the building's fresh air intake requirements. This is another example of how biophilic design principles can be applied in a variety of contexts. These examples prove that the built environment need does not interfere with biological human needs to commune with nature, nor with existing ecological systems.

H Ancient architects built for their cultures, which were almost always more in touch with the earth than western society of the present. They mimicked nature's forms, producing magnificent structures with which we are still awed – though biophilic design is a novel concept, they certainly employed some of its recommendations. Today, we can add another layer to this tradition and ensure maximal benefit for our planet and ourselves.

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Questions 14–19

Reading Passage 2 has eight paragraphs, **A–H**. *Which paragraph contains the following information? You may use any letter more than once.*

Q	Statement	Answer
14	a description of how rapid change has a negative effect on people	
15	a reference to an architect whose designs were uncharacteristic of biophilia	
16	a definition of the two main aims of biophilia	
17	a positive claim about early forms of architecture	
18	a reference to the fact that many architects are too focused on innovation	
19	a description of features which conserve energy in a biophilic design	

Questions 20–24

Complete the sentences below. Write **ONE WORD ONLY** *from the passage for each answer.*

In the modern world, (20) _____ advancements are happening so quickly that humans are unable to adjust to them.

Biophilia can help us deal with the (21) _____ caused by the difficulties of our current lifestyles.

People instinctively feel more comfortable in places which have plenty of (22) _____.

An experiment designed to test the effect of nature on patient recovery times found that those who could see a (23) _____ made a faster recovery.

There is a biophilic design feature in the University of Guelph-Humber building which improves the quality of the (24) _____.

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Questions 25–26

Choose **TWO** *letters, A–E. Which TWO of the following statements does the writer make about the Adam Joseph Lewis Center at Oberlin College?*

A It was the first educational establishment designed with biophilic principles.
B It is beneficial for both students and the environment.
C It has established new concepts for classroom design.
D It aims to inspire new ways of thinking about the environment.
E It provides solar power to other buildings in the surrounding area.

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Reading Passage 3

Biology of Bitterness

To many people, grapefruit is palatable only when doused in sugar. Bitter blockers like adenosine monophosphate could change that.

A There is a reason why grapefruit juice is served in little glasses: most people don't want to drink more than a few ounces at a time. Naringin, a natural chemical compound found in grapefruit, tastes bitter. Some people like that bitterness in small doses and believe it enhances the general flavor, but others would rather avoid it altogether. So juice packagers often select grapefruit with low naringin though the compound has antioxidant properties that some nutritionists contend may help prevent cancer and arteriosclerosis.

B It is possible, however, to get the goodness of grapefruit juice without the bitter taste. I found that out by participating in a test conducted at the Linguagen Corporation, a biotechnology company in Cranbury, New Jersey. Sets of two miniature white paper cups, labeled 304 and 305, were placed before five people seated around a conference table. Each of us drank from one cup and then the other, cleansing our palates between tastes with water and a soda cracker. Even the smallest sip of 304 had grapefruit's unmistakable bitter bite. But 305 was smoother; there was the sour taste of citrus but none of the bitterness of naringin. This juice had been treated with adenosine monophosphate, or AMP, a compound that blocks the bitterness in foods without making them less nutritious.

C Taste research is a booming business these days, with scientists delving into all five basics - sweet, bitter, sour, salty, and umami, the savory taste of protein. Bitterness is of special interest to industry because of its untapped potential in food. There are thousands of bitter-tasting compounds in nature. They defend plants by warning animals away and protect animals by letting them know when a plant may be poisonous. But the system isn't foolproof. Grapefruit and cruciferous vegetables like Brussels sprouts and kale are nutritious despite - and sometimes because of - their bitter-tasting components. Over time, many people have learned to love them, at least in small doses. "Humans are the only species that enjoys bitter taste," says Charles Zuker, a neuroscientist at the University of California School of Medicine at San Diego. "Every other species is averse to bitter because it means bad news. But we have learned to enjoy it. We drink coffee, which is bitter, and quinine [in tonic water] too. We enjoy having that spice in our lives." Because bitterness can be pleasing in small quantities but repellent when intense, bitter blockers like AMP could make a whole range of foods, drinks, and medicines more palatable - and therefore more profitable.

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D People have varying capacities for tasting bitterness, and the differences appear to be genetic. About 75 percent of people are sensitive to the taste of the bitter compounds phenylthiocarbamide and 6-n-propylthiouracil, and 25 percent are insensitive. Those who are sensitive to phenylthiocarbamide seem to be less likely than others to eat cruciferous vegetables, according to Stephen Wooding, a geneticist at the University of Utah. Some people, known as supertasters, are especially sensitive to 6-n-propylthiouracil because they have an unusually high number of taste buds. Supertasters tend to shun all kinds of bitter-tasting things, including vegetables, coffee, and dark chocolate. Perhaps as a result, they tend to be thin. They're also less fond of alcoholic drinks, which are often slightly bitter. Dewar's scotch, for instance, tastes somewhat sweet to most people. "But a supertaster tastes no sweetness at all, only bitterness," says Valerie Duffy, an associate professor of dietetics at the University of Connecticut at Storrs.

E In one recent study, Duffy found that supertasters consume alcoholic beverages, on average, only two to three times a week, compared with five or six times for the average nontasters. Each taste bud, which looks like an onion, consists of 50 to 100 elongated cells running from the top of the bud to the bottom. At the top is a little clump of receptors that capture the taste molecules, known as tastants, in food and drink. The receptors function much like those for sight and smell. Once a bitter signal has been received, it is relayed via proteins known as G proteins. The G protein involved in the perception of bitterness, sweetness, and umami was identified in the early 1990s by Linguagen's founder, Robert Margolskee, at Mount Sinai School of Medicine in New York City. Known as gustducin, the protein triggers a cascade of chemical reactions that lead to changes in ion concentrations within the cell. Ultimately, this delivers a signal to the brain that registers as bitter. "The signaling system is like a bucket brigade," Margolskee says. "It goes from the G protein to other proteins."

F In 2000 Zuker and others found some 30 different kinds of genes that code for bitter-taste receptors. "We knew the number would have to be large because there is such a large universe of bitter tastants," Zuker says. Yet no matter which tastant enters the mouth or which receptor it attaches to, bitter always tastes the same to us. The only variation derives from its intensity and the ways in which it can be flavored by the sense of smell. "Taste cells are like a light switch," Zuker says. "They are either on or off."

G Once they figured out the taste mechanism, scientists began to think of ways to interfere with it. They tried AMP, an organic compound found in breast milk and other substances, which is created as cells break down food. AMP has no bitterness of its own, but when put in foods, Margolskee and his colleagues discovered, it attaches to bitter-taste receptors. As effective as it is, AMP may not be able to dampen every type of bitter taste, because it probably doesn't attach to all 30 bitter-taste receptors. So Linguagen has scaled up the hunt for other bitter blockers with a technology called high-throughput screening.

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Researchers start by coaxing cells in culture to activate bitter-taste receptors. Then candidate substances, culled from chemical compound libraries, are dropped onto the receptors, and scientists look for evidence of a reaction.

H In time, some taste researchers believe, compounds like AMP will help make processed foods less unhealthy. Consider, for example, that a single cup of Campbell's chicken noodle soup contains 850 milligrams of sodium chloride, or table salt - more than a third of the recommended daily allowance. The salt masks the bitterness created by the high temperatures used in the canning process, which cause sugars and amino acids to react. Part of the salt could be replaced by another salt, potassium chloride, which tends to be scarce in some people's diets. Potassium chloride has a bitter aftertaste, but that could be eliminated with a dose of AMP. Bitter blockers could also be used in place of cherry or grape flavoring to take the harshness out of children's cough syrup, and they could dampen the bitterness of antihistamines, antibiotics, certain HIV drugs, and other medications.

I A number of foodmakers have already begun to experiment with AMP in their products, and other bitter blockers are being developed by rival firms such as Senomyx in La Jolla, California. In a few years, perhaps, after food companies have taken the bitterness from canned soup and TV dinners, they can set their sights on something more useful: a bitter blocker in a bottle that any of us can sprinkle on our brussels sprouts or stir into our grapefruit juice.

Questions 27–34

Reading Passage 3 has nine paragraphs, **A–I**. *Which paragraph contains the following information?*

Q	Statement	Answer
27	Experiment on bitterness conducted	
28	Look into the future application	
29	Bitterness means different information for humans and animals	
30	Spread process of bitterness inside of body	
31	How AMP blocks bitterness	
32	Some bitterness blocker may help lower unhealthy impact	
33	Bitterness introduced from a fruit	
34	Genetic feature determines sensitivity	

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Questions 35–38

Complete the following summary. Write **NO MORE THAN TWO WORDS** *for each answer.*

The reason why grapefruit tastes bitter is because a substance called (35) ———————— is contained in it. However, bitterness plays a significant role for plants. It gives a signal that a certain plant is (36) ————————. For human beings, different people carry various genetic abilities of tasting bitterness. According to a scientist at the University of Utah, (37) ———————— have exceptionally plenty of (38) ————————, which allows them to perceive bitter compounds.

Questions 39–40

Choose the correct answer.

(39) What is the main feature of AMP according to this passage?

A offset bitter flavour in food
B only exist in cup 304
C tastes like citrus
D chemical reaction when it meets a biscuit

(40) What is the main function of G protein?

A collecting taste molecules
B identifying different flavour elements
C resolving large molecules
D transmitting bitter signals to the brain

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PART TWO

Answer Keys

Complete answer keys for Tests 1–10 of Volume 7. Check your answers only after finishing each test.

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Answer Key & Explanations

Reading Test 1 (Volume 7)

Passage 1 — Carnivorous Plant

1. raindrops

QUESTION KEYWORD	did not respond to ____
PASSAGE EVIDENCE	"sundews... ignored **raindrops**."

2. stomach

QUESTION KEYWORD	acts like a ____
PASSAGE EVIDENCE	"formed itself into a temporary '**stomach**'."

3. pores

QUESTION KEYWORD	____ in cell walls open
PASSAGE EVIDENCE	"opens up **pores** in cell membranes."

4. vacuum

QUESTION KEYWORD	____ builds up inside
PASSAGE EVIDENCE	"creating a new **vacuum**."

5. frogs

QUESTION KEYWORD	capture and eat ____
PASSAGE EVIDENCE	"can consume whole **frogs**."

6. TRUE

QUESTION KEYWORD	pitcher plants grow bigger
PASSAGE EVIDENCE	"when scientists feed pitcher plants extra bugs, the plants **get bigger**."

7. TRUE

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QUESTION KEYWORD	enzymes from animal nutrients
PASSAGE EVIDENCE	"take nitrogen and phosphorus from the flesh in order to build **light-harvesting enzymes**."

8. FALSE

QUESTION KEYWORD	more efficient photosynthesis
PASSAGE EVIDENCE	"not very good at photosynthesis... do not have flat solar panels."

9. NOT GIVEN

QUESTION KEYWORD	flytraps vs other carnivorous plants
PASSAGE EVIDENCE	the passage discusses carnivorous plants generally in bogs, not a specific comparison to flytraps

10. FALSE

QUESTION KEYWORD	lack of sunlight in bogs
PASSAGE EVIDENCE	"bogs are often **flooded with sunshine**."

11. NOT GIVEN

QUESTION KEYWORD	scientists campaigned to reduce nitrogen
PASSAGE EVIDENCE	no campaign by scientists is mentioned

12. TRUE

QUESTION KEYWORD	exotic plants sold illegally
PASSAGE EVIDENCE	"The **black market trade** in exotic carnivorous plants is strong."

13. FALSE

QUESTION KEYWORD	preventing wildfires beneficial
PASSAGE EVIDENCE	"suppression of wildfires... allows other plants to grow quickly and **outcompete** the Venus flytraps."

Passage 2 — Drawing Lessons from History

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14. D

QUESTION KEYWORD	appeal of monumental ruins
PASSAGE EVIDENCE	"drawn to their **magnificent and haunting beauty**, as well as the enigmas they present."

15. C

QUESTION KEYWORD	recent findings
PASSAGE EVIDENCE	discoveries made by "**archaeologists, climatologists, historians, palaeontologists, and palynologists**."

16. A

QUESTION KEYWORD	how previous societies fell
PASSAGE EVIDENCE	"they experienced **rapid declines** after reaching their peak" across many societies.

17. NOT GIVEN

QUESTION KEYWORD	relics influenced by Shelley's sonnet
PASSAGE EVIDENCE	ruins are described as "reminiscent of" the sonnet — the reverse of the claim, and no causal influence is stated

18. YES

QUESTION KEYWORD	widespread belief re environment
PASSAGE EVIDENCE	"many individuals believe that environmental problems **eclipse other global threats**."

19. YES

QUESTION KEYWORD	toxic chemicals a modern issue
PASSAGE EVIDENCE	"plus four new ones: ...the **accumulation of toxic chemicals**."

20. NO

QUESTION KEYWORD	scientific consensus on severity
PASSAGE EVIDENCE	"vigorously **debated** among the scientific community."

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21. YES

QUESTION KEYWORD	reflect cautiously on past/present
PASSAGE EVIDENCE	"naive to assume that studying the past will yield **straightforward answers**."

22. C

QUESTION KEYWORD	evidence of achievements
PASSAGE EVIDENCE	"the grandeur of these ruins **attests to** the previous wealth and power."

23. A

QUESTION KEYWORD	individual/society analogies
PASSAGE EVIDENCE	"this metaphor proves **inaccurate for many past societies**."

24. F

QUESTION KEYWORD	environmental challenges today
PASSAGE EVIDENCE	"the same eight factors... **plus four new ones**."

25. D

QUESTION KEYWORD	powerful technology today
PASSAGE EVIDENCE	technology's "**beneficial effects**"... and "its **unintended destructive effects**."

26. E

QUESTION KEYWORD	reflections on ancient events
PASSAGE EVIDENCE	"naive to assume... **straightforward answers directly applicable** to today's world."

Passage 3 — Neanderthal Technology

27. v

QUESTION KEYWORD	heading for section A

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PASSAGE EVIDENCE	contrasts cave-dwelling west Europeans with **open-air** east European sites.

28. iii

QUESTION KEYWORD	heading for section B
PASSAGE EVIDENCE	Périgord was "an **exceptionally benign habitat** for humans."

29. i

QUESTION KEYWORD	heading for section C
PASSAGE EVIDENCE	"such **open-air sites** as have been discovered."

30. iv

QUESTION KEYWORD	heading for section D
PASSAGE EVIDENCE	detailed findings inside the **Grotte du Lazaret and Baume-Bonne caves**.

31. vi

QUESTION KEYWORD	heading for section E
PASSAGE EVIDENCE	"use of **fire** was... an already ancient accomplishment."

32. E — Krapina

QUESTION KEYWORD	burnt piece of wood
PASSAGE EVIDENCE	"charred piece of birch from **Krapina**."

33. D — The cave of Baume-Bonne

QUESTION KEYWORD	prevent pools of water
PASSAGE EVIDENCE	"cobbles... as though to **take care of a puddle area**."

34. C — The Grotte du Lazaret

QUESTION KEYWORD	remains of sea creatures

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PASSAGE EVIDENCE	"edged with small **marine mollusc shells**."

35. B — Moldova

QUESTION KEYWORD	circular arrangement of bones
PASSAGE EVIDENCE	"a **ring**... of mainly mammoth bones."

36. C — The Grotte du Lazaret

QUESTION KEYWORD	use of animal fur for warmth
PASSAGE EVIDENCE	"thought to indicate the use of **animal pelts as bed coverings**."

37. sub-arctic

QUESTION KEYWORD	conditions were ____
PASSAGE EVIDENCE	"survival in the **sub-arctic** conditions... is inconceivable without control of fire."

38. short

QUESTION KEYWORD	fires lasted a ____ time
PASSAGE EVIDENCE	"this fire was only of a **short duration**."

39. animals

QUESTION KEYWORD	protect from dangerous ____
PASSAGE EVIDENCE	fire gave "defence against predatory **animals**."

40. C

QUESTION KEYWORD	writer's purpose
PASSAGE EVIDENCE	the passage surveys evidence of **Neanderthal settlements** across cave and open-air sites throughout.

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Answer Key & Explanations

Reading Test 2 (Volume 7)

Passage 1 — Wood: a Valuable Resource in New Zealand's Economy

1. FALSE

QUESTION KEYWORD	wooden houses more dangerous
PASSAGE EVIDENCE	wooden houses were "**less likely to collapse** suddenly during earthquakes."

2. TRUE

QUESTION KEYWORD	exported wood for boat-building
PASSAGE EVIDENCE	kauri wood "sold to Australia and the UK" for "**constructing sailing vessels**."

3. NOT GIVEN

QUESTION KEYWORD	plantation wood better for construction
PASSAGE EVIDENCE	plantation wood provides "most of the country's sawn timber needs," but no quality comparison is made

4. FALSE

QUESTION KEYWORD	pine has narrow range of uses
PASSAGE EVIDENCE	"This amazing **versatility**."

5. NOT GIVEN

QUESTION KEYWORD	housing demand predicted to fall
PASSAGE EVIDENCE	only current annual house starts are given; no future prediction is made

6. TRUE

QUESTION KEYWORD	future expansion depends on exports
PASSAGE EVIDENCE	"the development of the **export market** is the key to the industry's growth."

7. shipping costs

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QUESTION KEYWORD	negative impact besides exchange rates
PASSAGE EVIDENCE	"affected by the exchange fluctuations and **shipping costs**."

8. export sector

QUESTION KEYWORD	ranks third in ____
PASSAGE EVIDENCE	"New Zealand's third largest **export sector**."

9. 60,000

QUESTION KEYWORD	workforce size by 2025
PASSAGE EVIDENCE	"generating $20 billion a year and employing **60,000** people."

10. softwood

QUESTION KEYWORD	available from renewable forests
PASSAGE EVIDENCE	"source large quantities of **softwood** from renewable forests."

11. sustainability

QUESTION KEYWORD	customers increasingly concerned about
PASSAGE EVIDENCE	"becoming more worried about **sustainability**."

12. Scandinavian countries

QUESTION KEYWORD	main competitors outside southern hemisphere
PASSAGE EVIDENCE	"in the northern hemisphere, **Scandinavian countries** have all expanded their forests."

13. wood substitutes

QUESTION KEYWORD	now competing with
PASSAGE EVIDENCE	"competition from goods such as **wood substitutes**."

Passage 2 — Intelligent Behaviour in Birds

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14. iv

QUESTION KEYWORD	heading for paragraph A
PASSAGE EVIDENCE	"many scholars maintained that humans were the only intelligent organism... **forcing us to reconsider**."

15. viii

QUESTION KEYWORD	heading for paragraph B
PASSAGE EVIDENCE	examples: choughs' rocks, kites' bait and fire stick.

16. i

QUESTION KEYWORD	heading for paragraph C
PASSAGE EVIDENCE	"only the **smartest** in each generation will live and reproduce."

17. ix

QUESTION KEYWORD	heading for paragraph D
PASSAGE EVIDENCE	New Caledonian crows' hooked tool, rakes, and probes.

18. vii

QUESTION KEYWORD	heading for paragraph E
PASSAGE EVIDENCE	"compared... the size of the birds' **forebrain**."

19. iii

QUESTION KEYWORD	heading for paragraph F
PASSAGE EVIDENCE	"the dynamic challenges of **communal living**... a distinctly 'political' brain."

20. vi

QUESTION KEYWORD	heading for paragraph G
PASSAGE EVIDENCE	"the best example of **deception** among birds."

21. C — New Caledonian crows

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QUESTION KEYWORD	keeping favourite tools
PASSAGE EVIDENCE	"carry their **favourite tool**... store their tools for later re-use."

22. B — Black kites

QUESTION KEYWORD	drawing prey by frightening
PASSAGE EVIDENCE	drops a burning stick, then "feasts on the small animals that **flee**."

23. A — White-winged choughs

QUESTION KEYWORD	remove outer covering from food
PASSAGE EVIDENCE	"use rocks as hammers to **crack open** the recalcitrant shells."

24. B — Black kites

QUESTION KEYWORD	using food to attract prey
PASSAGE EVIDENCE	"dropping **bait** into lakes to bring fish to the surface."

25. C — New Caledonian crows

QUESTION KEYWORD	unfamiliar materials for tools
PASSAGE EVIDENCE	"moulding a piece of **wire**, something they had never seen before."

26. A — White-winged choughs

QUESTION KEYWORD	activities for benefit of observers
PASSAGE EVIDENCE	"more likely to **preen the chicks** if another bird can see it do so."

Passage 3 — Jean Piaget 1896–1980

27. C

QUESTION KEYWORD	anthropology example illustrates
PASSAGE EVIDENCE	"a shift comparable to the way modern anthropology **displaced** stories..." — a turning point.

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28. A

QUESTION KEYWORD	point illustrated by wind dialogue
PASSAGE EVIDENCE	"Classifying them as 'true' or 'false' **misses the point**."

29. D

QUESTION KEYWORD	Piaget believed in importance of
PASSAGE EVIDENCE	"children have real understanding only of that which **they invent themselves**."

30. C

QUESTION KEYWORD	7th paragraph suggests
PASSAGE EVIDENCE	"tolerance for – indeed a **fascination with** – children's primitive laws of physics."

31. C

QUESTION KEYWORD	writer's main purpose
PASSAGE EVIDENCE	the passage traces Piaget's **role and influence across psychology** throughout.

32. G — logical

QUESTION KEYWORD	mental processes far more ____
PASSAGE EVIDENCE	"thought processes that had their own kind of order and their own special **logic**."

33. B — theories

QUESTION KEYWORD	tries out his own ____
PASSAGE EVIDENCE	"constantly creating and testing their own **hypotheses**."

34. F — psychology

QUESTION KEYWORD	impact on the area of ____
PASSAGE EVIDENCE	"his influence on **psychology** may be longer lasting."

35. C — brain

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QUESTION KEYWORD	knowledge associated exclusively with
PASSAGE EVIDENCE	"viewing knowledge as an intrinsic property of the **brain**."

36. E — teachers

QUESTION KEYWORD	professional development of ____
PASSAGE EVIDENCE	"every **teacher** in training still memorises Piaget's successive stages."

37. NOT GIVEN

QUESTION KEYWORD	innovative research techniques in Paris
PASSAGE EVIDENCE	only that he "noticed children of the same age made similar errors" — no characterisation of the technique

38. NO

QUESTION KEYWORD	clear guidelines for adults
PASSAGE EVIDENCE	"Piaget… **never laid down rules** about how to intervene."

39. YES

QUESTION KEYWORD	contribution to epistemology
PASSAGE EVIDENCE	epistemology "was considered a branch of philosophy until Piaget came along and **made it a science**."

40. YES

QUESTION KEYWORD	much still to learn
PASSAGE EVIDENCE	"the new discoveries… only **increase the mystery**."

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Answer Key & Explanations

Reading Test 3 (Volume 7)

Passage 1 — The Early History of Olive Oil

1. TRUE

QUESTION KEYWORD	period without rain advantageous
PASSAGE EVIDENCE	"produce better when subjected to these **stressful conditions**," e.g. hot, dry weather.

2. FALSE

QUESTION KEYWORD	most fertile fields chosen
PASSAGE EVIDENCE	"grown on land where **little else will survive**."

3. TRUE

QUESTION KEYWORD	divine origins
PASSAGE EVIDENCE	"a priceless gift from the goddess **Athena**."

4. NOT GIVEN

QUESTION KEYWORD	more costly than gold
PASSAGE EVIDENCE	"liquid gold" is a figurative nickname, not a literal price comparison

5. TRUE

QUESTION KEYWORD	Plato mentions food use
PASSAGE EVIDENCE	"cooking dates at least as far back... as described by the Greek philosopher **Plato**."

6. NOT GIVEN

QUESTION KEYWORD	North African farmers resisted
PASSAGE EVIDENCE	the passage doesn't describe any resistance to the trees' introduction

7. beating

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QUESTION KEYWORD	picking or ____ the trees
PASSAGE EVIDENCE	"by hand or by **beating** the fruit from the trees with long sticks."

8. bags

QUESTION KEYWORD	flesh placed in ____
PASSAGE EVIDENCE	"put in woven **bags** and pressed."

9. salt

QUESTION KEYWORD	____ used to aid the process
PASSAGE EVIDENCE	"a bit of **salt** was added to speed up the process."

10. pollution

QUESTION KEYWORD	thrown away, led to ____
PASSAGE EVIDENCE	"often simply discarded, causing serious **pollution**."

11. floors

QUESTION KEYWORD	used on ____ of buildings
PASSAGE EVIDENCE	"applied to the **floors** of grain storage buildings."

12. leather

QUESTION KEYWORD	making ____ into goods
PASSAGE EVIDENCE	"applied to **leather** to soften it... shape into articles of clothing."

13. pesticide

QUESTION KEYWORD	used as a ____
PASSAGE EVIDENCE	"as a fertiliser or as a **pesticide**."

Passage 2 — Playing with Science

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14. E

QUESTION KEYWORD	test showed evidence fake
PASSAGE EVIDENCE	Michel and Asaro "showed that the copper and zinc... were of a **higher quality**" than 1500s materials.

15. A

QUESTION KEYWORD	reasons unwilling to admit play
PASSAGE EVIDENCE	"a little self-conscious about it and therefore **try to hide it**."

16. F

QUESTION KEYWORD	result of trick going wrong
PASSAGE EVIDENCE	"things misfired after Bolton **went public** and announced that the relic was authentic."

17. B

QUESTION KEYWORD	skill in play/science similarity
PASSAGE EVIDENCE	"Guessing the solution of a scientific problem has many similarities to completing a **jigsaw puzzle**."

18. G

QUESTION KEYWORD	stopped colleague misusing results
PASSAGE EVIDENCE	the fake NMR spectrum "stopped him temporarily from taking our data and interpreting it."

19. D

QUESTION KEYWORD	playful scientific language
PASSAGE EVIDENCE	"the funny names... **tristane, windowpane, penguinone or megaphone**."

20. C

QUESTION KEYWORD	everyday object, scientific use
PASSAGE EVIDENCE	"a **perfume atomiser** bought at a local pharmacy."

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21. & 22. A & C

QUESTION KEYWORD	possible explanations for play
PASSAGE EVIDENCE	[A] "science presents them with too much **seriousness**"; [C] scientists "tend to be very **young**."

23. data

QUESTION KEYWORD	colleague stealing their ____
PASSAGE EVIDENCE	"had the habit of going through their **data** and rushing to Gray."

24. English navigator

QUESTION KEYWORD	famous ____ left plate
PASSAGE EVIDENCE	"supposedly left by the **English navigator** Sir Francis Drake."

25. copper and zinc

QUESTION KEYWORD	material the plate was made of
PASSAGE EVIDENCE	"the **copper and zinc** used in the plate were of a higher quality."

26. Bancroft Library

QUESTION KEYWORD	director of the ____
PASSAGE EVIDENCE	"Herbert E Bolton, who directed the **Bancroft Library**."

Passage 3 — Does Class Size Matter?

27. D

QUESTION KEYWORD	detailed methodology criticism
PASSAGE EVIDENCE	Section D: Hanushek's critique of STAR's design and conclusions.

28. F

QUESTION KEYWORD	comparison of data

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PASSAGE EVIDENCE	Section F: SAGE results "accord with those from STAR."

29. **A**

QUESTION KEYWORD	level of public interest
PASSAGE EVIDENCE	"a **policy juggernaut** with over twenty states."

30. **C**

QUESTION KEYWORD	action to protect validity
PASSAGE EVIDENCE	"teachers were **randomly assigned** to small and regular-size classrooms."

31. **D**

QUESTION KEYWORD	reasons class composition changed
PASSAGE EVIDENCE	"school personnel **caved in to parent demands**."

32. **A**

QUESTION KEYWORD	composition left to chance
PASSAGE EVIDENCE	STAR: students "**randomly assigned**" to class types.

33. **A**

QUESTION KEYWORD	long-term improvement claimed
PASSAGE EVIDENCE	Finn and Achilles: advantage "**persisted** even after students returned to larger classes."

34. **B**

QUESTION KEYWORD	similar results all groups
PASSAGE EVIDENCE	California: "the effect **did not vary** for students of different backgrounds."

35. **C**

QUESTION KEYWORD	preliminary to bigger study

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PASSAGE EVIDENCE	SAGE: "a five-year pilot study to do... groundwork for a **major project**."

36. A

QUESTION KEYWORD	several class types
PASSAGE EVIDENCE	STAR: "**one of three kinds** of classes."

37. C

QUESTION KEYWORD	special group of schools
PASSAGE EVIDENCE	SAGE: "targeting schools at which **30% of the students were below poverty** level."

38. A

QUESTION KEYWORD	classroom assistants used
PASSAGE EVIDENCE	STAR: a class type with "both a teacher and a full-time **teacher's aide**."

39. B

QUESTION KEYWORD	aggravating existing problems
PASSAGE EVIDENCE	California: "**exacerbating the disparity** in resources available to rich and poor schools."

40. A

QUESTION KEYWORD	certain groups benefited
PASSAGE EVIDENCE	STAR: "effect was **stronger for black and Hispanic minority groups**."

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Answer Key & Explanations

Reading Test 4 (Volume 7)

Passage 1 — The Development of Plastics

1. photographic film

QUESTION KEYWORD	other celluloid product
PASSAGE EVIDENCE	"the best-known celluloid product was **photographic film**."

2. Bakelite

QUESTION KEYWORD	material from 1907, USA
PASSAGE EVIDENCE	"Baekeland called it **Bakelite**."

3. electrical switches

QUESTION KEYWORD	Bakelite's use
PASSAGE EVIDENCE	"used in the manufacture of **electrical switches**."

4. Britain

QUESTION KEYWORD	polythene's country of origin
PASSAGE EVIDENCE	"chemists in **Britain** discovered that the gas ethylene would polymerize."

5. fireproof

QUESTION KEYWORD	rigid PVC property
PASSAGE EVIDENCE	"a **fireproof** plastic suitable for drains and gutters."

6. glass

QUESTION KEYWORD	polystyrene resembles ___
PASSAGE EVIDENCE	"a hard, clear material like **glass**."

7. foam

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QUESTION KEYWORD	polyurethane manufactured as
PASSAGE EVIDENCE	"commonly produced as a **foam**."

8. FALSE

QUESTION KEYWORD	rubber's structure very different
PASSAGE EVIDENCE	"the same bonding principle... is the **basis of the creation** of a huge range of plastics."

9. NOT GIVEN

QUESTION KEYWORD	Hyatt an industrial chemist
PASSAGE EVIDENCE	his profession is never specified, only that he won the competition

10. FALSE

QUESTION KEYWORD	same reaction to heat
PASSAGE EVIDENCE	celluloid is a "**thermoplastic**," Bakelite "**cannot be softened** by heat" once set.

11. TRUE

QUESTION KEYWORD	hard to break down and reuse
PASSAGE EVIDENCE	"different types of plastic are often found in the same items and **call for different treatments**."

12. FALSE

QUESTION KEYWORD	starch makes plastic more durable
PASSAGE EVIDENCE	starch is "attacked by bacteria and causes the plastic to **fall apart**."

13. TRUE

QUESTION KEYWORD	decomposable containers need storage conditions
PASSAGE EVIDENCE	"bottles... have to be **stored in the dark**."

Passage 2 — Lean Production Innovation

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14. ii

QUESTION KEYWORD	heading for paragraph A
PASSAGE EVIDENCE	traces the history from Ford/Sloan to **Toyota's** lean production.

15. iv

QUESTION KEYWORD	heading for paragraph B
PASSAGE EVIDENCE	compares craft, mass, and lean production's **advantages and disadvantages**.

16. vii

QUESTION KEYWORD	heading for paragraph C
PASSAGE EVIDENCE	"Lean production is **so called** because..."

17. viii

QUESTION KEYWORD	heading for paragraph D
PASSAGE EVIDENCE	"'good enough'" vs "**unwilling to compromise** standards."

18. x

QUESTION KEYWORD	heading for paragraph E
PASSAGE EVIDENCE	"none of these three objectives can be **achieved separately**."

19. vi

QUESTION KEYWORD	heading for paragraph F
PASSAGE EVIDENCE	"relies on two practical mechanisms: **just-in-time**."

20. iii

QUESTION KEYWORD	heading for paragraph G
PASSAGE EVIDENCE	"quality control must **coexist** with just-in-time procedures."

21. craft (production)

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QUESTION KEYWORD	unusual cars produced by ____
PASSAGE EVIDENCE	"customised production of a few exotic sports cars" via **craft production**.

22. multi-skilled

QUESTION KEYWORD	staff who are ____
PASSAGE EVIDENCE	"teams of **multi-skilled** workers to all levels."

23. manufacturing space

QUESTION KEYWORD	uses less ____, equipment, time
PASSAGE EVIDENCE	"half the **manufacturing space**."

24. just-in-time

QUESTION KEYWORD	storage not needed with ____
PASSAGE EVIDENCE	"if **just-in-time** is fully realised... stores or warehouses unnecessary."

25. abnormalities

QUESTION KEYWORD	spotting any ____
PASSAGE EVIDENCE	"automatic checking for **abnormalities**."

26. lights

QUESTION KEYWORD	suspended above production lines
PASSAGE EVIDENCE	"**lights**, which are hung so high in the factory."

Passage 3 — How Does Watching Sport Influence the Brain?

27. F

QUESTION KEYWORD	emotional satisfaction reasons
PASSAGE EVIDENCE	mirror neurons "directly associated with the **amygdala**," the emotion-processing area.

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28. B

QUESTION KEYWORD	why beginners struggle
PASSAGE EVIDENCE	"novices rely on **conscious effort**, utilising brain regions adjacent to the premotor cortex."

29. E

QUESTION KEYWORD	factor alongside mirroring
PASSAGE EVIDENCE	"supplemented by the **physical attributes** required to translate those movements into reality."

30. C

QUESTION KEYWORD	human vs primate mirror neurons
PASSAGE EVIDENCE	"Humans possess an **even more intricate** mirror neuron system."

31. E

QUESTION KEYWORD	integrating visual/motor processes
PASSAGE EVIDENCE	"align the **visual representation**... with our own **motor representation**."

32. D

QUESTION KEYWORD	first discovery of brain activity
PASSAGE EVIDENCE	"initially identified in **1954** by French physiologists Gastaut and Bert."

33. D

QUESTION KEYWORD	point about athletes, paragraph B
PASSAGE EVIDENCE	"instinctively execute perfectly balanced movements... **requiring no conscious control**."

34. B

QUESTION KEYWORD	golfer example illustrates
PASSAGE EVIDENCE	"watching a remarkable golfer **improves our own golfing skills**."

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35. C

QUESTION KEYWORD	principal motivation for watching
PASSAGE EVIDENCE	"most spectators tune in for the **emotional roller coaster**."

36. YES

QUESTION KEYWORD	Homer's epic same period as Olympics
PASSAGE EVIDENCE	"During the epoch when the poet Homer unveiled his grand epic, the ancient Greeks **initiated** [the Olympics]."

37. NO

QUESTION KEYWORD	no link to unrelated neurons
PASSAGE EVIDENCE	mirror neurons are "**directly associated with the amygdala**," an unrelated processing area.

38. NO

QUESTION KEYWORD	flawless swings without practice
PASSAGE EVIDENCE	watching "**improves**" skill — no claim of flawless performance without practice.

39. NOT GIVEN

QUESTION KEYWORD	still images under special conditions
PASSAGE EVIDENCE	only stated that static images leave mirror neurons "utterly silent" — no exception mentioned

40. YES

QUESTION KEYWORD	mu system suppressed during activity
PASSAGE EVIDENCE	the mu signal is "**dissipating** when we undertake active pursuits."

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Answer Key & Explanations

Reading Test 5 (Volume 7)

Passage 1 — Rubber

1. FALSE

QUESTION KEYWORD	grows only in certain regions
PASSAGE EVIDENCE	"spread right across the globe, and grow in **many different habitats**."

2. NOT GIVEN

QUESTION KEYWORD	extracted in Mexico by 6th century
PASSAGE EVIDENCE	the wall painting shows an "offering" of rubber, not a description of its extraction process

3. NOT GIVEN

QUESTION KEYWORD	Castilla elastica poorer quality
PASSAGE EVIDENCE	only that Castilla was used because "the Para plant... does not grow in the Mexican region" — no quality comparison

4. TRUE

QUESTION KEYWORD	French mathematician inspired interest
PASSAGE EVIDENCE	"no real interest in rubber was shown by any European until **Charles de la Condamine**, a French mathematician."

5. NOT GIVEN

QUESTION KEYWORD	discovered by accident
PASSAGE EVIDENCE	only that "Goodyear discovered that it was possible to stabilise rubber" — no mention of accident

6. FALSE

QUESTION KEYWORD	imports fell in 19th century
PASSAGE EVIDENCE	imports rose from "211 kg" to "10,000kg" and then to "**just under six times as much again**."

7. waterproof cloth

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QUESTION KEYWORD	produced using rubber, 1818
PASSAGE EVIDENCE	"James Syme first used rubber to make **waterproof cloth**."

8. waste strips

QUESTION KEYWORD	machine recycling ____
PASSAGE EVIDENCE	"a machine... which chewed up **waste strips** for re-use."

9. rigid

QUESTION KEYWORD	soft or ____ by temperature
PASSAGE EVIDENCE	"became pliant when warm and **rigid** when cold."

10. stabilise

QUESTION KEYWORD	process to ____ the substance
PASSAGE EVIDENCE	"possible to **stabilise** rubber by mixing it with sulphur."

11. steam engines

QUESTION KEYWORD	____ industry, Industrial Revolution
PASSAGE EVIDENCE	"employed in the **steam engines** found in factories, mills, mines and railways."

12. pneumatic tyre

QUESTION KEYWORD	developed in 1888
PASSAGE EVIDENCE	"Dunlop produced the first **pneumatic tyre**."

13. mass production

QUESTION KEYWORD	____ of motor car began
PASSAGE EVIDENCE	"with the **mass production** of cars just over the horizon."

Passage 2 — The Myth of the Eight-hour Sleep

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14. vii

QUESTION KEYWORD	heading for section 1
PASSAGE EVIDENCE	Wehr's darkened-room study demonstrates a "very distinct sleeping pattern" — a **natural, scientifically observed** divided pattern.

15. v

QUESTION KEYWORD	heading for section 2
PASSAGE EVIDENCE	Ekirch's "16 years of research" through "diaries, court recorders, medical books."

16. i

QUESTION KEYWORD	heading for section 3
PASSAGE EVIDENCE	Koslofsky's historical explanation: "Associations with night... were not good," giving no "prestige" to staying up late.

17. vi

QUESTION KEYWORD	heading for section 4
PASSAGE EVIDENCE	improved street/home lighting is an environmental factor: "the length of time people used for rest **declined**."

18. iii

QUESTION KEYWORD	heading for section 5
PASSAGE EVIDENCE	"sleep maintenance insomnia"; Foster: "**30% of the medical problems**... stem... from sleep."

19. C — Craig Koslofsky

QUESTION KEYWORD	criminal danger led to segmented sleep
PASSAGE EVIDENCE	"streets of cities and towns at night were often populated by **thieves**."

20. E — Russell Foster

QUESTION KEYWORD	physicians should learn more
PASSAGE EVIDENCE	"sleep has been **ignored in medical training**... He feels this needs to change."

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21. B — Roger Ekirch

QUESTION KEYWORD	used waking period for activities
PASSAGE EVIDENCE	"could be quite active... moved around the house... **read or wrote**."

22. B — Roger Ekirch

QUESTION KEYWORD	technological changes, whole-night sleep
PASSAGE EVIDENCE	"Ekirch attributes the initial shift to improvements in **street and home lighting**."

23. D — Gregg Jacobs

QUESTION KEYWORD	belief in continuous sleep harmful
PASSAGE EVIDENCE	"this **anxiety** can itself discourage sleep and is likely to affect waking life too."

24. sunset

QUESTION KEYWORD	went to bed early, after ___
PASSAGE EVIDENCE	"a first sleep which began not long following **sunset**."

25. dreams

QUESTION KEYWORD	consider meaning of ___
PASSAGE EVIDENCE	"attempt to analyse their **dreams**."

26. thieves

QUESTION KEYWORD	feared ___ on unlit streets
PASSAGE EVIDENCE	"streets... often populated by **thieves** or worse."

Passage 3 — Living Dunes

27. iv

QUESTION KEYWORD	heading for paragraph 1
PASSAGE EVIDENCE	"serious consequences if there is an **oil installation or a railway line** in their path."

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28. viii

QUESTION KEYWORD	heading for paragraph 2
PASSAGE EVIDENCE	"over half of that is classified as '**linear**' sand dunes."

29. vi

QUESTION KEYWORD	heading for paragraph 3
PASSAGE EVIDENCE	"Barchan dunes only tend to form where you have **one-directional winds**... near coastal areas."

30. vii

QUESTION KEYWORD	heading for paragraph 4
PASSAGE EVIDENCE	"it reappears at the back of the dune, ready to **repeat the process**."

31. v

QUESTION KEYWORD	heading for paragraph 5
PASSAGE EVIDENCE	"Even when two dunes **collide**, they quickly take on their distinctive shapes again."

32. ix

QUESTION KEYWORD	heading for paragraph 6
PASSAGE EVIDENCE	"Local legends attributed the sounds to dangerous spirits... Douady is **more pragmatic**."

33. ii

QUESTION KEYWORD	heading for paragraph 7
PASSAGE EVIDENCE	"able to take samples of the singing sand back to France and **replicate the sound at ENS**."

34. C

QUESTION KEYWORD	what told about linear dunes
PASSAGE EVIDENCE	"form in a **long curving wave**" — a recognisable shape.

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35. B

QUESTION KEYWORD	Bagnold's cause of movement
PASSAGE EVIDENCE	"dunes aren't solid, but **granular**."

36. D

QUESTION KEYWORD	compares to living organism
PASSAGE EVIDENCE	"the dune can regulate its own shape, and **maintain it as it moves**."

37. barchans

QUESTION KEYWORD	type of dune known as ____
PASSAGE EVIDENCE	"known as '**barchans**'."

38. shape

QUESTION KEYWORD	grains have similar ____
PASSAGE EVIDENCE	"the grains are a **uniform shape**."

39. tone

QUESTION KEYWORD	dimensions alter the ____
PASSAGE EVIDENCE	"the variations in size affect the **tone**."

40. minerals

QUESTION KEYWORD	mixture of different ____
PASSAGE EVIDENCE	"a precise combination of **minerals** from surrounding rocks."

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Answer Key & Explanations

Reading Test 6 (Volume 7)

Passage 1 — Reef Fish Study

1. FALSE

QUESTION KEYWORD	research concentrated on escape speed
PASSAGE EVIDENCE	"**body size** has received the most attention from researchers" — not escape speed.

2. TRUE

QUESTION KEYWORD	larger animals have better eyesight
PASSAGE EVIDENCE	"characteristics associated with this, such as overall strength and **visual development**."

3. FALSE

QUESTION KEYWORD	early-juvenile fish share similar characteristics
PASSAGE EVIDENCE	"these characteristics **vary from individual to individual** at the beginning of the juvenile stage."

4. NOT GIVEN

QUESTION KEYWORD	adult fish slower than juveniles
PASSAGE EVIDENCE	no comparison of adult vs. juvenile swimming speed is made

5. open ocean

QUESTION KEYWORD	larval stage takes place in ___
PASSAGE EVIDENCE	"an **open ocean** larval phase."

6. reef

QUESTION KEYWORD	juvenile stage close to ___
PASSAGE EVIDENCE	"a juvenile to adult phase near the **reef**."

7. mortality

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QUESTION KEYWORD	____ rates high after settlement
PASSAGE EVIDENCE	"increased levels of **mortality** in the first 48 hours following settlement."

8. new moon

QUESTION KEYWORD	time-change linked to ____
PASSAGE EVIDENCE	"settlement usually occurs within 2-3 days of a **new moon**."

9. laboratory

QUESTION KEYWORD	difficult to settle fish into the ____
PASSAGE EVIDENCE	"adapting them to the **laboratory**."

10. artificial light

QUESTION KEYWORD	trap with the aid of ____
PASSAGE EVIDENCE	"purpose-built traps that use **artificial light** to attract... fish."

11. predatory fish

QUESTION KEYWORD	____ also collected
PASSAGE EVIDENCE	"transported to an aquarium, along with **predatory fish**."

12. mouth size

QUESTION KEYWORD	physical features e.g. ____
PASSAGE EVIDENCE	"physical characteristics of predators, such as **mouth size**."

13. medium

QUESTION KEYWORD	most common prey choice ____ in size
PASSAGE EVIDENCE	"a preference for **medium**-sized prey."

Passage 2 — Will Eating Less Make You Live Longer?

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14. B

QUESTION KEYWORD	same % reduction, different animals
PASSAGE EVIDENCE	"In mice... a diet with 30% fewer calories... monkey trials... cut it by **30%**."

15. G

QUESTION KEYWORD	control monkeys reached older age than normal
PASSAGE EVIDENCE	"several of the NIA control monkeys have lived past the age of 40, far exceeding the **27-year average**."

16. E

QUESTION KEYWORD	food differences causing contrast in findings
PASSAGE EVIDENCE	"Wisconsin monkeys were given highly processed food... NIA diet was based on whole grains... may provide an **explanation** of why the two groups showed different results."

17. C

QUESTION KEYWORD	health problems less likely
PASSAGE EVIDENCE	"suffering less from the diseases of ageing: **diabetes, heart disease and brain diseases**."

18. A

QUESTION KEYWORD	negative opinion of calorie restriction
PASSAGE EVIDENCE	"'I find the idea **monumentally unattractive**!' she says."

19. D

QUESTION KEYWORD	life stage research began
PASSAGE EVIDENCE	"The Wisconsin study began with monkeys in **early adulthood**."

20. A — Rozalyn Anderson

QUESTION KEYWORD	good the studies took dissimilar approaches
PASSAGE EVIDENCE	"the two studies were set up differently, asking the same question in different ways... we will **gain maximally** from that."

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21. C — Leonie Heilbronn

QUESTION KEYWORD	neither study used typical diets
PASSAGE EVIDENCE	"neither group had the right notion of a **standard diet**."

22. A — Rozalyn Anderson

QUESTION KEYWORD	method to find out about ageing-related health issues
PASSAGE EVIDENCE	"a way to tease out what it is that creates increased **disease vulnerability** as a function of age."

23. B — Julie Mattison

QUESTION KEYWORD	positive effect for unhealthy diets
PASSAGE EVIDENCE	"an overweight person who goes to a **fast food restaurant** every day will obviously benefit."

24. ageing

QUESTION KEYWORD	diseases of the ____ process
PASSAGE EVIDENCE	"suffering less from the diseases of **ageing**."

25. lifespan

QUESTION KEYWORD	increases primates' ____
PASSAGE EVIDENCE	"whether calorie restriction extends **lifespan** in primates."

26. conflict

QUESTION KEYWORD	____ between the two data sets
PASSAGE EVIDENCE	"the current data **conflict** will ultimately provide deeper insights."

Passage 3 — Saving the Skink

27. J — small legs

QUESTION KEYWORD	scales and ____

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PASSAGE EVIDENCE	"long bodies and often surprisingly **short limbs**."

28. E — body heat

QUESTION KEYWORD	____ externally controlled
PASSAGE EVIDENCE	"need to **regulate their temperature**, and lie on sunny rocks to absorb heat."

29. F — winter sleep

QUESTION KEYWORD	do not require a long ____
PASSAGE EVIDENCE	"the only kinds that do **not hibernate** through winter."

30. B — location

QUESTION KEYWORD	identification depends on ____
PASSAGE EVIDENCE	"Identification depends on **where specimens are found**."

31. G — captivity

QUESTION KEYWORD	in ____, live for about 30 years
PASSAGE EVIDENCE	"in **zoos or research centres** they can live beyond thirty."

32. NO

QUESTION KEYWORD	cat removal helped stop decline
PASSAGE EVIDENCE	"skinks showed **no signs of recovery**; indeed, there were further declines."

33. NO

QUESTION KEYWORD	parasites affecting physical condition
PASSAGE EVIDENCE	"**no direct health implications** have been detected."

34. YES

QUESTION KEYWORD	best habitat unknown

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PASSAGE EVIDENCE	"we haven't a clue to the **habitat we should be recreating** for the skinks."

35. NOT GIVEN

QUESTION KEYWORD	used to have larger numbers of young
PASSAGE EVIDENCE	only current offspring numbers ("two offspring annually") are stated — no historical comparison given

36. YES

QUESTION KEYWORD	common species may soon be at risk
PASSAGE EVIDENCE	"even **common species may be facing a doubtful future**."

37. C

QUESTION KEYWORD	future of grand/Otago skinks (paragraph 6)
PASSAGE EVIDENCE	"No more than five years remain in which to turn things around - **a close call**."

38. A

QUESTION KEYWORD	problem with fencing project
PASSAGE EVIDENCE	"grands and Otagos have **subtly different environmental preferences**, both of which need catering for."

39. D

QUESTION KEYWORD	'a last resort', paragraph 8
PASSAGE EVIDENCE	"Captive management, however, is a last resort. The priority is to **manage and protect skinks in the wild**."

40. C

QUESTION KEYWORD	final paragraph
PASSAGE EVIDENCE	"with the generous support of a local farmer with grazing land there, has begun the construction of a mammal-proof **fence**."

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Answer Key & Explanations

Reading Test 7 (Volume 7)

Passage 1 — Deep Sea Discovery

1. TRUE

QUESTION KEYWORD	expected total darkness at 15m
PASSAGE EVIDENCE	"once he reached about 15 metres, he **should have been plunged into darkness**."

2. TRUE

QUESTION KEYWORD	saw markings without red filter
PASSAGE EVIDENCE	"Even with the red filter removed, Michiels could pick them out without much trouble."

3. FALSE

QUESTION KEYWORD	other divers assumed they'd see red
PASSAGE EVIDENCE	"no one saw it because **no one expected to see it**."

4. NOT GIVEN

QUESTION KEYWORD	all fish found came from Red Sea
PASSAGE EVIDENCE	later "50 species" identified with no statement restricting them all to the Red Sea

5. FALSE

QUESTION KEYWORD	first suspected fluorescence in Germany
PASSAGE EVIDENCE	"He suspected fluorescence" — stated while still describing the Red Sea reef, before returning to Germany.

6. TRUE

QUESTION KEYWORD	uncertain what creates fluorescence
PASSAGE EVIDENCE	"Michiels says they are **still not sure exactly what is fluorescing**."

7. head

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QUESTION KEYWORD	markings mainly near the ____
PASSAGE EVIDENCE	"markings tend to be on the body towards the **head**."

8. communication

QUESTION KEYWORD	used specifically for ____ purposes
PASSAGE EVIDENCE	"a very particular function: **communication** with other members of the species."

9. enemies

QUESTION KEYWORD	keep ____ away
PASSAGE EVIDENCE	"boost the visibility of body parts they use to signal, for example to ward off **enemies**."

10. location

QUESTION KEYWORD	gobies show their ____
PASSAGE EVIDENCE	"red-eyed gobies use signals to indicate their **location**."

11. species

QUESTION KEYWORD	variations among ____ which are similar
PASSAGE EVIDENCE	"closely related **species** do not have completely identical markings."

12. corals

QUESTION KEYWORD	backgrounds of ____ giving red light
PASSAGE EVIDENCE	"he noticed **corals** glow a dark but faint red too."

13. seahorse

QUESTION KEYWORD	only fish proven to see red
PASSAGE EVIDENCE	"One exception is the **seahorse**, whose eyes are sensitive to red."

Passage 2 — Jellyfish – The Dominant Species

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14. E

QUESTION KEYWORD	prediction of spread direction
PASSAGE EVIDENCE	"tropical species are moving towards the poles... could move into more densely populated subtropical and temperate regions."

15. A

QUESTION KEYWORD	physical characteristics of jellyfish
PASSAGE EVIDENCE	"they lack sharp claws, piercing teeth or even a brain... the stinging power of their tentacles."

16. F

QUESTION KEYWORD	consequences of jellyfish as lone survivors
PASSAGE EVIDENCE	"'monoculture of jellyfish'... an ecosystem that loses fish also loses the species that depend on fish for survival."

17. G

QUESTION KEYWORD	how to avoid further invasions
PASSAGE EVIDENCE	"propose focusing on '**prevention rather than cure**'."

18–19. C, E

QUESTION KEYWORD	effects of jellyfish mentioned
PASSAGE EVIDENCE	C: "devastated the Black Sea... feeding on anchovy eggs"; E: shrimp fishermen "resulting in millions of dollars in losses."

20–21. A, B

QUESTION KEYWORD	possible causes of increase
PASSAGE EVIDENCE	A: "once we remove fish, jellyfish can proliferate"; B: "fertilizer and sewage into the ocean" (Eutrophication).

22. eggs

QUESTION KEYWORD	jellyfish eat their ____
PASSAGE EVIDENCE	"feeding on anchovy **eggs** and the plankton that young fish rely on."

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23. monoculture

QUESTION KEYWORD	state has been named ____
PASSAGE EVIDENCE	"This state has been defined as a '**monoculture** of jellyfish'."

24. nets

QUESTION KEYWORD	damage ____ used for fishing
PASSAGE EVIDENCE	"a jellyfish boom that fills **nets**, causing them to break."

25. temperature

QUESTION KEYWORD	ideal water ____
PASSAGE EVIDENCE	"including their ideal water **temperature** and feeding habits."

26. prevention

QUESTION KEYWORD	focus on ____
PASSAGE EVIDENCE	"focusing on '**prevention** rather than cure'."

Passage 3 — The Pirahã People of Brazil

27. D

QUESTION KEYWORD	Everett, first paragraph
PASSAGE EVIDENCE	"has committed his career to studying their puzzling speech."

28. C

QUESTION KEYWORD	summary of second paragraph
PASSAGE EVIDENCE	"the Pirahã language departs from what some academics have long maintained are **essential and inalienable features** of all human languages."

29. D

QUESTION KEYWORD	why refer to subordinate clauses

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PASSAGE EVIDENCE	"it may be unique for **not employing subordinate clauses**" — given as an example of an unusual feature.

30. B

QUESTION KEYWORD	point about Whorf's work
PASSAGE EVIDENCE	"Everett's findings... have brought new life to a controversial theory by... Whorf."

31. D

QUESTION KEYWORD	why refer to Warlpiri
PASSAGE EVIDENCE	Warlpiri could count in English, but "not a single" Pirahã "could count to 10" in Portuguese — highlighting the Pirahã's distinctiveness.

32. B

QUESTION KEYWORD	Everett's point on intellectual capacities
PASSAGE EVIDENCE	"Everett is at pains to point out that the Pirahã are **not unintelligent**."

33. E — culture

QUESTION KEYWORD	language is a product of their ___
PASSAGE EVIDENCE	"Language... is created by a people's way of life, their belief system and values."

34. A — present

QUESTION KEYWORD	only place value on the ___
PASSAGE EVIDENCE	"the central tenet of the Pirahã culture is to live in the **here and now**."

35. F — grammar

QUESTION KEYWORD	no ___ to describe completed events
PASSAGE EVIDENCE	"the language has no means to **conjugate verbs** in order to describe 'yesterday'."

36. H — individuality

QUESTION KEYWORD	do not celebrate ___

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PASSAGE EVIDENCE	"Standing out or being different is **not encouraged** by the Pirahã."

37. NO

QUESTION KEYWORD	surprised by reaction
PASSAGE EVIDENCE	"Everett anticipated that these findings would be controversial and the reaction **came as expected**."

38. NOT GIVEN

QUESTION KEYWORD	Chomsky critical of methodology
PASSAGE EVIDENCE	only that "two of Chomsky's colleagues" will travel to check his claims — no statement of Chomsky's own criticism

39. YES

QUESTION KEYWORD	disproving recursion → Chomsky wrong
PASSAGE EVIDENCE	"if the Pirahã do not form subordinate clauses, then recursion cannot explain the uniqueness of human language, and this would **negate Chomsky's theories**."

40. NO

QUESTION KEYWORD	Pirahã will benefit from new status
PASSAGE EVIDENCE	"My concern is that soon the Pirahã will simply become one more scientific oddity with every aspect of their lives being **exploited** and analysed."

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Answer Key & Explanations

Reading Test 8 (Volume 7)

Passage 1 — The History of the Guitar

1. FALSE

QUESTION KEYWORD	world's oldest known harp
PASSAGE EVIDENCE	bowl harps existed "**for millennia**" before this "more advanced" harp appeared around 2500-2000 BC.

2. TRUE

QUESTION KEYWORD	Afghan panchtar similar to ancient instrument
PASSAGE EVIDENCE	"survived into modern times in almost unchanged form... such as the Turkish saz and **Afghan panchtar**."

3. NOT GIVEN

QUESTION KEYWORD	Har-Mose an excellent player
PASSAGE EVIDENCE	only his tanbur's construction is described — no comment on his playing skill

4. NOT GIVEN

QUESTION KEYWORD	Cairo museum contains many instruments
PASSAGE EVIDENCE	only Har-Mose's single tanbur is mentioned as being there

5. TRUE

QUESTION KEYWORD	Alaca Huyuk carving matches Kasha's definition
PASSAGE EVIDENCE	"a long neck and sides that clearly **curve inwards**" matches Kasha's definition.

6. FALSE

QUESTION KEYWORD	same number of strings in medieval literature
PASSAGE EVIDENCE	"variations with from **three to five strings**."

7. chartar

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QUESTION KEYWORD	chitarra developed from ____
PASSAGE EVIDENCE	"the four-stringed Persian **chartar** arrived in Spain... It became known as the chitarra."

8. body

QUESTION KEYWORD	larger ____ introduced by Torres
PASSAGE EVIDENCE	"increased the size of the **body**."

9. volume

QUESTION KEYWORD	greater ____
PASSAGE EVIDENCE	"radically increased the **volume** and improved the tone."

10. America

QUESTION KEYWORD	X-braced top first made in ____
PASSAGE EVIDENCE	"German immigrants to **America**... began making guitars with X-braced tops."

11. steel

QUESTION KEYWORD	strings made of ____
PASSAGE EVIDENCE	"**Steel** strings, which became widely available... in the early 1900s."

12. pickups

QUESTION KEYWORD	____ added to guitars
PASSAGE EVIDENCE	"**pickups** were fitted to Hawaiian and jazz guitars."

13. Gibson

QUESTION KEYWORD	well-known version by ____
PASSAGE EVIDENCE	"**Gibson** introduced its famous ES150 model."

Passage 2 — Playing Soccer

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14. E

QUESTION KEYWORD	contrast in learning positions
PASSAGE EVIDENCE	"assignment runs counter to the idea of youth soccer that every child needs to learn **every position**."

15. C

QUESTION KEYWORD	examples outside sport, individual safety
PASSAGE EVIDENCE	"Like the 1960s consumer movement which brought **seat belts and airbags to cars**."

16. F

QUESTION KEYWORD	selection for leadership
PASSAGE EVIDENCE	"Positions of responsibility are earned through competition... youngsters are guaranteed their time... to be **captain**."

17. B

QUESTION KEYWORD	urban changes discourage street soccer
PASSAGE EVIDENCE	"Congested traffic now dominates... requirement... for **official permits** to use public spaces."

18. G

QUESTION KEYWORD	lesson failure teaches
PASSAGE EVIDENCE	"players learn early on how to handle this. As a result... they acquire **resilience**."

19. D

QUESTION KEYWORD	why youth soccer needs coaches
PASSAGE EVIDENCE	"children need to be taught by experts... fear of falling behind that only **expert instruction** can prevent."

20–21. C, E

QUESTION KEYWORD	reasons for street soccer popularity
PASSAGE EVIDENCE	C: "no large space is needed... goal posts... typically absent"; E: "does not necessarily require standard 11-a-side teams."

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22–23. **B, D**

QUESTION KEYWORD	results of 1960s parenting style
PASSAGE EVIDENCE	B: "some youth leagues prohibited heading the ball altogether"; D: "handed out 'participation' prizes to every player."

24. **principle**

QUESTION KEYWORD	key ____ of youth soccer
PASSAGE EVIDENCE	"'Everyone's a winner; no one's a loser' is a guiding **principle** of youth soccer."

25. **reward**

QUESTION KEYWORD	get some kind of ____
PASSAGE EVIDENCE	"giving less than your best receives the same **reward** as giving your best."

26. **record**

QUESTION KEYWORD	keeps a ____ of results
PASSAGE EVIDENCE	"in street soccer a formal **record** is not kept."

Passage 3 — The History of Chocolate

27. **F**

QUESTION KEYWORD	part of cocoa tree used
PASSAGE EVIDENCE	"These pods contain the **beans** used in chocolate production."

28. **H**

QUESTION KEYWORD	annual consumption per capita US
PASSAGE EVIDENCE	"Americans consume over **five kilogrammes** of chocolate per person each year."

29. **F**

QUESTION KEYWORD	cultivated trees more vulnerable to pests

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PASSAGE EVIDENCE	"cocoa trees grown on farms are **more vulnerable to disease and pests** than their wild counterparts."

30. E

QUESTION KEYWORD	community preferred sweetened drinks
PASSAGE EVIDENCE	"Spanish elite first developed a taste for **sweetened chocolate drinks**."

31. D

QUESTION KEYWORD	how Aztecs acquired cocoa beans
PASSAGE EVIDENCE	"they sourced cocoa beans through **extensive trade networks**."

32. FALSE

QUESTION KEYWORD	ruling class forbade cocoa in rituals
PASSAGE EVIDENCE	"the aristocracy, particularly the royal families, **consumed chocolate during elaborate ceremonies**."

33. NOT GIVEN

QUESTION KEYWORD	Cortés made an investment
PASSAGE EVIDENCE	only that he "recognised the potential economic value" — no statement of an actual investment

34. FALSE

QUESTION KEYWORD	wild trees more susceptible than cultivated
PASSAGE EVIDENCE	reverse is stated: "cocoa trees grown on farms are **more vulnerable**... than their wild counterparts."

35. TRUE

QUESTION KEYWORD	some seeds discarded during production
PASSAGE EVIDENCE	"extracting the **outer shells**, leaving only the inner nibs."

36. TRUE

QUESTION KEYWORD	more prestige in Europe/US than elsewhere

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PASSAGE EVIDENCE	"its reputation remains **most prominent in Europe and the United States**."

37. outer shells

QUESTION KEYWORD	stripping their ____
PASSAGE EVIDENCE	"extracting the **outer shells**, leaving only the inner nibs."

38. chocolate liquor

QUESTION KEYWORD	producing ____
PASSAGE EVIDENCE	"ground into a dense paste known as **chocolate liquor**."

39. cocoa butter

QUESTION KEYWORD	mixing with ____
PASSAGE EVIDENCE	"blend the chocolate liquor with precise ratios of sugar, milk, and **cocoa butter**."

40. moulds

QUESTION KEYWORD	manufacturing shapes in ____
PASSAGE EVIDENCE	"placed into **moulds** to create various forms."

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Answer Key & Explanations

Reading Test 9 (Volume 7)

Passage 1 — Dust and the American West

1. TRUE

QUESTION KEYWORD	dramatic rise in dust, late 19th century
PASSAGE EVIDENCE	"from about 1860 to 1900, **dust deposition rates shot up**."

2. FALSE

QUESTION KEYWORD	Anasazi disappeared due to dust
PASSAGE EVIDENCE	collapse was due to "a near-permanent drought"; "Even as their civilization was collapsing, the Anasazi seem to have **protected their soils from erosion**" — i.e. not dusty.

3. FALSE

QUESTION KEYWORD	bison in American Southwest before cattle
PASSAGE EVIDENCE	"there were few grazers in the American Southwest until the Europeans came. **No bison** and few antelope or deer."

4. NOT GIVEN

QUESTION KEYWORD	bison decline due to hunting
PASSAGE EVIDENCE	no mention of hunting or bison population decline anywhere in the passage

5. NOT GIVEN

QUESTION KEYWORD	railroads more expensive than expected
PASSAGE EVIDENCE	only that it was "funded by a bubble of speculative investment" — no cost-overrun comparison given

6. FALSE

QUESTION KEYWORD	Aztec company protected the land
PASSAGE EVIDENCE	"it was only interested in quick profits and had **little incentive to protect the soils** from overgrazing."

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7. TRUE

QUESTION KEYWORD	land remains infertile today
PASSAGE EVIDENCE	"even now **few of the pastures have recovered**."

8. cattle herds

QUESTION KEYWORD	laws to control the size of ____
PASSAGE EVIDENCE	"federal authorities finally sought to limit **cattle herds**."

9. winds

QUESTION KEYWORD	soil not affected by high ____
PASSAGE EVIDENCE	"These crusts can survive **winds** of up to a hundred miles an hour, but cattle hooves break the crust."

10. lake sediments

QUESTION KEYWORD	____ in Colorado contained dust
PASSAGE EVIDENCE	"Neff first discovered dust in Colorado **lake sediments** laid down in the 19th century."

11. size

QUESTION KEYWORD	chemical content as well as its ____
PASSAGE EVIDENCE	"investigating the **size** and chemical composition of the dust."

12. nutrients

QUESTION KEYWORD	bringing in ____ not normally found there
PASSAGE EVIDENCE	"They carry **nutrients** with them into areas which previously evolved and survived without them."

13. glaciers

QUESTION KEYWORD	disappearance of some ____ and snow
PASSAGE EVIDENCE	"the shrinking of **glaciers** across the American West."

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Passage 2 — Urban Regeneration in Berlin

14. iv

QUESTION KEYWORD	heading, Paragraph A
PASSAGE EVIDENCE	"Tracks, sheds and warehouses belonging to **three old railway stations**."

15. viii

QUESTION KEYWORD	heading, Paragraph B
PASSAGE EVIDENCE	"became increasingly **run down and abandoned**... used as a rubbish tip."

16. i

QUESTION KEYWORD	heading, Paragraph C
PASSAGE EVIDENCE	"stimulate the development of sixteen new hectares... capable of integrating **different generations and social groups**."

17. vii

QUESTION KEYWORD	heading, Paragraph D
PASSAGE EVIDENCE	"a concrete **footpath**... a pair of **railway lines**... a train slowly travels along these lines."

18. iii

QUESTION KEYWORD	heading, Paragraph E
PASSAGE EVIDENCE	describes the terrace, swings, "a nursery, sports fields, concave surfaces for skateboards" etc.

19. v

QUESTION KEYWORD	heading, Paragraph F
PASSAGE EVIDENCE	"The heart of Berlin now has a new green lung... fits neatly into a large-scale... **robust general order**."

20–21. B, C

QUESTION KEYWORD	reasons Gleisdreieck chosen for park

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PASSAGE EVIDENCE	B: "German Museum of Technology was established there, attracting large numbers of visitors"; C: "unification of the formerly divided city gave the zone more centrality."

22–23. **A, D**

QUESTION KEYWORD	what State of Berlin hoped to achieve
PASSAGE EVIDENCE	A: "capable of integrating different generations and social groups"; D: "reconcile these goals with the conservation of railway heritage."

24. **meadow**

QUESTION KEYWORD	middle of the park is a ____
PASSAGE EVIDENCE	"planned around a large central **meadow**, intersected... by a concrete footpath... by a pair of railway lines."

25. **museum**

QUESTION KEYWORD	journey between base and a ____
PASSAGE EVIDENCE	"a train slowly travels along these lines from its parking shed to the German **Museum** of Technology."

26. **edges**

QUESTION KEYWORD	sports/leisure areas around the ____
PASSAGE EVIDENCE	"The **edges** of the park are finished with a collection of distinctive, functional spaces."

Passage 3 — The Fluoridation Controversy

27. **A**

QUESTION KEYWORD	optimum fluorine amount calculated partly by
PASSAGE EVIDENCE	"depending on the **regional temperature** and hence the amount of water people are likely to drink."

28. **C**

QUESTION KEYWORD	reason for opposing fluoridation
PASSAGE EVIDENCE	"fluoridation is thought to be an infringement on **individual rights** because it is compulsory medication."

29. **D**

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QUESTION KEYWORD	why mention Kuhn
PASSAGE EVIDENCE	"the shift from one scientific way of thinking to another... can include **social factors**."

30. A

QUESTION KEYWORD	Sutton's discovery about earlier studies
PASSAGE EVIDENCE	"found that each showed significant **methodological shortcomings**."

31. B

QUESTION KEYWORD	last paragraph, about scientists
PASSAGE EVIDENCE	"not an issue on which **declarations by scientific experts ought to be considered the final word**."

32. NO

QUESTION KEYWORD	science should be kept separate from values
PASSAGE EVIDENCE	the sociological challenge shows "social values may also be involved" throughout scientific knowledge.

33. YES

QUESTION KEYWORD	sociologists disregarded scientists' doubts
PASSAGE EVIDENCE	"sociologists have also downplayed... by ignoring the number of eminent scientists who have questioned aspects of fluoridation."

34. NOT GIVEN

QUESTION KEYWORD	ignored by scientists outside North America
PASSAGE EVIDENCE	only that "Sutton's book is not cited... by any sociologist" — no statement specific to scientists outside North America

35. YES

QUESTION KEYWORD	valid arguments on both sides
PASSAGE EVIDENCE	"not between rationality and irrationality but is a **legitimate difference in values**."

36. B

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QUESTION KEYWORD	traditional view of science
PASSAGE EVIDENCE	"established through scientific methods and objectively applied by members of a scientific community." *(Note: the source file's option B text was corrupted/nonsensical; reconstructed here to match paragraph D so the given answer letter B is usable.)*

37. **G**

QUESTION KEYWORD	sociological view of science
PASSAGE EVIDENCE	"scientific knowledge is socially negotiated, and inevitably linked to the values of the relevant parties."

38. **A**

QUESTION KEYWORD	Collins's opinion
PASSAGE EVIDENCE	"the outcome of experiments was not something whose meaning could be **immediately comprehended**."

39. **D**

QUESTION KEYWORD	supporter's conclusion
PASSAGE EVIDENCE	"the evidence for harm is limited and dubious. I think the likely benefits outweigh the possible dangers."

40. **C**

QUESTION KEYWORD	opponent's conclusion
PASSAGE EVIDENCE	"I oppose fluoridation of water supplies and favor the **voluntary use** of fluoride tablets by those who want to take them."

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Answer Key & Explanations

Reading Test 10 (Volume 7)

Passage 1 — Dolls Through the Ages

1. clay

QUESTION KEYWORD	____ used for the hair
PASSAGE EVIDENCE	"'hair' made of **clay**."

2. goddesses

QUESTION KEYWORD	dolls given to ____
PASSAGE EVIDENCE	"offered their wooden dolls to **goddesses**."

3. limbs

QUESTION KEYWORD	separate clothes and ____
PASSAGE EVIDENCE	"dolls with movable **limbs** and removable garments."

4. wax

QUESTION KEYWORD	dolls made of ____
PASSAGE EVIDENCE	"**wax** dolls grew in popularity in the 17th and 18th centuries."

5. plaster

QUESTION KEYWORD	moulds made of ____
PASSAGE EVIDENCE	"cover it with **plaster** to create a mould."

6. composition

QUESTION KEYWORD	mixtures known as ____
PASSAGE EVIDENCE	"'**Composition**' is a collective term for mixtures of pulped wood or paper."

7. FALSE

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QUESTION KEYWORD	bisque less realistic than china
PASSAGE EVIDENCE	"Bisque... looked **more like skin** than china did."

8. TRUE

QUESTION KEYWORD	French dolls cost more than German
PASSAGE EVIDENCE	"German bisque dolls became quite popular because they were **not as expensive**."

9. FALSE

QUESTION KEYWORD	first rag dolls made in 1850s
PASSAGE EVIDENCE	"For **many centuries**, rag dolls were made by mothers" — only commercial production began in the 1850s.

10. TRUE

QUESTION KEYWORD	only cotton/linen = cloth dolls
PASSAGE EVIDENCE	"'Cloth doll' refers to a subset of rag dolls made of **linen or cotton**."

11. TRUE

QUESTION KEYWORD	celluloid dolls lost colour
PASSAGE EVIDENCE	"propensity to **fade in bright light**."

12. FALSE

QUESTION KEYWORD	composition lasted longer than 1940s plastic
PASSAGE EVIDENCE	hard plastic dolls "resembled composition dolls, but they were **much more durable**."

13. NOT GIVEN

QUESTION KEYWORD	collectors prefer original clothing
PASSAGE EVIDENCE	not discussed anywhere in the passage

Passage 2 — Biophilic Design

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14. C

QUESTION KEYWORD	rapid change, negative effect
PASSAGE EVIDENCE	"rates of technological progress are far exceeding rates of psychological evolution... increased stress."

15. B

QUESTION KEYWORD	architect uncharacteristic of biophilia
PASSAGE EVIDENCE	"the box-like blocks of flats designed by his contemporary **Le Corbusier**."

16. A

QUESTION KEYWORD	two main aims of biophilia
PASSAGE EVIDENCE	"seeks both to avoid or minimize harmful impacts... and... to **provide and restore beneficial contacts**."

17. H

QUESTION KEYWORD	positive claim, early architecture
PASSAGE EVIDENCE	"producing magnificent structures with which we are **still awed**."

18. B

QUESTION KEYWORD	architects focused on innovation
PASSAGE EVIDENCE	"architects have put **creative originality** before the needs of the people."

19. G

QUESTION KEYWORD	features which conserve energy
PASSAGE EVIDENCE	the bio-wall "has the potential to fulfill the building's **fresh air intake requirements**" via natural filtration, reducing reliance on mechanical systems.

20. technological

QUESTION KEYWORD	____ advancements happening quickly
PASSAGE EVIDENCE	"rates of **technological** progress are far exceeding rates of psychological evolution."

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21. stress

QUESTION KEYWORD	deal with the ____
PASSAGE EVIDENCE	"which in turn leads to increased **stress**."

22. sunlight

QUESTION KEYWORD	plenty of ____
PASSAGE EVIDENCE	"one crucial element of the natural landscape to human health is **sunlight**."

23. tree

QUESTION KEYWORD	could see a ____
PASSAGE EVIDENCE	"One group had a view of **tree**."

24. air

QUESTION KEYWORD	improves quality of the ____
PASSAGE EVIDENCE	"The wall purifies the **air**."

25–26. B, D

QUESTION KEYWORD	Oberlin Lewis Center statements
PASSAGE EVIDENCE	B: "engineered a healthy and comfortable space for students, while ensuring the surrounding environment was undamaged"; D: "help to redefine the relationship between humankind and the environment."

Passage 3 — Biology of Bitterness

27. B

QUESTION KEYWORD	experiment on bitterness conducted
PASSAGE EVIDENCE	"a test conducted at the Linguagen Corporation... Sets of two miniature white paper cups."

28. I

QUESTION KEYWORD	future application

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PASSAGE EVIDENCE	"In a few years, perhaps... a bitter blocker in a bottle."

29. C

QUESTION KEYWORD	different meaning for humans/animals
PASSAGE EVIDENCE	"Humans are the only species that enjoys bitter taste... Every other species is averse to bitter because it means bad news."

30. E

QUESTION KEYWORD	spread of bitterness inside body
PASSAGE EVIDENCE	"relayed via proteins known as G proteins... triggers a cascade of chemical reactions... delivers a signal to the brain."

31. G

QUESTION KEYWORD	how AMP blocks bitterness
PASSAGE EVIDENCE	"it attaches to **bitter-taste receptors**."

32. H

QUESTION KEYWORD	blocker lowers unhealthy impact
PASSAGE EVIDENCE	"compounds like AMP will help make processed foods **less unhealthy**."

33. A

QUESTION KEYWORD	bitterness introduced from a fruit
PASSAGE EVIDENCE	"Naringin, a natural chemical compound found in **grapefruit**, tastes bitter."

34. D

QUESTION KEYWORD	genetic feature determines sensitivity
PASSAGE EVIDENCE	"the differences appear to be **genetic**."

35. naringin

QUESTION KEYWORD	substance called ____

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PASSAGE EVIDENCE	"**Naringin**, a natural chemical compound found in grapefruit, tastes bitter."

36. poisonous

QUESTION KEYWORD	plant is ____
PASSAGE EVIDENCE	"letting them know when a plant may be **poisonous**."

37. supertasters

QUESTION KEYWORD	____ have plenty of ____
PASSAGE EVIDENCE	"Some people, known as **supertasters**, are especially sensitive... because they have an unusually high number of taste buds."

38. taste buds

QUESTION KEYWORD	plenty of ____
PASSAGE EVIDENCE	"an unusually high number of **taste buds**."

39. A

QUESTION KEYWORD	main feature of AMP
PASSAGE EVIDENCE	"a compound that **blocks the bitterness** in foods without making them less nutritious."

40. D

QUESTION KEYWORD	main function of G protein
PASSAGE EVIDENCE	"triggers a cascade of chemical reactions... delivers a signal to the **brain** that registers as bitter."

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